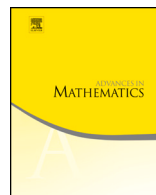




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Unified theory on V-states structures for active scalar equations [☆]

Taoufik Hmidi ^{a,*}, Liutang Xue ^b, Zhilong Xue ^b^a *New York University Abu Dhabi, PO Box 129188, Abu Dhabi, United Arab Emirates*^b *Laboratory of Mathematics and Complex Systems (MOE), School of Mathematical Sciences, Beijing Normal University, Beijing 100875, PR China*

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ABSTRACT

This paper revolves around the existence of V-states close to Rankine vortices for active scalar equations with completely monotone kernels. This allows to unify various results on this topic related to geophysical flows. A key ingredient is a new factorization formula for the spectrum using a universal function which is independent of the model. This function admits several interesting properties allowing to track the spectrum distribution.

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* Corresponding author.

E-mail addresses: th2644@nyu.edu (T. Hmidi), xuelt@bnu.edu.cn (L. Xue), zhilongxue@mail.bnu.edu.cn (Z. Xue).

Geophysical flows
 Completely monotone kernels
 Bifurcation theory

Contents

1.	Introduction	2
2.	Time periodic patches and linearization	11
2.1.	Boundary equation	11
2.2.	Linearization	12
2.3.	Spectrum factorization	15
3.	Analysis of the universal function ϕ_n	16
3.1.	Comparison theorem	17
3.2.	Positivity and monotonicity of ϕ_n	18
3.3.	Asymptotic structure of ϕ_n	20
3.4.	The decay rate of $\phi_n - \phi_{n+1}$	25
3.5.	Spectrum convexity	30
4.	Proof of the main theorems	32
4.1.	Strong regularity	33
4.2.	Spectral study	45
5.	Applications to geophysical flows	50
5.1.	2D Euler equation in the whole space	50
5.2.	gSQG equation in the whole space	52
5.3.	QGSW equation in the whole space	55
5.4.	Euler- α equation in the whole space	57
5.5.	2D Euler equation in the unit disc	58
5.6.	gSQG equation in the unit disc	59
5.7.	QGSW equation in the unit disc	61
6.	Tools	63
6.1.	Completely monotone functions	64
6.2.	Boundedness property of some operators on the torus	68
6.3.	Bessel functions and Hankel transform	71
6.4.	Crandall-Rabinowitz's theorem	73
	References	74

1. Introduction

In this paper, we consider the Cauchy problem of the following two-dimensional (abbr. 2D) active scalar equation

$$\begin{cases} \partial_t \omega + (v \cdot \nabla) \omega = 0, & (t, \mathbf{x}) \in (0, +\infty) \times \mathbf{D}, \\ v = \nabla^\perp \psi, & (t, \mathbf{x}) \in (0, +\infty) \times \mathbf{D}, \\ \omega(0, \mathbf{x}) = \omega_0(\mathbf{x}), & \mathbf{x} \in \mathbf{D}, \end{cases} \quad (1.1)$$

where $\mathbf{D}$ is either the whole space $\mathbb{R}^2$ or the unit disc $\mathbb{D}$, $\nabla^\perp = (\partial_2, -\partial_1)$, $v = (v_1, v_2)$ refers to the velocity field, ω is a scalar field understood as vorticity or temperature or buoyancy of the fluid, and the stream function ψ is prescribed through the following relation

$$\psi(\mathbf{x}) = \int_{\mathbf{D}} K(\mathbf{x}, \mathbf{y}) \omega(\mathbf{y}) d\mathbf{y}. \quad (1.2)$$

In the following, we identify the complex plane $\mathbb{C}$ with $\mathbb{R}^2$. We will focus on the kernel K that fulfills the following symmetry conditions

$$K(\mathbf{x}, \mathbf{y}) = K(\mathbf{y}, \mathbf{x}), \quad K(\bar{\mathbf{x}}, \bar{\mathbf{y}}) = K(\mathbf{x}, \mathbf{y}), \quad (1.3)$$

with $\bar{\mathbf{x}} \triangleq (x_1, -x_2)$ the reflection of $\mathbf{x} = (x_1, x_2)$, and

$$K(e^{i\theta} \mathbf{x}, e^{i\theta} \mathbf{y}) = K(\mathbf{x}, \mathbf{y}), \quad \theta \in \mathbb{R}. \quad (1.4)$$

Indeed, given assumptions (1.3) and (1.4), $\omega_0(\mathbf{x}) = \mathbf{1}_{b\mathbb{D}}(\mathbf{x})$, where $b > 0$ ($b\mathbb{D} \subset \mathbf{D}$), serves as a stationary solution to the system (1.1)-(1.2); refer to Section 2.

Taking different forms of the kernel K , the equation (1.1)-(1.2) includes several important hydrodynamic models as special cases.

▷ Case $\mathbf{D} = \mathbb{R}^2$,

$$K(\mathbf{x}, \mathbf{y}) = -\frac{1}{2\pi} \log |\mathbf{x} - \mathbf{y}|, \quad \text{that is,} \quad \psi(\mathbf{x}) = (-\Delta)^{-1} \omega(\mathbf{x}),$$

(1.1)-(1.2) becomes the 2D Euler equation in the vorticity form, which describes the motion of a 2D inviscid incompressible fluid and is a fundamental model in fluid dynamics.

▷ Case $\mathbf{D} = \mathbb{R}^2$,

$$K(\mathbf{x}, \mathbf{y}) = c_\beta |\mathbf{x} - \mathbf{y}|^{-\beta}, \quad \beta \in (0, 2), \quad \text{that is,} \quad \psi(\mathbf{x}) = (-\Delta)^{-1+\frac{\beta}{2}} \omega(\mathbf{x}),$$

with $c_\beta \triangleq \frac{\Gamma(\frac{\beta}{2})}{\pi^{2-\beta} \Gamma(1-\frac{\beta}{2})}$, (1.1)-(1.2) is the inviscid generalized surface quasi-geostrophic (abbr. gSQG) equation. In particular, for $\beta = 1$, it is the surface quasi-geostrophic (abbr. SQG) equation, which is a simplified model to track the atmospheric circulation near the tropopause [52] and the ocean dynamics in the upper layers [71]. This model in the range $0 < \beta < 1$ was introduced in [19] as an interpolation between the 2D Euler equation and the SQG equation.

▷ Case $\mathbf{D} = \mathbb{R}^2$,

$$K(\mathbf{x}, \mathbf{y}) = \frac{1}{2\pi} \mathbf{K}_0(\varepsilon |\mathbf{x} - \mathbf{y}|), \quad \varepsilon > 0, \quad \text{that is,} \quad \psi(\mathbf{x}) = (-\Delta + \varepsilon^2)^{-1} \omega(\mathbf{x}),$$

with $\mathbf{K}_0$ the modified Bessel function (see Subsection 6.3), (1.1)-(1.2) is the quasi-geostrophic shallow water (abbr. QGSW) equation. This model is derived asymptotically from the rotating shallow water equations in the limit of fast rotation and small variation of free surface [89]. The parameter ε is known as the inverse ‘Rossby deformation length’, and small ε physically corresponds to a nearly rigid free surface.

▷ Case $\mathbf{D} = \mathbb{D}$, consider the kernel

$$K(\mathbf{x}, \mathbf{y}) = -\frac{1}{2\pi} \log \left(\frac{|\mathbf{x} - \mathbf{y}|}{|1 - \overline{\mathbf{x}}\mathbf{y}|} \right),$$

which describes the Green function associated with the spectral Laplace operator $-\Delta$ in the unit disc $\mathbb{D}$ with Dirichlet boundary condition, and (1.1)-(1.2) becomes the 2D Euler equation (in vorticity form) in the unit disc.

Besides these examples, we refer to Section 5 for more active scalar equations (1.1)-(1.2).

Due to their substantial physical relevance and formal simplicity, active scalar equations (1.1)-(1.2) have garnered considerable attention over the past decades. Significant progress has been achieved across multiple fronts. The global well-posedness of classical solutions for the 2D Euler equation in the whole space $\mathbb{R}^2$ or in any smooth bounded domain $\mathbf{D}$ has been well-known for a long time, see, for instance [77], while it is still an open problem for the gSQG equation with $\beta \in (0, 2)$ in the whole space or torus $\mathbb{T}^2$. As to the local well-posedness issue in the framework of Sobolev spaces, it was explored in [15] for the whole space, in [62,92] for the half space, and in [17] for smooth bounded domains. On the other hand, L^2 -weak solutions for the gSQG equation are known to exist globally in time; see [72,76,82] for $\mathbb{R}^2$ and [18,79] for smooth bounded domains. Recently, their non-uniqueness aspect in the plane has been investigated in [7,61].

Another significant class of solutions extensively studied in the literature involves the *patch solutions*, which are weak solutions to (1.1)-(1.2) with initial data in the form of the characteristic function of a bounded domain D , that is, $\omega_0(\mathbf{x}) = \mathbf{1}_D(\mathbf{x})$. According to Yudovich [91], the vorticity patch solution for 2D Euler equation in whole space is globally well-defined keeping during the motion the form of the patch structure. The patch problem initiated in 1980s revolves around the regularity persistence of the boundary. It aims to determine whether the initial regularity, for instance of type $C^{k,\gamma}$ with $k \in \mathbb{N}^*$ and $0 < \gamma < 1$, can persist for all time. This problem was successfully tackled by Chemin [16], see also Bertozzi and Constantin [6] for another proof. Similar results in the half plane or within the smooth bounded domain were also obtained in [28,67,64]. The situation becomes more involved for the gSQG equation with $\beta \in (0, 2)$. Here, only local-in-time persistence in Sobolev spaces has been established, as proved in [15,31,83]. In this case, some numerical experiments show strong evidence for the formation of finite-time singularities; see [19,87,88]. Finite-time singularity results with multi-signed patches in the half plane in various ranges of β have been achieved by Kiselev et al. [67,68] and also [32,92]. Very recently, ill-posedness results in various Hölder and Sobolev spaces, associated with the boundary of patches or with the initial data, have been established in [20,21,65,66].

The main goal of this paper is to construct time periodic solutions in the patch form for active scalar equations (1.1)-(1.2) with a general kernel form K that will cover most of the equations that arise in geophysical flows. This type of patch solutions is commonly known as *V-states*, or *relative equilibria*, or *rotating patches*. Their shape is not altered

during the motion and can be described through a rigid body transformation. Identifying $\mathbb{R}^2$ with the complex plane $\mathbb{C}$ and assuming that the center of rotation is the origin, the V-states take the form

$$\omega(\mathbf{x}, t) = \mathbf{1}_{D_t}(\mathbf{x}), \quad \text{with} \quad D_t = e^{i\Omega t} D,$$

where $D \subset \mathbb{R}^2$ is a bounded domain. The real number Ω is called the *angular velocity* of the rotating domain and will play the role of a bifurcation parameter.

The V-states study for active scalar equations (1.1)-(1.2) has a long history and is still an active area with intensive research. Over the last few decades, significant contributions have shaped this field at both the analytical and numerical levels. The first example of rotating patches for Euler equations dates back to Kirchhoff [63], who proved that any ellipse with the semi-axis a and b rotates uniformly with the angular velocity $\Omega = \frac{ab}{a^2+b^2}$, see also [69, p. 232] or [75, p. 306]. About a century later, Deem and Zabusky [27] conducted numerical computations that showcase the existence of implicit V-states with m -fold symmetry. This was analytically justified by Burbea [8] using bifurcation theory and conformal parametrization. In fact, the bifurcation from the Rankine vortices (radial case) occurs at the angular velocities $\Omega = \frac{m-1}{2m}$ ($m \geq 2$). Later, Hmidi, Mateu, and Verdera [56] revisited this construction and show the C^∞ boundary regularity and convexity of the bifurcated V-states close to Rankine vortices. The analyticity of the boundary has been recently explored by Castro, Córdoba and Gómez-Serrano in [12], and its global version has been discussed by Hassainia, Masmoudi and Wheeler in [49].

The V-states for the gSQG model in the whole plane was first investigated by Hassainia and Hmidi in [45] and confirmed a similar result to Burbea for all $\beta \in (0, 1)$. Later, Castro, Córdoba and Gómez-Serrano [11] extended the construction for the range $\beta \in [1, 2)$ and proved the C^∞ boundary regularity; see also [12] for the real analyticity of the V-states boundary.

Similar rigid time periodic solutions for the QGSW equation were studied by Dritschel, Hmidi, and Renault [29]. The topic of V-states in radial domains with rigid boundary was initiated by De la Hoz, Hassainia, Hmidi, and Mateu for the 2D Euler equation in [25] and by the authors of this paper for the gSQG equation [58].

In addition to the above results, there are abundant papers in the recent literature on the mathematical study of V-states for the active scalar equation (1.1)-(1.2) from various aspects. For instance, a second family of countable branches bifurcate from Kirchhoff's ellipses was proved in [12,53]; the existence of doubly connected V-states close to the annulus was established in [24,26,40,54,84]; concentrated multi vortices centered at regular n -gons or distributed according to suitable periodic spatial patterns are analyzed in [10,33,34,36,46,51,55]. Very recently, the exploration of time quasi-periodic vortex patches for some active scalar equations (1.1)-(1.2) has been conducted by employing advanced tools from the KAM theory, we refer to [4,5,41,47,48,50,57,85]. For other related topics, one can see [2,11,13,14,24,35,37–39,42,44,55] and the references therein.

In this paper, we intend to develop a unified approach on the construction of V-states for the active scalar equation (1.1)-(1.2) near Rankine vortices. More precisely, we shall apply the local bifurcation theory to construct time periodic patch solutions around the Rankine vortices of type $\mathbf{1}_{b\mathbb{D}}$, with $b > 0$ and $b\mathbb{D} \subset \mathbf{D}$, for the system (1.1)-(1.2) by imposing general assumptions on the kernel K , which include all the aforementioned important models as special examples. It should be emphasized that the explicit expression of K plays a crucial role in the analysis of previous work, especially along the spectrum study where we need the monotonicity of spectrum sequence.

Before describing our primary contributions, we need to introduce the equations that govern rotating simply-connected patches. As we will see in Section 2, we find it more convenient to parametrize the boundary of the V-states close to the stationary solution $\mathbf{1}_{b\mathbb{D}}$ in terms of polar coordinates $\theta \in \mathbb{R} \mapsto \sqrt{b^2 + 2r(\theta)}e^{i\theta}$ with $b > 0$, so that $b\mathbb{D} \subset \mathbf{D}$. The contour dynamics equation can be formulated as a nonlinear integro-differential equation $F(\Omega, r) = 0$ with

$$\begin{aligned} F(\Omega, r) &\triangleq \Omega r'(\theta) + \partial_\theta \left(\int_{\mathbb{T}} \int_0^{R(\eta)} K(R(\theta)e^{i\theta}, \rho e^{i\eta}) \rho d\rho d\eta \right), \quad R(\theta) \triangleq \sqrt{b^2 + 2r(\theta)} \\ &\triangleq \Omega r'(\theta) + F_1(r). \end{aligned} \quad (1.5)$$

One can easily show that $F(\Omega, 0) = 0$ for all $\Omega \in \mathbb{R}$ and therefore the next task is to check that the local bifurcation tools such as Crandall-Rabinowitz's theorem (see Theorem 6.2 below) apply in this framework.

The first main result concerns the stream function ψ associated with a convolution kernel

$$K(\mathbf{x}, \mathbf{y}) = K_0(|\mathbf{x} - \mathbf{y}|), \quad \forall \mathbf{x}, \mathbf{y} \in \mathbf{D}, \quad (1.6)$$

where the function $t \in (0, \infty) \mapsto K_0(t)$ satisfies the following assumptions,

(A1) *Complete monotonicity*: the function $-K'_0$ is a nonzero completely monotone function (see Definition 6.1), equivalently, there exists a non-negative measure μ on $[0, \infty)$ such that

$$-K'_0(t) = \int_0^\infty e^{-tx} d\mu(x), \quad \forall t > 0. \quad (1.7)$$

(A2) *Integrability assumption*: there exists a constant $a_0 > 0$ and some $\alpha \in (0, 1)$ such that

$$\int_0^{a_0} |K_0(t)| t^{-\alpha+\alpha^2} dt < \infty. \quad (1.8)$$

It is important to emphasize that the parameter α introduced in (A2) characterizes the singularity of the kernel $t \in (0, \infty) \mapsto K_0(t)$ near $t = 0$. This parameter will appear in the boundary regularity of the V-states that will be constructed subsequently. Actually, the boundary belongs to the Hölder class $C^{2-\alpha}$, we refer to (4.7) and (4.8) for detailed function spaces.

Note that assumptions (A1)-(A2) encompass as special examples the classical equations: 2D Euler, gSQG and QGSW equations; see Section 5 for more discussion.

Our first main result is as follows.

Theorem 1.1. *Consider the case (1.6) with K_0 satisfying the conditions (A1)-(A2). Then for any $m \in \mathbb{N}^*$, there exists a family of m -fold symmetric V-states for the active scalar equation (1.1)-(1.2) bifurcating from the Rankine vortices $\mathbf{1}_{b\mathbb{D}}(\mathbf{x})$, provided that $b\mathbb{D} \subset \mathbf{D}$, at the angular velocity*

$$\Omega_{m,b}^0 = \int_{\mathbb{T}} K_0(|2b \sin \frac{\eta}{2}|) \cos \eta \, d\eta - \int_{\mathbb{T}} K_0(|2b \sin \frac{\eta}{2}|) \cos(m\eta) \, d\eta. \quad (1.9)$$

Motivated by the papers [25,58] on the V-states in radial domains, our second main result considers the perturbative case where the kernel involved in the stream function takes a more general form

$$K(\mathbf{x}, \mathbf{y}) = K_0(|\mathbf{x} - \mathbf{y}|) + K_1(\mathbf{x}, \mathbf{y}), \quad (1.10)$$

where K_0 satisfies (A1)-(A2), while K_1 satisfies

(A3) *Regularity assumption:* $K_1 \in C_{\text{loc}}^k(\mathbf{D}^2)$ for some $k \geq 4$.

(A4) *Symmetry assumption:* we assume that for any $\mathbf{x}, \mathbf{y} \in \mathbf{D}$,

$$K_1(\mathbf{x}, \mathbf{y}) = K_1(\mathbf{y}, \mathbf{x}), \quad K_1(\bar{\mathbf{x}}, \bar{\mathbf{y}}) = K_1(\mathbf{x}, \mathbf{y}), \quad K_1(e^{i\theta}\mathbf{x}, e^{i\theta}\mathbf{y}) = K_1(\mathbf{x}, \mathbf{y}), \quad \forall \theta \in \mathbb{R},$$

where $\bar{\mathbf{x}} = (x_1, -x_2)$ is the reflection of $\mathbf{x} = (x_1, x_2)$.

We note that under assumption (A4) the kernel K defined in (1.10) directly satisfies the conditions (1.3)-(1.4) imposed before.

Theorem 1.2. *Consider the general case (1.10) with the assumptions (A1)-(A4). Then there exists a sufficiently large number $m_0 \in \mathbb{N}^*$, such that for any $m \geq m_0$, the equation (1.1)-(1.2) admits a family of m -fold symmetric V-states bifurcating from the trivial solution $\mathbf{1}_{b\mathbb{D}}(\mathbf{x})$, provided that $b\mathbb{D} \subset \mathbf{D}$, at some angular velocity $\Omega_{m,b}$.*

Remark 1.1 (Formula of $\Omega_{m,b}$). The angular velocity $\Omega_{m,b}$ in Theorem 1.2 can be explicitly linked to the kernel as follows

$$\Omega_{m,b} = -b^{-1} \int_{\mathbb{T}} \int_0^b \left(\nabla_{\mathbf{x}} K(b e^{i\theta}, \rho e^{i\eta}) \cdot e^{i\theta} \right) \rho d\rho d\eta - \int_{\mathbb{T}} K(b, b e^{i\eta}) e^{im\eta} d\eta.$$

In particular, with the notation $G_1(\rho_1, \theta, \rho_2, \eta) \triangleq K_1(\rho_1 e^{i\theta}, \rho_2 e^{i\eta})$, we also have

$$\Omega_{m,b} = \Omega_{m,b}^0 - b^{-1} \int_{\mathbb{T}} \int_0^b \partial_{\rho_1} G_1(b, 0, \rho, \eta) \rho d\rho d\eta - \int_{\mathbb{T}} K_1(b, b e^{i\eta}) \cos(m\eta) d\eta. \quad (1.11)$$

Remark 1.2. The assumption of C^4 -regularity for K_1 will be crucially employed in proving (4.51), specifically in deriving the estimate of $\mathcal{I}_{12}$ as defined in (4.36).

In the proof of Theorem 1.1, our primary challenge lies in exploring the spectrum distribution of the linearized operator to the functional F_1 defined by (1.5) in the equilibrium state. One of the crucial ingredients is the strict monotonicity of the spectrum $(\Omega_{m,b}^0)_{m \in \mathbb{N}^*}$ with respect to m , needed to get a one-dimensional kernel, which is a requisite condition stipulated in Crandall-Rabinowitz's theorem; see Theorem 6.2.

Note that $\Omega_{m,b}^0$ has the expression (1.9) according to the analysis implemented in Subsection 2.2. Given this representation involving oscillating trigonometric functions, it is not obvious whether this sequence exhibits a monotonic behavior with a general kernel function K_0 . A crucial discovery is that when K_0 satisfies the assumption (A1), then we find an interesting factorization of the spectrum as follows, see Lemma 2.1,

$$\Omega_{m,b}^0 = 2 \int_0^\infty \left(\phi_1(bx) - \phi_m(bx) \right) \frac{d\mu(x)}{x}, \quad \text{with} \quad \phi_m(x) \triangleq \int_0^\pi e^{-2\sin(\eta)x} e^{i2m\eta} d\eta,$$

where μ is a nonnegative Borel measure. In this factorization, we appeal to the universal function ϕ_m , which is completely independent of the model and will encode the key feature of the spectral distribution. Especially, we show in Proposition 3.1 that for each $x > 0$, $\phi_m(x)$ is positive and the sequence $m \in \mathbb{N}^* \mapsto \phi_m(x)$ is strictly decreasing, which yields in turn the monotonicity of the spectrum. These properties in ϕ_m are not obvious and do not directly result from the definition of ϕ_m because the integrand undergoes oscillations with changes in sign. The crucial point here is that ϕ_m solves a second-order linear differential equation with variable coefficients given by (3.3). Then applying a result of *ad hoc* comparison theorem outlined in Lemma 3.1 allows us to show that ϕ_m is positive and strictly decreasing in m . Another serious difficulty lies in the proof of the strong regularity properties of $F(\Omega, r)$ needed in Crandall-Rabinowitz's theorem. Since we only impose an integrability condition on K_0 by the assumption (A2), the boundedness results in [29,45,56,58] related to singular kernel integrals with pointwise assumptions on the kernels cannot be used directly. To circumvent this difficulty, we establish suitable results, see Lemma 6.4 and Lemma 6.3, dealing with integral operators (6.7) in the torus and use some persistence regularity estimates employed several times to

infer the required regularity for $F(\Omega, r)$ as detailed in Subsection 4.1. The third delicate point in the proof is to check that $\partial_r F(\Omega_{m,b}^0, 0)$ is of co-dimension one. To this end, we shall use a Mikhlin type multiplier theorem stated in Lemma 6.5 on the periodic framework, as described in Proposition 4.2. Another interesting result is summarized in Proposition 3.2 and Corollary 3.1, where we derive the following spectrum expansion: for each $N \in \mathbb{N}$ and $n \geq 1$,

$$\Omega_{n,b}^0 = 2 \int_0^\infty \phi_1(bx) \frac{d\mu(x)}{x} - 2 \sum_{k=0}^N \frac{1}{n^{2k+1}} \int_0^\infty \Psi_k\left(\frac{bx}{n}\right) \frac{d\mu(x)}{x} + \varepsilon_{n,N},$$

where

$$\Psi_0(x) = \frac{x}{1+x^2}, \quad \Psi_{k+1}(x) = \frac{x^2}{4(1+x^2)} \left(\Psi_k''(x) + \frac{1}{x} \Psi_k'(x) \right), \quad \forall k \geq 0,$$

and

$$|\varepsilon_{n,N}| \leq \frac{C_{N,\delta}}{n^{2N+\frac{5}{3}}} \int_0^\infty \frac{x^{\delta-1}}{1+\frac{bx}{n}} d\mu(x), \quad \forall \delta \in [0, \frac{1}{3}).$$

This has significant consequences in classical analysis, illustrated in Section 5 through several examples stemming from geophysical flows, see Section 5. The proof of the foregoing expansion results from a rescaling argument coupled with an application of the Hankel transform.

Regarding the proof of Theorem 1.2, the main challenge is still to show the monotonicity of the spectrum sequence $(\Omega_{m,b})_{m \in \mathbb{N}^*}$, which takes the form (1.11) as shown in Subsection 2.2. The idea is to perform perturbative arguments, where from the regularity assumption on K_1 defined in (1.10) we derive that the last term $\int_{\mathbb{T}} K_1(b, be^{i\eta}) \cos(m\eta) d\eta$ involved in (1.11) decays in m as $O(m^{-k})$ with some $k \in \mathbb{N}^*$ that can be chosen large enough. Thus, to obtain the monotonicity property of the sequence $(\Omega_{m,b})_{m \geq 1}$, it is enough to analyze the spectrum repartition and show an algebraic lower bound decay for $\Omega_{m+1,b}^0 - \Omega_{m,b}^0$. To this end, we need a more careful quantitative study of the sequence $(\phi_m(x))_{m \geq 1}$. In Proposition 3.22 we show that for all $m \geq 1$ and $x > 0$,

$$\frac{1}{2} \frac{(2m+1)x}{(m^2+x^2)((m+1)^2+x^2)} \leq \phi_m(x) - \phi_{m+1}(x) \leq 4 \frac{(2m+1)x}{(m^2+x^2)((m+1)^2+x^2)}.$$

From this we find according to (4.57) a constant $c_* > 0$ such that

$$\Omega_{m+1,b}^0 - \Omega_{m,b}^0 \geq \frac{c_*}{m^3}.$$

This is the key point to get the spectrum monotonicity for large modes. Notice that, as a byproduct of the spectral analysis, we also present a discussion in Section 3.5 about the convexity of the spectrum $(\Omega_{m,b}^0)_{m \geq 1}$.

In Section 5, we will go into some applications of Theorem 1.1 and Theorem 1.2. The 2D Euler, gSQG and QGSW equations in the whole space align seamlessly with Theorem 1.1. We point out that the spectral study of the gSQG and QGSW equations as detailed in [29,45] involves the intricate analysis of special functions. However, with our approach, these results can be easily derived, yielding new identities and estimates such as (5.7), (5.11), (5.13), (5.14), (5.16). The V-states for 2D Euler, gSQG and QGSW equations within the unit disc $\mathbb{D}$ with rigid boundary condition fall under the scope of Theorem 1.2, allowing us to get the results outlined in [25,58]. In particular, the application to the QGSW equation in $\mathbb{D}$ with a rigid boundary condition is a new contribution.

The remainder of this paper is organized as follows. In the next section, we introduce the boundary equation modeling the V-states, consider the linearization around the equilibrium state, and give an important factorization formula of the spectrum in terms of the universal function ϕ_n . In Section 3, we focus on the analysis of some crucial properties of ϕ_n . We first prove a useful comparison theorem in Subsection 3.1 allowing us to derive the positivity and monotonicity of ϕ_n and its asymptotic behavior; see Subsections 3.2 - 3.3, respectively. This approach offers suitable tools in Subsections 3.4 - 3.5 to track the decay rate of $\phi_n - \phi_{n+1}$ and the convexity of the spectrum. In Section 4, we give detailed proofs for Theorem 1.1 and Theorem 1.2 by checking the required conditions of Crandall-Rabinowitz's theorem. In Section 5, we present various examples that follow from Theorems 1.1 and 1.2, and naturally deduce some interesting properties of the associated spectrum (most are new). In Section 6, we compile the tools used in the paper: completely monotone functions, Bessel functions and Hankel transform, boundedness property of some integral operators on the torus, and Crandall-Rabinowitz's theorem.

Notation. Throughout this paper, the following notation and convention will be used.

- The symbol C denotes a positive constant that may change its value from line to line.
- We denote the unit disc by $\mathbb{D}$.
- Denote by $\mathbb{T} = \mathbb{R}/(2\pi)\mathbb{Z}$ the torus (the unit circle). For a (2π) -periodic function f , we often denote $\int_0^{2\pi} f(x)dx$ by $\int_{\mathbb{T}} f(x)dx$.
- The set $\mathbb{N} = \{0, 1, 2, \dots\}$ is made up of nonnegative integers and $\mathbb{N}^* = \{1, 2, \dots\}$ only includes positive integers.
- Let $\mathbf{X}$ and $\mathbf{Y}$ be two Banach spaces. We denote by $\mathcal{L}(\mathbf{X}, \mathbf{Y})$ the space of all continuous linear maps $T : \mathbf{X} \rightarrow \mathbf{Y}$ endowed with its usual strong topology.
- Although $t \in (0, +\infty) \mapsto K_0(t)$ is a single-variable function, it is acceptable within the context to employ

$$\text{the notation } \nabla_{\mathbf{x}} K_0(|\mathbf{x} - \mathbf{y}|) \text{ to denote } \nabla_{\mathbf{x}}(K_0(|\mathbf{x} - \mathbf{y}|)), \quad (1.12)$$

and similarly, use $\nabla_{\mathbf{y}} K_0(|\mathbf{x} - \mathbf{y}|)$ to denote $\nabla_{\mathbf{y}}(K_0(|\mathbf{x} - \mathbf{y}|))$.

2. Time periodic patches and linearization

In this section, we have multiple goals. First, in the context of the vortex patches, we describe the contour dynamics in polar coordinates. Then, we will describe the linearized operator around Rankine vortices, which are radial equilibrium states. This operator takes the form of a Fourier multiplier, and its spectrum within the framework of completely monotone kernels will be factorized based on a Bessel-type universal function.

2.1. Boundary equation

Our primary focus lies on the motion of vortex patches concerning the active scalar equation (1.1)-(1.2). Specifically, the solution takes the form $\omega(t, \mathbf{x}) = \mathbf{1}_{D_t}(\mathbf{x})$, where the domain $D_t \subset \subset \mathbf{D}$ is a smooth perturbation of the disc $b\mathbb{D}$, with $b > 0$. Note that when the domain $\mathbf{D} = \mathbb{D}$ is the unit disc, we impose $0 < b < 1$ as in [25,58].

Our analysis will be centered on a specific patch solution within rotating domains, defined by

$$D_t = e^{it\Omega}D$$

with some angular velocity $\Omega \in \mathbb{R}$. Clearly, this generates a time periodic solution with a period $T = \frac{2\pi}{\Omega}$. In this section, the kernel K involved in the stream function (1.2) satisfies properties (1.3) and (1.4). Now, we will parameterize the boundary ∂D_t using the polar coordinates, as follows

$$\begin{aligned} \mathbf{z}(t, \cdot) : \mathbb{T} &\mapsto \partial D_t, \\ \theta &\mapsto e^{it\Omega}\mathbf{z}(\theta) \triangleq e^{it\Omega}\sqrt{b^2 + 2r(\theta)}e^{i\theta}, \end{aligned} \quad (2.1)$$

where $\mathbf{z}(\theta) \in \partial D$. Denote by $\mathbf{n}(t, \mathbf{z}(t, \theta)) \triangleq i\partial_\theta \mathbf{z}(t, \theta)$ an inward normal vector to the boundary ∂D_t at the point $\mathbf{z}(t, \theta)$. According to [56, p. 174], the vortex patch equation writes

$$\begin{aligned} \partial_t \mathbf{z}(t, \theta) \cdot \mathbf{n} &= v(t, \mathbf{z}(t, \theta)) \cdot \mathbf{n} \\ &= -\partial_\theta [\psi(t, \mathbf{z}(t, \theta))], \end{aligned}$$

where ψ is the stream function defined by (1.2). Then making a change of variables and using the symmetry property (1.4), we deduce that

$$\begin{aligned} \psi(t, \mathbf{z}(t, \theta)) &= \int_D K(e^{it\Omega}\mathbf{z}(\theta), e^{it\Omega}\mathbf{y})d\mathbf{y} \\ &= \int_D K(\mathbf{z}(\theta), \mathbf{y})d\mathbf{y}. \end{aligned}$$

In addition,

$$\partial_t \mathbf{z}(t, \theta) = i\Omega \mathbf{z}(t, \theta) = i\Omega e^{it\Omega} \sqrt{b^2 + 2r(\theta)} e^{i\theta},$$

and

$$\begin{aligned} \partial_t \mathbf{z}(t, \theta) \cdot \mathbf{n}(t, \mathbf{z}(t, \theta)) &= \operatorname{Im} \left(\partial_t \mathbf{z}(t, \theta) \overline{\partial_\theta \mathbf{z}(t, \theta)} \right) \\ &\stackrel{(2.1)}{=} \Omega r'(\theta). \end{aligned}$$

Thus, we obtain the equation characterizing the boundary ∂D ,

$$\Omega r'(\theta) = -\partial_\theta \left(\int_D K(\mathbf{z}(\theta), \mathbf{y}) d\mathbf{y} \right). \quad (2.2)$$

Using the polar coordinates gives

$$\begin{aligned} \int_D K(z(\theta), \mathbf{y}) d\mathbf{y} &= \int_{\mathbb{T}} \int_0^{R(\eta)} K(R(\theta) e^{i\theta}, \rho e^{i\eta}) \rho d\rho d\eta \\ &\triangleq F_0[r](\theta), \quad \text{with } R(\theta) \triangleq \sqrt{b^2 + 2r(\theta)}, \end{aligned} \quad (2.3)$$

and thus, we arrive at

$$F(\Omega, r) \triangleq \Omega r'(\theta) + \partial_\theta F_0[r](\theta) = 0. \quad (2.4)$$

Notice that Rankine vortices $\mathbf{1}_{b\mathbb{D}}(\mathbf{x})$ are stationary solutions of equation (2.4), that is,

$$F(\Omega, 0) \equiv 0, \quad \forall \Omega \in \mathbb{R}.$$

This property easily follows from the fact that $F_0[0]$ is rotationally invariant (that is, independent of θ) according to (1.4).

2.2. Linearization

In this section, the kernel K in (1.2) satisfies (1.10) together with the properties (A1)-(A2) and (A3)-(A4). To linearize the rotating patch equation (2.4) around the equilibrium, we may assume that the functional is smooth enough and compute the Gâteaux derivative, that will coincide with its Fréchet derivative. We proceed as follows,

$$\begin{aligned} \partial_r F(\Omega, r) h(\theta) &= \partial_s (F(\Omega, r + sh))|_{s=0} = \partial_s \partial_\theta (\Omega r + \Omega sh + F_0[r + sh])|_{s=0} \\ &= \partial_\theta \partial_s (\Omega r + \Omega sh + F_0[r + sh])|_{s=0} \\ &= \Omega h'(\theta) + \partial_\theta (\partial_s (F_0[r + sh])|_{s=0}). \end{aligned}$$

Next, using straightforward computations yields

$$\begin{aligned} \partial_r F(\Omega, r)h(\theta) &= \Omega h'(\theta) + \partial_\theta \left[\frac{h(\theta)}{R(\theta)} \int_{\mathbb{T}} \int_0^{R(\eta)} (\nabla_{\mathbf{x}} K(R(\theta)e^{i\theta}, \rho e^{i\eta}) \cdot e^{i\theta}) \rho \, d\rho d\eta \right] \\ &\quad + \partial_\theta \left(\int_{\mathbb{T}} K(R(\theta)e^{i\theta}, R(\eta)e^{i\eta}) h(\eta) d\eta \right) \\ &\triangleq \partial_\theta \left((\Omega + V[r](\theta)) h(\theta) + \mathcal{L}[r](h)(\theta) \right). \end{aligned} \quad (2.5)$$

From (1.6) we infer that

$$\nabla_{\mathbf{x}}(K_0(|\mathbf{x} - \mathbf{y}|)) = -\nabla_{\mathbf{y}}(K_0(|\mathbf{x} - \mathbf{y}|)),$$

and using the simplified notation (1.12), we find

$$\begin{aligned} V[r](\theta) &= \frac{1}{R(\theta)} \int_{\mathbb{T}} \int_0^{R(\eta)} (\nabla_{\mathbf{x}} K_0(|R(\theta)e^{i\theta} - \rho e^{i\eta}|) \cdot e^{i\theta}) \rho \, d\rho d\eta + V_1[r](\theta) \\ &= -\frac{1}{R(\theta)} \int_{\mathbb{T}} \int_0^{R(\eta)} (\nabla_{\mathbf{y}} K_0(|R(\theta)e^{i\theta} - \rho e^{i\eta}|) \cdot e^{i\theta}) \rho \, d\rho d\eta + V_1[r](\theta) \\ &= -\frac{1}{R(\theta)} \iint_D (\nabla_{\mathbf{y}} K_0(|R(\theta)e^{i\theta} - \mathbf{y}|) \cdot e^{i\theta}) \, d\mathbf{y} + V_1[r](\theta), \end{aligned}$$

with

$$V_1[r](\theta) \triangleq \frac{1}{R(\theta)} \int_{\mathbb{T}} \int_0^{R(\eta)} (\nabla_{\mathbf{x}} K_1(R(\theta)e^{i\theta}, \rho e^{i\eta}) \cdot e^{i\theta}) \rho \, d\rho d\eta.$$

Using the Gauss-Green theorem, we rewrite $V[r](\theta)$ as

$$\begin{aligned} V[r](\theta) &= -\frac{1}{R(\theta)} \int_{\mathbb{T}} K_0(|R(\theta)e^{i\theta} - R(\eta)e^{i\eta}|) (-i\partial_\eta(R(\eta)e^{i\eta})) \cdot e^{i\theta} d\eta + V_1[r](\theta) \\ &\triangleq V_0[r](\theta) + V_1[r](\theta). \end{aligned} \quad (2.6)$$

Hence, setting $G_1(\rho_1, \theta, \rho_2, \eta) \triangleq K_1(\rho_1 e^{i\theta}, \rho_2 e^{i\eta})$ and using (1.4), (2.6), (4.2), (4.5), at the equilibrium state $r = 0$ one finds that $V[0]$ is a constant independent of θ and

$$\begin{aligned}
V[0](\theta) &= b^{-1} \int_{\mathbb{T}} \int_0^b (\nabla_{\mathbf{x}} K(b e^{i\theta}, \rho e^{i\eta}) \cdot e^{i\theta}) \rho \, d\rho d\eta \\
&= - \int_{\mathbb{T}} K_0(|b e^{i\theta} - b e^{i\eta}|) (e^{i\eta} \cdot e^{i\theta}) \, d\eta + b^{-1} \int_{\mathbb{T}} \int_0^b (\nabla_{\mathbf{x}} K_1(b e^{i\theta}, \rho e^{i\eta}) \cdot e^{i\theta}) \rho \, d\rho d\eta \\
&= - \int_{\mathbb{T}} K_0(|b - b e^{i\eta}|) \cos(\eta) \, d\eta + b^{-1} \int_{\mathbb{T}} \int_0^b \partial_{\rho_1} G_1(b, \theta, \rho, \eta) \rho \, d\rho d\eta \\
&= - \int_{\mathbb{T}} K_0(|b - b e^{i\eta}|) e^{i\eta} \, d\eta + b^{-1} \int_{\mathbb{T}} \int_0^b \partial_{\rho_1} G_1(b, 0, \rho, \eta) \rho \, d\rho d\eta. \tag{2.7}
\end{aligned}$$

In addition, by the assumption (A4) we get

$$\mathcal{L}[0](h)(\theta) = \int_{\mathbb{T}} K(b e^{i\theta}, b e^{i\eta}) h(\eta) \, d\eta = \int_{\mathbb{T}} K(b, b e^{i\eta}) h(\theta + \eta) \, d\eta. \tag{2.8}$$

It is easy to check that the operator $\mathcal{L}[0]$ is a Fourier multiplier. Actually, for every smooth function $h(\theta) = \sum_{n \in \mathbb{Z}} h_n e^{in\theta}$,

$$\mathcal{L}[0](h)(\theta) = \sum_{n \in \mathbb{Z}} \Lambda_{n,b} h_n e^{in\theta}, \quad \Lambda_{n,b} \triangleq \int_{\mathbb{T}} K(b, b e^{i\eta}) e^{in\eta} \, d\eta. \tag{2.9}$$

Notice that $\Lambda_{n,b} = \Lambda_{-n,b}$ (owing to (1.3)) and the spectrum of $\mathcal{L}[0]$ is discrete and given by

$$\text{sp}(\mathcal{L}[0]) = \{\Lambda_{n,b}, n \in \mathbb{N}\}.$$

Denoting by

$$\begin{aligned}
d_r \mathcal{L}[r](h, w) &\triangleq \left(\frac{d}{ds} \mathcal{L}[r + sw](h) \right) \Big|_{s=0} \\
&= \int_{\mathbb{T}} \left(\nabla_{\mathbf{x}} K(R(\theta) e^{i\theta}, R(\eta) e^{i\eta}) \cdot \left(\frac{w(\theta) e^{i\theta}}{R(\theta)} \right) + \nabla_{\mathbf{y}} K(R(\theta) e^{i\theta}, R(\eta) e^{i\eta}) \cdot \left(\frac{w(\eta) e^{i\eta}}{R(\eta)} \right) \right) d\eta,
\end{aligned}$$

and using the chain rule, we deduce that

$$\begin{aligned}
\partial_{\theta} \left(\mathcal{L}[r](h)(\theta) \right) \Big|_{r=0} &= \left(d_r \mathcal{L}[r](h, r')(\theta) \right) \Big|_{r=0} + \partial_{\theta} \left(\mathcal{L}[0](h)(\theta) \right) \\
&= d_r \mathcal{L}[0](h, 0)(\theta) + \mathcal{L}[0](h')(\theta) = \mathcal{L}[0](h')(\theta). \tag{2.10}
\end{aligned}$$

Similarly, we obtain

$$\partial_\theta \left(V[r](\theta)h(\theta) \right) \Big|_{r=0} = \left(d_r V[r](\theta)r'(\theta)h(\theta) \right) \Big|_{r=0} + \partial_\theta \left(V[0](\theta)h(\theta) \right) = V[0]h'(\theta). \quad (2.11)$$

Consequently, provided that $\{\Lambda_{n,b}\}_{n \in \mathbb{N}^*}$ is strictly monotone with respect to n , the kernel of $\partial_r F(\Omega, 0)$ is nontrivial if and only if (see Subsection 4.2 for more discussion)

$$\Omega \in \left\{ -V[0] - \Lambda_{n,b}, n \in \mathbb{N}^* \right\}. \quad (2.12)$$

In the particular case where $K(\mathbf{x}, \mathbf{y}) = K_0(|\mathbf{x} - \mathbf{y}|)$, one gets

$$\Lambda_{n,b} = \lambda_{n,b} \triangleq \int_{\mathbb{T}} K_0(|b - be^{i\eta}|) e^{in\eta} d\eta = 2 \int_0^\pi K_0(2b \sin \eta) e^{i2n\eta} d\eta. \quad (2.13)$$

By a change of variables $\eta \mapsto \eta' = \pi - \eta$, we see that $\int_0^\pi K_0(2b \sin \eta) \sin(2n\eta) d\eta = 0$, and thus $\lambda_{n,b}$ is a real number with

$$\lambda_{n,b} = 2 \int_0^\pi K_0(2b \sin \eta) \cos(2n\eta) d\eta = \int_0^{2\pi} K_0(2b \sin \frac{\eta}{2}) \cos(n\eta) d\eta. \quad (2.14)$$

2.3. Spectrum factorization

The main goal is to factorize the spectrum $\lambda_{n,b}$ given by (2.13) using a universal function when the kernel $-K'_0$ is completely monotone as in the assumption (A1). More precisely, we have the following key result.

Lemma 2.1. *Assume that $K(\mathbf{x}, \mathbf{y}) = K_0(|\mathbf{x} - \mathbf{y}|)$ with the assumption (A1) being satisfied. Then for every $n \in \mathbb{N}^*$, $\lambda_{n,b}$ given by (2.13) satisfies*

$$\lambda_{n,b} = 2 \int_0^\infty \phi_n(bx) \frac{d\mu(x)}{x}, \quad (2.15)$$

with

$$\phi_n(x) \triangleq \int_0^\pi e^{-2x \sin \eta} e^{i2n\eta} d\eta. \quad (2.16)$$

Proof of Lemma 2.1. Under the assumption (A1), and according to Theorem 6.1, we infer the existence of a Borel measure μ on $[0, \infty)$ such that

$$-K'_0(t) = \int_0^\infty e^{-tx} d\mu(x), \quad \forall t > 0. \quad (2.17)$$

Integrating (2.17) with respect to t -variable, and using Fubini's theorem we obtain

$$\begin{aligned} K_0(t) &= K_0(2b) - \int_{2b}^t \int_0^\infty e^{-\tau x} d\mu(x) d\tau \\ &= K_0(2b) - \int_0^\infty \int_{2b}^t e^{-\tau x} d\tau d\mu(x) \\ &= K_0(2b) + \int_0^\infty \frac{e^{-tx} - e^{-2bx}}{x} d\mu(x). \end{aligned} \quad (2.18)$$

By virtue of Fubini's theorem again and (2.13), we can rewrite the spectrum $\lambda_{n,b}$ as

$$\begin{aligned} \lambda_{n,b} &= 2 \int_0^\pi \int_0^\infty \left(\frac{e^{-2bx \sin \eta} - e^{-2bx}}{x} d\mu(x) \right) e^{i2n\eta} d\eta \\ &= 2 \int_0^\pi \int_0^\infty (e^{-2bx \sin \eta} - e^{-2bx}) e^{i2n\eta} d\eta \frac{d\mu(x)}{x} \\ &= 2 \int_0^\infty \phi_n(bx) \frac{d\mu(x)}{x}. \end{aligned}$$

This achieves the proof of the desired result. $\square$

3. Analysis of the universal function ϕ_n

In this section, we study various properties of the real-valued function ϕ_n , which is defined by (2.16). In (2.15), we encounter the universal function ϕ_n that naturally emerges in the analysis of the spectrum $\lambda_{n,b}$ of the linearized operator $\partial_r F(\Omega, 0)$. The positivity and monotonicity of ϕ_n , together with its asymptotic behavior and the decay rate of $\phi_n - \phi_{n+1}$ are pivotal elements in the spectral study. We plan to explore these aspects along Subsections 3.2-3.4. Additionally, we leverage some of the properties of ϕ_n to introduce a lemma regarding the convexity of $(\lambda_{n,b})_{n \in \mathbb{N}^*}$ in Subsection 3.5. This lemma pertains to a specific class of nonnegative measures.

Defining ϕ_n as in (2.16) through an integral featuring oscillating trigonometric functions in the integrand makes it challenging to establish the aforementioned properties, such as positivity or monotonicity. Fortunately, we discover that ϕ_n obeys an ordinary

differential equation (3.3), which significantly helps us to establish the desired properties of ϕ_n . We basically employ suitable comparison principles to (3.3) as will be stated in Subsection 3.1.

3.1. Comparison theorem

We intend to detail a comparison principle that serves as the cornerstone in establishing several qualitative and quantitative properties of ϕ_n .

Lemma 3.1. *Let $a : (0, \infty) \rightarrow \mathbb{R}$ and $b : (0, \infty) \rightarrow (0, \infty)$ be two given continuous functions. The following statements hold true.*

(1) *Suppose that $f \in C^2((0, \infty))$ is a function satisfying the conditions*

$$\begin{cases} f''(x) + a(x)f'(x) - b(x)f(x) \leq 0, & \forall x > 0, \\ f(0) \geq 0, \quad \lim_{x \rightarrow \infty} f(x) \geq 0. \end{cases}$$

Then f is non-negative on $(0, \infty)$, that is, $f(x) \geq 0$. Moreover, if f additionally satisfies

$$f''(x) + a(x)f'(x) - b(x)f(x) < 0, \quad \forall x > 0,$$

then f is strictly positive on $(0, \infty)$.

(2) *Suppose that $f \in C^2((0, \infty))$ solves the inequality*

$$\begin{cases} f''(x) + a(x)f'(x) - b(x)f(x) \geq 0, & \forall x > 0, \\ f(0) \leq 0, \quad \lim_{x \rightarrow \infty} f(x) \leq 0. \end{cases}$$

Then f is non-positive on $(0, \infty)$, that is, $f(x) \leq 0$. Moreover, if f additionally satisfies

$$f''(x) + a(x)f'(x) - b(x)f(x) > 0, \quad \forall x > 0,$$

then f is strictly negative on $(0, \infty)$.

Proof of Lemma 3.1. (1) We will start by proving the first assertion. For this aim, we shall argue by contradiction. Assume that f takes strictly negative values at some points of $(0, \infty)$. Then in light of the assumptions $f(0) \geq 0$ and $\lim_{x \rightarrow \infty} f(x) \geq 0$, one can find some $x_0 > 0$ such that

$$\inf_{x > 0} f(x) = f(x_0) < 0.$$

Hence,

$$f'(x_0) = 0, \quad f''(x_0) \geq 0. \quad (3.1)$$

Coming back to the differential inequality we find

$$f''(x_0) \leq b(x_0)f(x_0) < 0,$$

which is a contradiction.

For the second assertion, we assume that f takes non-positive values at some points of $(0, \infty)$, then there exists some $x_0 > 0$ so that $\inf_{x>0} f(x) = f(x_0) \leq 0$ satisfies (3.1), but using the strict differential inequality gives $f''(x_0) < b(x_0)f(x_0) \leq 0$, and it yields a contradiction. This concludes the proof of the desired result.

(2) It follows from applying statement (1) to the function $-f$. $\square$

3.2. Positivity and monotonicity of ϕ_n

This subsection is dedicated to exploring the application of the comparison theorem to establish some qualitative properties of ϕ_n introduced in (2.16). We shall show the following result.

Proposition 3.1. *For every $n \geq 1$ and $x > 0$, $\phi_n(x) > 0$ and the map $n \mapsto \phi_n(x)$ is strictly decreasing.*

Proof of Proposition 3.1. For $\mathbf{z} \in \mathbb{C}$, define

$$\Phi_n(\mathbf{z}) \triangleq \frac{1}{\pi} \int_0^\pi e^{i(-\mathbf{z} \sin \eta + 2n\eta)} d\eta.$$

Recall the Anger and Weber functions defined successively by, see 8.580 in [43],

$$\mathbf{J}_\nu(\mathbf{z}) = \frac{1}{\pi} \int_0^\pi \cos(\nu\eta - \mathbf{z} \sin \eta) d\eta \quad \text{and} \quad \mathbf{E}_\nu(\mathbf{z}) = \frac{1}{\pi} \int_0^\pi \sin(\nu\eta - \mathbf{z} \sin \eta) d\eta.$$

Then we find

$$\Phi_n(\mathbf{z}) = \mathbf{J}_{2n}(\mathbf{z}) + i\mathbf{E}_{2n}(\mathbf{z})$$

and

$$\phi_n(x) = \pi \Phi_n(-2ix). \quad (3.2)$$

Now, it is a classical fact that the functions $\mathbf{J}_{2n}(\mathbf{z})$ and $\mathbf{E}_{2n}(\mathbf{z})$ satisfy the following ODEs, for instance see 8.584 in [43],

$$\mathbf{J}_{2n}''(\mathbf{z}) + \mathbf{z}^{-1}\mathbf{J}_{2n}'(\mathbf{z}) + \left(1 - \frac{4n^2}{\mathbf{z}^2}\right)\mathbf{J}_{2n}(\mathbf{z}) = 0,$$

and

$$\mathbf{E}_{2n}''(\mathbf{z}) + \mathbf{z}^{-1}\mathbf{E}_{2n}'(\mathbf{z}) + \left(1 - \frac{4n^2}{\mathbf{z}^2}\right)\mathbf{E}_{2n}(\mathbf{z}) = -\frac{2}{\pi\mathbf{z}}.$$

It follows that

$$\Phi_n''(\mathbf{z}) + \mathbf{z}^{-1}\Phi_n'(\mathbf{z}) + \left(1 - \frac{4n^2}{\mathbf{z}^2}\right)\Phi_n(\mathbf{z}) = -\frac{2i}{\pi\mathbf{z}}.$$

This implies by virtue of (3.2) that

$$\phi_n''(x) + x^{-1}\phi_n'(x) - 4\left(1 + \frac{n^2}{x^2}\right)\phi_n(x) = -\frac{4}{x}, \quad x > 0. \quad (3.3)$$

On the other hand, one may get from Riemann-Lebesgue's lemma applied with (2.16) that

$$\forall n \geq 1, \quad \phi_n(0) = 0, \quad \text{and} \quad \lim_{x \rightarrow \infty} \phi_n(x) = 0. \quad (3.4)$$

Hence, Lemma 3.1 guarantees that

$$\forall x > 0, \quad \phi_n(x) > 0. \quad (3.5)$$

Now we show that for any $x > 0$ the sequence $n \mapsto \phi_n(x)$ is strictly decreasing. For this aim, we define

$$\chi_n(x) \triangleq \phi_n(x) - \phi_{n+1}(x).$$

Then using the equation (3.3) we find

$$\chi_n''(x) + x^{-1}\chi_n'(x) - 4\left(1 + \frac{n^2}{x^2}\right)\chi_n(x) = -4\frac{2n+1}{x^2}(x)\phi_{n+1}(x), \quad x > 0, \quad (3.6)$$

with

$$\chi_n(0) = 0, \quad \lim_{x \rightarrow \infty} \chi_n(x) = 0.$$

Thus, (3.6) and Lemma 3.1 ensure that

$$\forall x > 0, \quad \chi_n(x) > 0,$$

which implies the strict monotonicity of ϕ_n . This concludes the proof of the desired results. $\square$

3.3. Asymptotic structure of ϕ_n

The next goal is to explore the asymptotic behavior of ϕ_n with respect to n . This will be the crucial step in describing the asymptotic behavior of the spectrum given through (2.15). For this purpose, we shall rescale the function ϕ_n as follows,

$$\phi_n(x) \triangleq \frac{1}{n} \varphi_n\left(\frac{x}{n}\right). \quad (3.7)$$

Then from (3.3) we easily find that

$$\frac{1}{n^2} (\varphi_n''(x) + \frac{1}{x} \varphi_n'(x)) - 4 \left(1 + \frac{1}{x^2}\right) \varphi_n(x) = -\frac{4}{x}, \quad x > 0. \quad (3.8)$$

In the following, we plan to provide an expansion formula of $\varphi_n(x)$ in terms of $\frac{1}{n}$.

Proposition 3.2. *For every $x > 0$, $n \geq 1$, $N \in \mathbb{N}$, we have*

$$\varphi_n(x) = \sum_{k=0}^N \frac{1}{n^{2k}} \Psi_k(x) + g_{n,N}(x), \quad (3.9)$$

with

$$\Psi_0(x) = \frac{x}{1+x^2}, \quad (3.10)$$

$$\Psi_{k+1}(x) = \frac{x^2}{4(1+x^2)} \left(\Psi_k''(x) + \frac{1}{x} \Psi_k'(x) \right), \quad \forall k \in \mathbb{N}, \quad (3.11)$$

and

$$|g_{n,N}(x)| \leq \frac{C}{n^{2N+\frac{2}{3}-\delta}} \frac{x^\delta}{1+x}, \quad \forall \delta \in [0, \frac{1}{3}], \quad (3.12)$$

where $C = C(N, \delta) > 0$ is independent of n and x .

Proof of Proposition 3.2. We define the second order differential operators

$$\mathbf{L}_0 f(x) \triangleq f''(x) + \frac{1}{x} f'(x), \quad (3.13)$$

and

$$\mathbf{L} f(x) \triangleq \frac{1}{n^2} (f''(x) + \frac{1}{x} f'(x)) - 4 \left(1 + \frac{1}{x^2}\right) f(x). \quad (3.14)$$

Putting this ansatz (3.9) into equation (3.8), we obtain that

$$\sum_{k=0}^N \frac{1}{n^{2k+2}} \mathbf{L}_0 \Psi_k(x) - \frac{4(x^2+1)}{x^2} \sum_{k=0}^N \frac{1}{n^{2k}} \Psi_k(x) + \mathbf{L} g_{n,N}(x) = -\frac{4}{x},$$

that is,

$$\begin{aligned} & \sum_{k=0}^{N-1} \frac{1}{n^{2k+2}} \left(\mathbf{L}_0 \Psi_k(x) - \frac{4(x^2+1)}{x^2} \Psi_{k+1}(x) \right) + \frac{\mathbf{L}_0 \Psi_N(x)}{n^{2N+2}} + \mathbf{L}g_{n,N}(x) \\ &= \frac{4(x^2+1)}{x^2} \Psi_0(x) - \frac{4}{x}. \end{aligned}$$

Taking advantage of the relations (3.10)-(3.11) gives the error equation

$$\mathbf{L}g_{n,N}(x) = -\frac{1}{n^{2N+2}} \mathbf{L}_0 \Psi_N(x). \quad (3.15)$$

Now, let us consider

$$\widetilde{\mathbf{L}}_0 f(x) \triangleq \frac{x^2}{4(1+x^2)} \mathbf{L}_0 f(x),$$

then we write

$$\Psi_{k+1}(x) = \widetilde{\mathbf{L}}_0 \Psi_k(x), \quad \text{and} \quad \mathbf{L}g_{n,N}(x) = -\frac{1}{n^{2N+2}} \mathbf{L}_0 \widetilde{\mathbf{L}}_0^N \Psi_0(x) \triangleq \frac{1}{n^{2N+2}} F_N(x).$$

By straightforward computations, using for instance an induction argument, we obtain

$$|\Psi_n(x)| \leq \frac{C_n |x|}{(1+x^2)^{n+1}}, \quad \text{and} \quad |F_N(x)| \leq \frac{C_N}{|x|(1+x^2)^{N+1}}. \quad (3.16)$$

Concerning the equation of $g_{n,N}$, it can be written in the form

$$g''_{n,N}(x) + \frac{1}{x} g'_{n,N}(x) - \frac{4n^2}{x^2} g_{n,N}(x) - 4n^2 g_{n,N}(x) = \frac{1}{n^{2N}} F_N(x). \quad (3.17)$$

In view of (3.9) and the relation $\varphi_n(x) = n \phi_n(nx)$, we claim that

$$\lim_{x \rightarrow 0} g_{n,N}(x) = 0, \quad \lim_{x \rightarrow \infty} x^{\frac{1}{2}} (|g_{n,N}(x)| + |g'_{n,N}(x)|) = 0, \quad \forall n, N \in \mathbb{N}.$$

Indeed, this can be directly justified by the dominated convergence theorem, noticing that for every $x > 0$,

$$|\phi_n(x)| + |\phi'_n(x)| \leq 3 \int_0^\pi e^{-2 \sin(\eta)x} d\eta \leq 6 \int_0^{\frac{\pi}{2}} e^{-\frac{4}{\pi} \eta x} d\eta \leq \min \left\{ 3\pi, \frac{6}{x} \right\},$$

and

$$\forall k \in \mathbb{N}, \quad \lim_{x \rightarrow 0} \Psi_k(x) = 0, \quad \lim_{x \rightarrow \infty} x^{\frac{1}{2}} (|\Psi_k(x)| + |\Psi'_k(x)|) = 0.$$

To estimate this error function, we find it convenient to use the Hankel transform; for more details, see Subsection 6.3. Then applying the Hankel transform $\mathcal{H}_{2n}$ to the equation (3.17) and using (6.26), we get

$$(-r^2 - 4n^2)(\mathcal{H}_{2n} g_{n,N})(r) = \frac{1}{n^{2N}} \mathcal{H}_{2n} F_N(r).$$

In light of the inverse formula (6.27) we deduce that

$$\forall x > 0, \quad g_{n,N}(x) = -\frac{1}{n^{2N}} \int_0^\infty \frac{r}{r^2 + 4n^2} (\mathcal{H}_{2n} F_N)(r) J_{2n}(xr) dr. \quad (3.18)$$

Via a change of variable and integration by parts, we find that

$$\begin{aligned} g_{n,N}(x) &= -\frac{1}{n^{2N}} \int_0^\infty \frac{r}{r^2 + 4n^2 x^2} (\mathcal{H}_{2n} F_N)\left(\frac{r}{x}\right) J_{2n}(r) dr \\ &= \frac{1}{n^{2N}} \int_0^\infty \left(\frac{r^{-2n}}{r^2 + 4n^2 x^2} (\mathcal{H}_{2n} F_N)\left(\frac{r}{x}\right) \right)' r^{2n+1} J_{2n+1}(r) dr \\ &= -\frac{1}{n^{2N}} \int_0^\infty \frac{2n (\mathcal{H}_{2n} F_N)\left(\frac{r}{x}\right)}{r^2 + 4n^2 x^2} J_{2n+1}(r) dr - \frac{1}{n^{2N}} \int_0^\infty \frac{2r^2 (\mathcal{H}_{2n} F_N)\left(\frac{r}{x}\right)}{(r^2 + 4n^2 x^2)^2} J_{2n+1}(r) dr \\ &\quad + \frac{1}{n^{2N}} \int_0^\infty \frac{r (\mathcal{H}_{2n} F_N)'\left(\frac{r}{x}\right)}{(r^2 + 4n^2 x^2)x} J_{2n+1}(r) dr \\ &\triangleq \text{I} + \text{II} + \text{III}, \end{aligned} \quad (3.19)$$

where in the second line we have used the classical identity $r^{2n+1} J_{2n}(r) = (r^{2n+1} J_{2n+1}(r))'$.

We point out that in [70,80], it was proved the existence of an absolute constant $C_0 > 0$ independent of n, x so that

$$|J_n(x)| \leq C_0 \min \left\{ n^{-\frac{1}{3}}, x^{-\frac{1}{3}} \right\}, \quad \forall n \in \mathbb{N}, x > 0.$$

Combining this estimate with (6.24) and (3.16) yields

$$\begin{aligned} |(\mathcal{H}_{2n} F_N)(r)| &\leq \int_0^\infty x |F_N(x)| |J_{2n+1}(xr)| dx \\ &\leq C_0 \min \left\{ r^{-\frac{1}{3}} \int_0^\infty x^{\frac{2}{3}} |F_N(x)| dx, n^{-\frac{1}{3}} \int_0^\infty x |F_N(x)| dx \right\} \end{aligned}$$

$$\begin{aligned}
&\leq C_N \min \left\{ r^{-\frac{1}{3}} \int_0^\infty \frac{1}{x^{\frac{1}{3}}(1+x^2)^{N+1}} dr, n^{-\frac{1}{3}} \int_0^\infty \frac{1}{(1+x^2)^{N+1}} dx \right\} \\
&\leq C_N \min \left\{ r^{-\frac{1}{3}}, n^{-\frac{1}{3}} \right\}.
\end{aligned} \tag{3.20}$$

Similarly, using the relation (6.13) allows to get

$$\begin{aligned}
|(\mathcal{H}_{2n}F_N)'(r)| &\leq \int_0^\infty x^2 |F_N(x)| |J'_{2n}(xr)| dx \\
&\leq \int_0^\infty x^2 |F_N(x)| \left(|J_{2n+1}(xr)| + |J_{2n-1}(xr)| \right) dx \\
&\leq C_N r^{-\frac{1}{3}}.
\end{aligned}$$

Using the interpolation inequality and (6.13),

$$|J_n(x)| \leq |J_n(x)|^{1-\delta} |J_n(x) - J_n(0)|^\delta \leq C_0 n^{-\frac{1}{3}} x^\delta, \quad \delta \in [0, 1], n \geq 1, \tag{3.21}$$

together with a change of variables, we infer that for every $\delta \in [0, \frac{1}{3})$,

$$\begin{aligned}
|\text{I}| &\leq C_N \frac{n^{1-\frac{1}{3}}}{n^{2N}} \int_0^\infty \frac{x^{\frac{1}{3}} r^{-\frac{1}{3}+\delta}}{r^2 + 4n^2 x^2} dr \\
&\leq \frac{C_N}{n^{2N+\frac{2}{3}-\delta}} \frac{1}{x^{1-\delta}} \int_0^\infty \frac{s^{\delta-\frac{1}{3}}}{1+s^2} ds \\
&\leq \frac{C_{N,\delta}}{n^{2N+\frac{2}{3}-\delta}} \frac{1}{x^{1-\delta}}.
\end{aligned}$$

Proceeding in the same way, we successively get

$$|\text{II}| \leq C_N \frac{n^{-\frac{1}{3}}}{n^{2N}} \int_0^\infty \frac{x^{\frac{1}{3}} r^{-\frac{1}{3}+\delta}}{r^2 + 4n^2 x^2} dr \leq \frac{C_{N,\delta}}{n^{2N+\frac{2}{3}-\delta}} \frac{1}{x^{1-\delta}}$$

and

$$|\text{III}| \leq C_N \frac{1}{n^{2N} x} \int_0^\infty \frac{r^{1-\frac{1}{3}+\delta} x^{\frac{1}{3}}}{r^2 + 4n^2 x^2} n^{-\frac{1}{3}} dr \leq \frac{C_{N,\delta}}{n^{2N+\frac{2}{3}-\delta}} \frac{1}{x^{1-\delta}}.$$

Combining these estimates with (3.19), we obtain

$$|g_{n,N}(x)| \leq \frac{C_{N,\delta}}{n^{2N+\frac{2}{3}-\delta}} \frac{1}{x^{1-\delta}}.$$

In addition, using (3.18) and (3.21), we also have that for every $\delta \in [0, \frac{1}{3})$,

$$\begin{aligned} |g_{n,N}(x)| &\leq \frac{1}{n^{2N}} \int_0^\infty \frac{r}{r^2 + 4n^2} |\mathcal{H}_{2n} F_N(r)| |J_{2n}(xr)| dr \\ &\leq \frac{C_N}{n^{2N}} \int_0^\infty \frac{r^{\frac{2}{3}} n^{-\frac{1}{3}} (xr)^\delta}{r^2 + 4n^2} dr \\ &\leq C_N \frac{x^\delta}{n^{2N+\frac{2}{3}-\delta}} \int_0^\infty \frac{r^{\frac{2}{3}+\delta}}{r^2 + 1} dr \\ &\leq C_{N,\delta} \frac{x^\delta}{n^{2N+\frac{2}{3}-\delta}}. \end{aligned}$$

Therefore, collecting the above two estimates leads to the last point of the proposition as desired. Hence, the proof is completed. $\square$

As an immediate consequence of Proposition 3.2 and (3.7), (2.15), we have the following results on the asymptotic representation of the universal function ϕ_n and the spectrum.

Corollary 3.1. *Let $b \in (0, 1)$ and $\delta \in [0, \frac{1}{3})$. Then, for any $n \geq 1$ and $N \in \mathbb{N}$, the following statements hold true.*

(1) *We have*

$$\forall x > 0, \quad \phi_n(x) = \sum_{k=0}^N \frac{1}{n^{2k+1}} \Psi_k\left(\frac{x}{n}\right) + \frac{1}{n} g_{n,N}\left(\frac{x}{n}\right),$$

with

$$\left| \frac{1}{n} g_{n,N}\left(\frac{x}{n}\right) \right| \leq \frac{C_{N,\delta}}{n^{2N+\frac{5}{3}}} \frac{x^\delta}{1 + \frac{x}{n}}.$$

(2) *We have*

$$\lambda_{n,b} = 2 \sum_{k=0}^N \frac{1}{n^{2k+1}} \int_0^\infty \Psi_k\left(\frac{bx}{n}\right) \frac{d\mu(x)}{x} + \varepsilon_{n,N},$$

with

$$|\varepsilon_{n,N}| \leq \frac{C_{N,\delta}}{n^{2N+\frac{5}{3}}} \int_0^\infty \frac{x^{\delta-1}}{1+\frac{bx}{n}} d\mu(x).$$

3.4. The decay rate of $\phi_n - \phi_{n+1}$

Our main goal in this section is to provide an explicit lower/upper bound for $\chi_n \triangleq \phi_n - \phi_{n+1}$, which is useful in handling the perturbative argument employed in the proof of Theorem 1.2.

According to the differential equation (3.6), the function ϕ_{n+1} contributes to the source term, and thus we shall need some pointwise controls for ϕ_{n+1} or φ_{n+1} in order to estimate χ_n . We note that, by choosing $N = 0$ in Proposition 3.2, we obtain

$$\forall n \geq 1, x > 0, \quad \varphi_n(x) = \frac{x}{1+x^2} + g_{n,0}(x),$$

with $\lim_{n \rightarrow \infty} \|g_{n,0}\|_{L^\infty(\mathbb{R}^+)} = 0$. Thus, for sufficiently large n , $\varphi_n(x)$ remains close to $\frac{x}{1+x^2}$. We will see that by analyzing carefully the differential equation governing $\varphi_n(x) - \frac{x}{1+x^2}$, we can show the following lower/upper bound of $\varphi_n(x)$, which gives a more precise version of that result.

Lemma 3.2. *For every $x > 0$ and $n \geq 1$, the following inequalities hold true:*

$$\frac{4n^2}{4n^2+1} \frac{x}{1+x^2} \leq \varphi_n(x) \leq \frac{4n^2}{4n^2-1} \frac{x}{1+x^2}$$

and

$$\frac{4n^2}{4n^2+1} \frac{x}{n^2+x^2} \leq \phi_n(x) \leq \frac{4n^2}{4n^2-1} \frac{x}{n^2+x^2}.$$

Remark 3.1. One may expect that there exist some constants $c_1, c_2 > 0$ such that at least for sufficiently large n ,

$$\frac{c_1}{n^2} \Psi_1(x) \leq \varphi_n(x) - \frac{x}{1+x^2} \leq \frac{c_2}{n^2} \Psi_1(x), \quad x > 0.$$

However, numerical experiments indicate that such inequalities do not hold even for very small c_1 or very large c_2 . On the other hand, if we define

$$w_n(x) \triangleq \varphi_n(x) - \frac{x}{1+x^2} - \frac{c}{n^2} \Psi_1(x), \quad c > 0,$$

we have

$$\frac{1}{n^2} (w_n''(x) + \frac{1}{x} w_n'(x)) - 4(1 + \frac{1}{x^2}) w_n(x) = \frac{c-1}{n^2} \frac{(x^2-3)^2-8}{(1+x^2)^3 x} - \frac{c}{n^4} (\Psi_1''(x) + \frac{1}{x} \Psi_1'(x)).$$

The leading term on the right-hand side $\frac{c-1}{n^2} \frac{(x^2-3)^2-8}{(1+x^2)^3x} = \frac{4(c-1)}{n^2} (1 + \frac{1}{x^2}) \Psi_1(x)$ does not have a definite sign, so that the comparison test seen in Lemma 3.1 does not apply in this context.

Proof of Lemma 3.2. Define

$$f_n(x) \triangleq \varphi_n(x) - \frac{cx}{1+x^2},$$

where $c > 0$ is a constant that will be chosen later according to n . From (3.8) and (3.4) we deduce by straightforward computations that

$$\frac{1}{n^2} \left(f_n''(x) + \frac{1}{x} f_n'(x) \right) - 4 \left(1 + \frac{1}{x^2} \right) f_n(x) = -(1-c) \frac{4}{x} - \frac{c}{n^2} \frac{(x^2-3)^2-8}{(1+x^2)^3x},$$

with

$$f_n(0) = 0, \quad \text{and} \quad \lim_{x \rightarrow \infty} f_n(x) = 0.$$

Next, we shall use the following bounds

$$\forall x \geq 0, \quad -1 \leq -6 \frac{x^2}{(1+x^2)^3} \leq \frac{(x^2-3)^2-8}{(1+x^2)^3} \leq \frac{x^4+1}{(1+x^2)^2} \leq 1,$$

leading to

$$-\frac{c}{n^2} \frac{1}{x} \leq -\frac{c}{n^2} \frac{(x^2-3)^2-8}{(1+x^2)^3x} \leq \frac{c}{n^2} \frac{1}{x},$$

and

$$\left(c - 1 - \frac{c}{4n^2} \right) \frac{4}{x} \leq -(1-c) \frac{4}{x} - \frac{c}{n^2} \frac{(x^2-3)^2-8}{(1+x^2)^3x} \leq -\left(1 - c - \frac{c}{4n^2} \right) \frac{4}{x}.$$

Hence, by choosing $c = \frac{4n^2}{4n^2+1}$ and $c = \frac{4n^2}{4n^2-1}$ and applying Lemma 3.1, we obtain the lower and upper bounds for $\varphi_n(x)$, respectively. Combined with the relation (3.7), it gives the required lower/upper bounds for $\phi_n(x)$. $\square$

The next goal is to estimate the difference $\phi_n - \phi_{n+1}$ that will be used later to explore the spectrum distribution.

Proposition 3.3. *For every $x > 0$ and $n \geq 1$, we have*

$$\frac{1}{2} \frac{(2n+1)x}{(n^2+x^2)((n+1)^2+x^2)} \leq \phi_n(x) - \phi_{n+1}(x) \leq 4 \frac{(2n+1)x}{(n^2+x^2)((n+1)^2+x^2)}. \quad (3.22)$$

Proof of Proposition 3.3. We shall first prove the following result: for any $x > 0$, $n \geq 1$, we have

$$\frac{4(n+1)^2}{4(n+1)^2+1}r_n(x) \leq \phi_n(x) - \phi_{n+1}(x) \leq \frac{4(n+1)^2}{4(n+1)^2-1}r_n(x), \quad (3.23)$$

where $r_n(x)$ is a solution to the equation

$$r_n''(x) + \frac{1}{x}r_n'(x) - 4\left(1 + \frac{n^2}{x^2}\right)r_n(x) = -\frac{4(2n+1)}{x((n+1)^2+x^2)}, \quad (3.24)$$

supplemented with the boundary conditions

$$r_n(0) = 0, \quad \text{and} \quad \lim_{x \rightarrow \infty} r_n(x) = 0. \quad (3.25)$$

The construction of r_n can be done by using the Hankel transform. Indeed, applying (6.26) to (3.24) yields

$$-s^2\mathcal{H}_{2n}r_n(s) - 4\mathcal{H}_{2n}r_n(s) = -\mathcal{H}_{2n}\left(\frac{4(2n+1)}{x((n+1)^2+x^2)}\right).$$

Thus,

$$\mathcal{H}_{2n}r_n(s) = \frac{1}{s^2+4}\mathcal{H}_{2n}\left(\frac{4(2n+1)}{x((n+1)^2+x^2)}\right)(s).$$

Then in view of (6.27), we find

$$r_n(x) = \mathcal{H}_{2n}\left(\frac{1}{s^2+4}\mathcal{H}_{2n}\left(\frac{4(2n+1)}{x((n+1)^2+x^2)}\right)(s)\right)(x). \quad (3.26)$$

By the definition of $\mathcal{H}_{2n}$ in (6.24) and arguing to get the estimate (3.20) we obtain (3.25).

Now we define $h_n(x) \triangleq \phi_n(x) - \phi_{n+1}(x) - \frac{4(n+1)^2}{4(n+1)^2+1}r_n(x)$ and

$$\mathbf{T}_n f(x) \triangleq f''(x) + \frac{1}{x}f'(x) - 4\left(1 + \frac{n^2}{x^2}\right)f(x).$$

Then thanks to (3.6), $h_n(x)$ satisfies

$$\mathbf{T}_n h_n(x) = -4\frac{2n+1}{x^2}\phi_{n+1}(x) - \frac{4(n+1)^2}{4(n+1)^2+1}\mathbf{T}_n r_n(x),$$

with

$$h_n(0) = 0, \quad \text{and} \quad \lim_{x \rightarrow \infty} h_n(x) = 0.$$

Lemma 3.2 ensures that

$$-4 \frac{2n+1}{x^2} \phi_{n+1}(x) \leq -\frac{4(n+1)^2}{4(n+1)^2+1} \frac{4(2n+1)}{x((n+1)^2+x^2)}.$$

Thus

$$-4 \frac{2n+1}{x^2} \phi_{n+1}(x) - \frac{4(n+1)^2}{4(n+1)^2+1} \mathbf{T}_n r_n(x) \leq 0.$$

Taking advantage of Lemma 3.1, we find that $h_n(x) \geq 0$ for every $x > 0$ and $n \geq 1$, which leads to the desired inequality

$$\phi_n(x) - \phi_{n+1}(x) \geq \frac{4(n+1)^2}{4(n+1)^2+1} r_n(x).$$

Performing a similar argument, that we shall omit here, one can prove the other inequality

$$\phi_n(x) - \phi_{n+1}(x) \leq \frac{4(n+1)^2}{4(n+1)^2-1} r_n(x).$$

This achieves the proof of (3.23).

Next, we investigate some lower and upper bounds for r_n . We shall first deal with the following rescaled function $R_n(x) \triangleq n r_n(nx)$, which satisfies the equation, see (3.14),

$$\mathbf{L}R_n(x) = \frac{1}{n^2} (R_n''(x) + \frac{1}{x} R_n'(x)) - 4(1 + \frac{1}{x^2}) R_n(x) = -\frac{4(2n+1)}{x((n+1)^2+n^2x^2)}.$$

Our primary goal is to derive the pointwise lower/upper bound of $R_n(x)$. To this end, we define

$$H_n(x) \triangleq \frac{x}{1+x^2} \frac{2n+1}{(n+1)^2+n^2x^2}.$$

We plan to show the following result

$$\frac{16}{25} H_n(x) \leq R_n(x) \leq \frac{8}{3} H_n(x), \quad (3.27)$$

which implies in turn that

$$\frac{16}{25} \frac{x}{n^2+x^2} \frac{2n+1}{(n+1)^2+x^2} \leq r_n(x) \leq \frac{8}{3} \frac{x}{n^2+x^2} \frac{2n+1}{(n+1)^2+x^2}.$$

Then combining this estimate with (3.23) gives the desired result of Proposition 3.3.

Now, let us move to the proof of (3.27). Set

$$\tilde{f}_n(x) \triangleq R_n(x) - c H_n(x),$$

with some constant $c > 0$ that will be carefully chosen later. Then, direct computations, using the notation (3.13), imply

$$\mathbf{L}\tilde{f}_n(x) = -(1-c)\frac{4(2n+1)}{x((n+1)^2+n^2x^2)} - c\frac{1}{n^2}\mathbf{L}_0H_n(x),$$

and we note that

$$\tilde{f}_n(0) = 0, \quad \text{and} \quad \lim_{x \rightarrow \infty} \tilde{f}_n(x) = 0.$$

According to Lemma 3.1, in order to obtain that $\tilde{f}_n(x) \geq 0$ or $\tilde{f}_n(x) \leq 0$, we only need to let the right-hand side of the above equation be non-positive or non-negative. Next, we plan to compute

$$\frac{1}{n^2}\mathbf{L}_0H_n(x) = \frac{1}{n^2}(H_n''(x) + \frac{1}{x}H_n'(x)).$$

From straightforward computations, we get for every $x > 0$ and $n \geq 1$,

$$\begin{aligned} \frac{1}{n^2}H_n'(x) &= \frac{1}{n^2} \frac{1-x^2}{(1+x^2)^2} \frac{2n+1}{(n+1)^2+n^2x^2} - \frac{x}{1+x^2} \frac{(2n+1)2x}{((n+1)^2+n^2x^2)^2} \\ &\geq -\frac{2n+1}{(n+1)^2+n^2x^2} \left(\frac{x^2}{(1+x^2)^2} + \frac{2x^2}{(1+x^2)} \frac{1}{(n+1)^2+n^2x^2} \right) \\ &\geq -\frac{3}{4} \frac{2n+1}{(n+1)^2+n^2x^2}. \end{aligned} \quad (3.28)$$

Direct computations yield

$$\begin{aligned} \frac{1}{n^2}H_n''(x) &= \frac{1}{n^2} \frac{(-2x)(3-x^2)}{(1+x^2)^3} \frac{2n+1}{(n+1)^2+n^2x^2} - \frac{1-x^2}{(1+x^2)^2} \frac{(2n+1)2x}{((n+1)^2+n^2x^2)^2} \\ &\quad - \frac{x}{(1+x^2)^2} \frac{4(2n+1)}{((n+1)^2+n^2x^2)^2} + \frac{x^3}{1+x^2} \frac{2(2n+1)4n^2}{((n+1)^2+n^2x^2)^3}. \end{aligned} \quad (3.29)$$

It follows that

$$\begin{aligned} \frac{1}{n^2}H_n''(x) &\geq -\frac{4(2n+1)}{x((n+1)^2+n^2x^2)} \left(\frac{3x^2}{2n^2(1+x^2)^3} + \frac{3x^2}{2(1+x^2)^2} \frac{1}{(n+1)^2+n^2x^2} \right) \\ &\geq -\frac{3}{8} \frac{4(2n+1)}{x((n+1)^2+n^2x^2)}, \end{aligned}$$

where we have used the inequalities $\frac{x^2}{(1+x^2)^2} \leq \frac{1}{4}$ and $\frac{x^2}{(1+x^2)^3} \leq \frac{4}{27}$. Hence, we find

$$\mathbf{L}\tilde{f}_n(x) \leq \frac{4(2n+1)}{x((n+1)^2 + n^2x^2)} \left(-(1-c) + \frac{9c}{16} \right),$$

and choosing $c = \frac{16}{25}$ gives $\mathbf{L}\tilde{f}_n \leq 0$. Then, Lemma 3.1 implies that $\tilde{f}_n(x) \geq 0$, that is,

$$\forall x \geq 0, \quad R_n(x) \geq \frac{16}{25} H_n(x).$$

Now, we turn to the proof of the second estimate of (3.27). First, we observe from (3.28) that

$$\frac{1}{n^2} H_n''(x) \leq \frac{1}{4} \frac{4(2n+1)}{(n+1)^2 + n^2x^2}.$$

In addition, we deduce from (3.29)

$$\begin{aligned} \frac{1}{n^2} H_n''(x) &\leq \frac{4(2n+1)}{x((n+1)^2 + n^2x^2)} \left(\frac{x^4}{2n^2(1+x^2)^3} + \frac{x^4}{2(1+x^2)^2} \frac{1}{(n+1)^2 + n^2x^2} \right) \\ &\quad + \frac{4(2n+1)}{x((n+1)^2 + n^2x^2)} \times \frac{2x^2}{1+x^2} \frac{n^2x^2}{((n+1)^2 + n^2x^2)^2} \\ &\leq \frac{3}{8} \frac{4(2n+1)}{x((n+1)^2 + n^2x^2)}, \end{aligned}$$

where we have used the fact that $\frac{x^4}{(1+x^2)^3} \leq \frac{4}{27}$ and $\frac{n^2x^2}{((n+1)^2 + n^2x^2)^2} \leq \frac{1}{16}$. Then we have

$$\mathbf{L}\tilde{f}_n(x) \geq \frac{4(2n+1)}{x((n+1)^2 + n^2x^2)} \left(-(1-c) - \frac{5c}{8} \right).$$

Choosing $c = \frac{8}{3}$ guarantees that $\mathbf{L}\tilde{f}_n \geq 0$. Therefore, in view of Lemma 3.1 we conclude that $\tilde{f}_n(x) \leq 0$, that is,

$$R_n(x) \leq \frac{8}{3} H_n(x),$$

as stated in (3.27). This achieves the proof of Proposition 3.3. $\square$

3.5. Spectrum convexity

In the forthcoming lemma, we intend to discuss a result concerning the convexity of the spectrum $(\lambda_{n,b})_{n \in \mathbb{N}^*}$ associated with a class of measures μ with suitable densities. Our result reads as follows.

Lemma 3.3. *Let $\lambda_{n,b}$ be given by (2.15) with $d\mu(x) = xf(x)dx$, $f(x) \geq 0$ and $f \in C^2(\mathbb{R}^+)$. Suppose that there exists some constant $C > 0$ such that the following conditions hold:*

- (1) $\limsup_{x \rightarrow 0^+} x|f(x)| + \limsup_{x \rightarrow 0^+} x^2|f'(x)| \leq C$,
 (2) $\lim_{x \rightarrow +\infty} f(x) = 0$ and $\lim_{x \rightarrow +\infty} xf'(x) = 0$,
 (3) $\forall x > 0, \quad f''(x) \geq 0$,

then we have

$$\forall n \geq 2, \quad \lambda_{n+1,b} + \lambda_{n-1,b} - 2\lambda_{n,b} \geq 0.$$

Proof of Lemma 3.3. First, recalling that ϕ_n is defined by (2.16) and using the fact that

$$\forall \eta \in \mathbb{R}, \quad -e^{i2\eta} + 2 - e^{-i2\eta} = 4 \sin^2 \eta,$$

we get the following identity,

$$\forall n \geq 2, \quad \frac{d^2 \phi_n(x)}{dx^2} = -\phi_{n+1}(x) + 2\phi_n(x) - \phi_{n-1}(x). \quad (3.30)$$

Combining together (2.15) with (3.30) allows us to get

$$\begin{aligned} \lambda_{n+1,b} + \lambda_{n-1,b} - 2\lambda_{n,b} &= 2 \int_0^\infty (\phi_{n+1}(bx) + \phi_{n-1}(bx) - 2\phi_n(bx)) f(x) dx \\ &= -2 \int_0^\infty \phi_n''(bx) f(x) dx \\ &= -\frac{2}{b^2} \int_0^\infty (\phi_n(bx) - \phi_n'(0)bx)'' f(x) dx. \end{aligned}$$

Integration by parts, using the above assumptions on f and the fact that $|\phi_n'(x)| \leq C$,

$$\forall n \geq 2, \quad \phi_n''(0) = \int_0^\pi 4(\sin \eta)^2 e^{i2n\eta} d\eta = 2 \int_0^\pi (1 - \cos 2\eta) \cos(2n\eta) d\eta = 0,$$

we obtain

$$\begin{aligned} \lambda_{n+1,b} + \lambda_{n-1,b} - 2\lambda_{n,b} &= \frac{2}{b^2} \int_0^\infty (\phi_n(bx) - \phi_n'(0)bx)' f'(x) dx \\ &= -\frac{2}{b^2} \int_0^\infty (\phi_n(bx) - \phi_n'(0)bx) f''(x) dx. \end{aligned}$$

Now, define the function

$$\tilde{h}_n(x) \triangleq \phi'_n(0)x - \phi_n(x) = \frac{4x}{4n^2 - 1} - \phi_n(x).$$

We intend to prove $\tilde{h}_n(x) \geq 0$ for any $x \geq 0$, which implies in turn the desired result of Lemma 3.3. By straightforward computations we find

$$\tilde{h}_n''(x) + \frac{1}{x}\tilde{h}_n'(x) - 4\left(1 + \frac{n^2}{x^2}\right)\tilde{h}_n(x) = -\frac{16x}{4n^2 - 1} \leq 0,$$

and

$$\tilde{h}_n(0) = \tilde{h}_n'(0) = 0, \quad \text{and} \quad \lim_{x \rightarrow \infty} \tilde{h}_n(x) = \infty.$$

Applying Lemma 3.1 yields

$$\forall x \geq 0, \quad \tilde{h}_n(x) \geq 0.$$

This concludes the proof of the positivity of $\tilde{h}_n(x)$ and achieves the desired result. $\square$

4. Proof of the main theorems

In this section, we apply Crandall-Rabinowitz's theorem to prove the existence of a time-periodic solution for the active scalar equation (1.1)-(1.2). We consider the kernel $K(\mathbf{x}, \mathbf{y}) = K_0(|\mathbf{x} - \mathbf{y}|) + K_1(\mathbf{x}, \mathbf{y})$, and if $K_1 \equiv 0$, it corresponds to the case treated in Theorem 1.1, and if $K_1 \not\equiv 0$, it is the case studied in Theorem 1.2. Below, we always identify the complex plane $\mathbb{C}$ with $\mathbb{R}^2$.

Before proceeding with the proofs, we collect some useful facts in polar coordinates. Denote by

$$G_1(\rho_1, \theta, \rho_2, \eta) \triangleq K_1(\rho_1 e^{i\theta}, \rho_2 e^{i\eta}), \quad G(\rho_1, \theta, \rho_2, \eta) \triangleq K(\rho_1 e^{i\theta}, \rho_2 e^{i\eta}), \quad (4.1)$$

then thanks to (A4) we have

$$\begin{aligned} G_1(\rho_1, -\theta, \rho_2, -\eta) &= G_1(\rho_1, \theta, \rho_2, \eta), \\ G_1(\rho_1, \theta + \theta', \rho_2, \eta + \eta') &= G_1(\rho_1, \theta, \rho_2, \eta), \quad \forall \theta' \in \mathbb{R}. \end{aligned} \quad (4.2)$$

Hence, in particular, we get

$$G_1(\rho_1, 0, \rho_2, -\eta) = G_1(\rho_1, 0, \rho_2, \eta), \quad (4.3)$$

and differentiating at $\theta' = 0$ the second identity in (4.2) yields

$$\begin{aligned}\partial_\theta G_1(\rho_1, \theta, \rho_2, \eta) &= -\partial_\eta G_1(\rho_1, \theta, \rho_2, \eta), \\ \partial_\theta G(\rho_1, \theta, \rho_2, \eta) &= -\partial_\eta G(\rho_1, \theta, \rho_2, \eta).\end{aligned}\quad (4.4)$$

By setting $\mathbf{x} = \rho_1 e^{i\theta}$ and $\mathbf{y} = \rho_2 e^{i\eta}$, and using straightforward computations, we obtain

$$\nabla_{\mathbf{x}} K(\mathbf{x}, \mathbf{y}) = \begin{pmatrix} \partial_{\rho_1} G \cos \theta - \partial_\theta G \frac{\sin \theta}{\rho_1} \\ \partial_{\rho_1} G \sin \theta + \partial_\theta G \frac{\cos \theta}{\rho_1} \end{pmatrix}, \quad (4.5)$$

$$\nabla_{\mathbf{y}} K(\mathbf{x}, \mathbf{y}) = \begin{pmatrix} \partial_{\rho_2} G \cos \eta - \partial_\eta G \frac{\sin \eta}{\rho_2} \\ \partial_{\rho_2} G \sin \eta + \partial_\eta G \frac{\cos \eta}{\rho_2} \end{pmatrix}. \quad (4.6)$$

We note that the above identities hold when (K, G) is replaced by (K_1, G_1) .

Next, we introduce the function spaces that will be used in the bifurcation arguments. For $m \in \mathbb{N}^*$ and $\alpha \in (0, 1)$, we define

$$\mathbf{X} = \mathbf{X}_m \triangleq \left\{ f \in C^{2-\alpha}(\mathbb{T}) : f(\theta) = \sum_{n \geq 1} b_n \cos(nm\theta), b_n \in \mathbb{R}, \theta \in \mathbb{T} \right\}, \quad (4.7)$$

and

$$\mathbf{Y} = \mathbf{Y}_m \triangleq \left\{ f \in C^{1-\alpha}(\mathbb{T}) : f(\theta) = \sum_{n \geq 1} b_n \sin(nm\theta), b_n \in \mathbb{R}, \theta \in \mathbb{T} \right\}, \quad (4.8)$$

equipped with the usual norms. For $\epsilon_0 > 0$, we denote by $\mathbf{B}_{\epsilon_0}$ the open ball of $\mathbf{X}_m$ centered at 0 and of radius ϵ_0 , that is,

$$\mathbf{B}_{\epsilon_0} \triangleq \{ f \in \mathbf{X}_m : \|f\|_{\mathbf{X}_m} < \epsilon_0 \}.$$

4.1. Strong regularity

The aim of this part is to explore the strong regularity of the functional F described by (2.4). We have the following result.

Proposition 4.1. *Let $m \geq 1$, $\alpha \in (0, 1)$ and $\mathbf{X}_m$ and $\mathbf{Y}_m$ be the spaces given by (4.7)-(4.8). There exists $\epsilon_0 > 0$ small enough so that the following statements hold true:*

- (1) $F : \mathbb{R} \times \mathbf{B}_{\epsilon_0} \rightarrow \mathbf{Y}_m$ is well-defined.
- (2) $F : \mathbb{R} \times \mathbf{B}_{\epsilon_0} \rightarrow \mathbf{Y}_m$ is of class C^1 .
- (3) The partial derivative $\partial_\Omega \partial_r F : \mathbb{R} \times \mathbf{B}_{\epsilon_0} \rightarrow \mathcal{L}(\mathbf{X}_m, \mathbf{Y}_m)$ exists and is continuous.

Proof of Proposition 4.1. (1) Using the Gauss-Green theorem (similarly as deriving (2.6)), we can rewrite $\partial_\theta F_0[r]$ as

$$\partial_\theta F_0[r] = (F_{00}[r] + F_{01}[r]) \cdot \partial_\theta (R(\theta) e^{i\theta}), \quad (4.9)$$

where

$$\begin{aligned} F_{00}[r](\theta) &= \int_{\mathbb{T}} \int_0^{R(\eta)} \nabla_{\mathbf{x}} K_0(|R(\theta)e^{i\theta} - \rho e^{i\eta}|) \rho \, d\rho d\eta \\ &= i \int_{\mathbb{T}} K_0(|R(\theta)e^{i\theta} - R(\eta)e^{i\eta}|) \partial_{\eta}(R(\eta)e^{i\eta}) d\eta, \end{aligned}$$

and

$$F_{01}[r](\theta) \triangleq \int_{\mathbb{T}} \int_0^{R(\eta)} \nabla_{\mathbf{x}} K_1(R(\theta)e^{i\theta}, \rho e^{i\eta}) \rho \, d\rho d\eta.$$

Since $\partial_{\theta}(R(\theta)e^{i\theta}) = \left(\frac{r'(\theta)}{R(\theta)}e^{i\theta} + R(\theta)ie^{i\theta}\right) \in C^{1-\alpha}(\mathbb{T})$, from (2.4) and (4.9) and in order to show $F(\Omega, r) \in C^{1-\alpha}(\mathbb{T})$, we only need to check that

$$\theta \in \mathbb{T} \mapsto F_{00}[r], F_{01}[r] \in C^{1-\alpha}(\mathbb{T}). \quad (4.10)$$

Next, we plan to prove (4.10). First, letting $\epsilon_0 > 0$ small enough, we have that for every $r \in \mathbf{B}_{\epsilon_0}$ and for every $\theta, \eta \in \mathbb{R}$,

$$b \left| \sin \frac{\theta - \eta}{2} \right| \leq |R(\theta)e^{i\theta} - R(\eta)e^{i\eta}| \leq 3b \left| \sin \frac{\theta - \eta}{2} \right|. \quad (4.11)$$

Indeed, this is quite similar to [58, Eq. (59)]: according to the following estimates

$$R(\theta)e^{i\theta} - R(\eta)e^{i\eta} = be^{i\eta}(e^{i(\theta-\eta)} - 1) + \left((R(\theta)e^{i\theta} - be^{i\theta}) - (R(\eta)e^{i\eta} - be^{i\eta}) \right),$$

and $|\partial_{\theta}(R(\theta)e^{i\theta} - be^{i\theta})| \leq \frac{2\epsilon_0}{\sqrt{b^2 - 2\epsilon_0}}$, and

$$\frac{2|\theta|}{\pi} \leq |e^{i\theta} - 1| = |2 \sin \frac{\theta}{2}| \leq |\theta|, \quad \text{for } |\theta| \leq \pi, \quad (4.12)$$

we can get (4.11) for every $|\theta - \eta| \leq \pi$ by letting $\epsilon_0 > 0$ be sufficiently small; and the general case follows from the periodicity property. Now, define

$$\mathbf{k}_1(\theta, \eta) \triangleq K_0(|R(\theta)e^{i\theta} - R(\eta)e^{i\eta}|).$$

Since $-K'_0(t) > 0$ on $(0, \infty)$, it follows that

$$|K_0(t)| \leq \max\{|K_0(t_1)|, |K_0(t_2)|\}, \quad \forall t \in [t_1, t_2] \subset (0, \infty). \quad (4.13)$$

We remind that K_0 may change the sign. Using (4.11) and (4.13), we deduce that

$$|\mathbf{k}_1(\theta, \theta + \eta)| \leq \max \left\{ |K_0(b|\sin \frac{\eta}{2}|)|, |K_0(3b|\sin \frac{\eta}{2}|)| \right\} \triangleq H_1(|\sin \frac{\eta}{2}|).$$

From (1.8), which implies in part that $\int_0^{a_0} |K_0(t)|dt < \infty$, (4.12) and (4.13), we get

$$\begin{aligned} \int_{\mathbb{T}} H_1(|\sin \frac{\eta}{2}|)d\eta &\leq 2 \int_0^\pi H_1(|\sin \frac{\eta}{2}|)d\eta \\ &\leq 2 \max \left\{ \int_0^\pi |K_0(\frac{b}{\pi}\eta)|d\eta, \int_0^\pi |K_0(\frac{3b}{2}\eta)|d\eta \right\} \\ &\leq \frac{C_0}{b} \int_0^{\frac{3\pi}{2}b} |K_0(t)|dt < \infty. \end{aligned} \quad (4.14)$$

Noticing that $\partial_\eta(R(\eta)e^{i\eta}) = (\frac{r'(\eta)}{R(\eta)}e^{i\eta} + R(\eta)ie^{i\eta}) \in C^{1-\alpha}(\mathbb{T})$, and the estimate

$$\left| \partial_\theta \left(R(\theta)e^{i\theta} - R(\theta + \eta)e^{i(\theta+\eta)} \right) \right| \leq C|\eta|^{1-\alpha} \leq C\pi^{1-\alpha} \left| \sin \frac{\eta}{2} \right|^{1-\alpha}, \quad \forall |\eta| \leq \pi,$$

which combined with the periodic property of R leads to

$$\left| \partial_\theta \left(R(\theta)e^{i\theta} - R(\theta + \eta)e^{i(\theta+\eta)} \right) \right| \leq C \left| \sin \frac{\eta}{2} \right|^{1-\alpha}, \quad \forall \eta \in \mathbb{R}. \quad (4.15)$$

Then, we use (4.11) to deduce that

$$\begin{aligned} |\partial_\theta(\mathbf{k}_1(\theta, \theta + \eta))| &\leq C|K'_0(|R(\theta)e^{i\theta} - R(\theta + \eta)e^{i(\theta+\eta)}|)| \left| \partial_\theta \left(R(\theta)e^{i\theta} - R(\theta + \eta)e^{i(\theta+\eta)} \right) \right| \\ &\leq C|K'_0(b|\sin \frac{\eta}{2}|)| \left| \sin \frac{\eta}{2} \right|^{1-\alpha} \triangleq H_2(|\sin \frac{\eta}{2}|). \end{aligned}$$

Thanks to (4.13) and the decreasing property of the function $t \in (0, \infty) \mapsto |K'_0(t)|$, it follows from (4.12) that

$$H_1(|\sin \frac{\eta}{2}|) \leq |K_0(\frac{b}{\pi}|\eta|)| + |K_0(\frac{3b}{2}|\eta|)|, \quad \text{for } |\eta| \leq \pi, \quad (4.16)$$

and

$$H_2(|\sin \frac{\eta}{2}|) \leq C|K'_0(\frac{b}{\pi}|\eta|)| |\eta|^{1-\alpha}, \quad \text{for } |\eta| \leq \pi. \quad (4.17)$$

Then, in view of (4.16)-(4.17), Lemma 6.3 (with $\beta = -\alpha(1 - \alpha)$) and (1.8), we have

$$\begin{aligned}
\int_{\mathbb{T}} (H_1(|\sin \frac{\eta}{2}|))^{\alpha} (H_2(|\sin \frac{\eta}{2}|))^{1-\alpha} d\eta &\leq 2 \int_0^{\pi} (H_1(|\sin \frac{\eta}{2}|))^{\alpha} (H_2(|\sin \frac{\eta}{2}|))^{1-\alpha} d\eta \\
&\leq C \int_0^{\pi} \left(|K_0(\frac{b}{\pi}\eta)|^{\alpha} + |K_0(\frac{3b}{2}\eta)|^{\alpha} \right) |K'_0(\frac{b}{\pi}\eta)|^{1-\alpha} \eta^{(1-\alpha)^2} d\eta \quad (4.18) \\
&\leq C \int_0^{(\frac{3b}{2} \vee 1)\pi} |K_0(t)| t^{-\alpha+\alpha^2} dt + C < \infty,
\end{aligned}$$

with $\frac{3b}{2} \vee 1 \triangleq \max\{\frac{3b}{2}, 1\}$. Hence, gathering (4.14)-(4.18) with Lemma 6.4 implies

$$\|F_{00}[r]\|_{C^{1-\alpha}} \leq C \|\partial_{\eta}(R(\eta)e^{i\eta})\|_{C^{1-\alpha}} \leq C. \quad (4.19)$$

For the remaining result in (4.10), using the assumption (A3), one can easily show that

$$\begin{aligned}
\|F_{01}[r]\|_{C^1(\mathbb{T})} &\leq C \sup_{\mathbf{x}, \mathbf{y} \in B(0, b + \sqrt{2}\epsilon_0)} \left(|\nabla_{\mathbf{x}} K_1(\mathbf{x}, \mathbf{y})| + |\nabla_{\mathbf{x}}^2 K_1(\mathbf{x}, \mathbf{y})| \|\partial_{\theta}(R(\theta)e^{i\theta})\|_{L^{\infty}(\mathbb{T})} \right) \\
&\leq C, \quad (4.20)
\end{aligned}$$

which guarantees that $F_{01}[r]$ belongs to $C^{1-\alpha}(\mathbb{T})$, as desired.

Now, we prove that $F(\Omega, r)$ given by (2.4) has the series expansion as in $\mathbf{Y}_m$. Indeed, noting that under the assumption (A4), the kernel K satisfies

$$K(\bar{\mathbf{x}}, \bar{\mathbf{y}}) = K(\mathbf{x}, \mathbf{y}), \quad K(e^{i\theta}\mathbf{x}, e^{i\theta}\mathbf{y}) = K(\mathbf{x}, \mathbf{y}), \quad \forall \theta \in \mathbb{R},$$

we can argue as in [58, p. 27] to deduce that

$$F_0[r](-\theta) = F_0[r](\theta), \quad F_0[r](\theta + \frac{2\pi}{m}) = F_0[r](\theta),$$

which leads to

$$F_0[r](\theta) = \sum_{n=0}^{\infty} c_{nm} \cos(nm\theta), \quad c_{nm} \in \mathbb{R},$$

and consequently, $F(\Omega, r)$ has the desired expansion formula. Therefore, taking $\epsilon_0 > 0$ small enough, we conclude that $F(\Omega, r) \in \mathbf{Y}_m$.

(2) It is obvious that $\partial_{\Omega} F(\Omega, r) = r'$ is a continuous mapping. So we only need to show that $\partial_r F(\Omega, r)$ is continuous with respect to r . In view of (2.5)-(2.6), we have

$$\begin{aligned}
\partial_r F(\Omega, r)h(\theta) &= \partial_{\theta} \left((\Omega + V[r](\theta))h(\theta) + \mathcal{L}[r](h)(\theta) \right) \\
&= \partial_{\theta} \left((\Omega + V_0[r](\theta) + V_1[r](\theta))h(\theta) + \mathcal{L}[r](h)(\theta) \right) \\
&= \Omega h'(\theta) + \partial_{\theta} \left(\mathcal{I}_0[r](h) + \mathcal{I}_1[r](h) \right), \quad (4.21)
\end{aligned}$$

where

$$\begin{aligned}\mathcal{I}_0[r](h)(\theta) &\triangleq \int_{\mathbb{T}} K_0(|R(\theta)e^{i\theta} - R(\eta)e^{i\eta}|)h(\eta)d\eta \\ &\quad + \int_{\mathbb{T}} K_0(|R(\theta)e^{i\theta} - R(\eta)e^{i\eta}|)(i\partial_\eta(R(\eta)e^{i\eta})) \cdot \left(\frac{h(\theta)}{R(\theta)}e^{i\theta}\right)d\eta,\end{aligned}$$

and

$$\begin{aligned}\mathcal{I}_1[r](h)(\theta) &\triangleq \int_{\mathbb{T}} \int_0^{R(\eta)} \nabla_{\mathbf{x}} K_1(R(\theta)e^{i\theta}, \rho e^{i\eta}) \cdot \left(\frac{h(\theta)}{R(\theta)}e^{i\theta}\right) \rho d\rho d\eta \\ &\quad + \int_{\mathbb{T}} K_1(R(\theta)e^{i\theta}, R(\eta)e^{i\eta})h(\eta)d\eta.\end{aligned}\tag{4.22}$$

For the term $\partial_\theta \mathcal{I}_0$, using the notation (1.12), we decompose it as follows

$$\begin{aligned}\partial_\theta \mathcal{I}_0[r](h)(\theta) &= \int_{\mathbb{T}} K_0(|R(\theta)e^{i\theta} - R(\eta)e^{i\eta}|)(i\partial_\eta(R(\eta)e^{i\eta})) \cdot \partial_\theta \left(\frac{h(\theta)}{R(\theta)}e^{i\theta}\right) d\eta \\ &\quad + \partial_\theta(R(\theta)e^{i\theta}) \cdot \int_{\mathbb{T}} \nabla_{\mathbf{x}} K_0(|R(\theta)e^{i\theta} - R(\eta)e^{i\eta}|) \mathbf{w}(\theta, \eta) d\eta \\ &\triangleq \mathcal{I}_{00}[r](h)(\theta) + \mathcal{I}_{01}[r](h)(\theta),\end{aligned}\tag{4.23}$$

with

$$\begin{aligned}\mathbf{w}(\theta, \eta) &= h(\eta) + (i\partial_\eta(R(\eta)e^{i\eta})) \cdot \left(\frac{h(\theta)}{R(\theta)}e^{i\theta}\right) \\ &= h(\eta) + \frac{h(\theta)}{R(\theta)}\partial_\eta(R(\eta)\sin(\theta - \eta)) \\ &= h(\eta) - \underbrace{\frac{h(\theta)}{R(\theta)}R(\eta)\cos(\theta - \eta)}_{\triangleq \mathbf{w}_2(\theta, \eta)} + \underbrace{\frac{h(\theta)}{R(\theta)}\frac{r'(\eta)}{R(\eta)}\sin(\theta - \eta)}_{\triangleq \mathbf{w}_3(\theta, \eta)}.\end{aligned}$$

The estimate of $\mathcal{I}_{00}[r](h)(\theta)$ is similar to that of $F_{00}[r]$ in (4.19). Actually, using the product laws in $C^{1-\alpha}(\mathbb{T})$, we have

$$\|\mathcal{I}_{00}[r](h)\|_{C^{1-\alpha}} \leq C \|\partial_\eta(R(\eta)e^{i\eta})\|_{C^{1-\alpha}} \|\partial_\theta \left(\frac{h(\theta)}{R(\theta)}e^{i\theta}\right)\|_{C^{1-\alpha}} \leq C \|h\|_{C^{2-\alpha}}.\tag{4.24}$$

For $\mathcal{I}_{01}[r](h)$, since $\theta \in \mathbb{T} \mapsto \partial_\theta(R(\theta)e^{i\theta}) \in C^{1-\alpha}(\mathbb{T})$, then using the product laws we get

$$\begin{aligned}
\|\mathcal{I}_{01}[r](h)\|_{C^{1-\alpha}} &\leq C \left\| \underbrace{\int_{\mathbb{T}} \nabla_{\mathbf{x}} K_0(|R(\theta)e^{i\theta} - R(\eta)e^{i\eta}|) \mathbf{w}_2(\theta, \eta) d\eta}_{\triangleq \mathcal{I}_{02}[r](h)} \right\|_{C^{1-\alpha}} \\
&\quad + C \left\| \underbrace{\int_{\mathbb{T}} \nabla_{\mathbf{x}} K_0(|R(\theta)e^{i\theta} - R(\eta)e^{i\eta}|) \mathbf{w}_3(\theta, \eta) d\eta}_{\triangleq \mathcal{I}_{03}[r](h)} \right\|_{C^{1-\alpha}}.
\end{aligned} \tag{4.25}$$

In light of $\nabla_{\mathbf{x}} K_0(|\mathbf{x} - \mathbf{y}|) = K'_0(|\mathbf{x} - \mathbf{y}|) \frac{\mathbf{x} - \mathbf{y}}{|\mathbf{x} - \mathbf{y}|}$ (recalling (1.12)), we define

$$\begin{aligned}
\mathbf{k}_2(\theta, \eta) &\triangleq \nabla_{\mathbf{x}} K_0(|R(\theta)e^{i\theta} - R(\eta)e^{i\eta}|) \mathbf{w}_2(\theta, \eta) \\
&= K'_0(|R(\theta)e^{i\theta} - R(\eta)e^{i\eta}|) \frac{R(\theta)e^{i\theta} - R(\eta)e^{i\eta}}{|R(\theta)e^{i\theta} - R(\eta)e^{i\eta}|} \mathbf{w}_2(\theta, \eta).
\end{aligned}$$

Notice that

$$\mathbf{w}_2(\theta, \theta + \eta) = h(\theta + \eta) - h(\theta) + h(\theta) \frac{R(\theta) - R(\theta + \eta) \cos \eta}{R(\theta)},$$

and

$$\begin{aligned}
\partial_{\theta}(\mathbf{w}_2(\theta, \theta + \eta)) &= h'(\theta + \eta) - h'(\theta) + \partial_{\theta} \left(\frac{h(\theta)}{R(\theta)} \right) (R(\theta) - R(\theta + \eta) \cos \eta) \\
&\quad + \frac{h(\theta)}{R(\theta)} \left(\frac{r'(\theta)}{R(\theta)} - \frac{r'(\theta + \eta) \cos \eta}{R(\theta + \eta)} \right).
\end{aligned}$$

Arguing as for (4.15), we deduce that

$$|\mathbf{w}_2(\theta, \theta + \eta)| \leq C \|h\|_{C^1} \left| \sin \frac{\eta}{2} \right|, \tag{4.26}$$

and

$$|\partial_{\theta}(\mathbf{w}_2(\theta, \theta + \eta))| \leq C \|h\|_{C^{2-\alpha}} \left| \sin \frac{\eta}{2} \right|^{1-\alpha}. \tag{4.27}$$

Thanks to the non-increasing property of $|K'_0|$ and (4.11), we deduce that

$$|\mathbf{k}_2(\theta, \theta + \eta)| \leq C |K'_0(\frac{b}{2} \sin \frac{\eta}{2})| \left| \sin \frac{\eta}{2} \right| \|h\|_{C^1} \triangleq CH_3 \left(\left| \sin \frac{\eta}{2} \right| \right) \|h\|_{C^1}. \tag{4.28}$$

Applying the estimates (6.6) and (6.2) allows us to obtain

$$\int_{\mathbb{T}} H_3 \left(\left| \sin \frac{\eta}{2} \right| \right) d\eta \leq 2 \int_0^{\pi} |K'_0(\frac{b}{2\pi} \eta)| \eta d\eta \leq C \int_0^{\pi} |K_0(t)| dt + C. \tag{4.29}$$

By using (4.11), (4.15), (4.26)-(4.27) and the non-increasing property of $|K'_0|$, $|K''_0|$, together with Lemma 6.1-(1), we infer that

$$\begin{aligned}
|\partial_\theta(\mathbf{k}_2(\theta, \theta + \eta))| &\leq C \left(|K'_0(b|\sin \frac{\eta}{2})| |\sin \frac{\eta}{2}|^{-\alpha} + |K''_0(b|\sin \frac{\eta}{2})| |\sin \frac{\eta}{2}|^{1-\alpha} \right) \|h\|_{C^1} |\sin \frac{\eta}{2}| \\
&\quad + C \|h\|_{C^{2-\alpha}} |K'_0(b|\sin \frac{\eta}{2})| |\sin \frac{\eta}{2}|^{1-\alpha} \\
&\leq C \|h\|_{C^{2-\alpha}} \left(|K'_0(b|\sin \frac{\eta}{2})| + |K''_0(b|\sin \frac{\eta}{2})| |\sin \frac{\eta}{2}| \right) |\sin \frac{\eta}{2}|^{1-\alpha} \\
&\leq C \|h\|_{C^{2-\alpha}} |K'_0(\frac{1}{2}b|\sin \frac{\eta}{2})| |\sin \frac{\eta}{2}|^{1-\alpha} \triangleq C \|h\|_{C^{2-\alpha}} H_4(|\sin \frac{\eta}{2}|).
\end{aligned} \tag{4.30}$$

Lemma 6.3, estimates (6.2), (6.6) and assumption (A2) ensure that

$$\begin{aligned}
\int_{\mathbb{T}} (H_3(|\sin \frac{\eta}{2}|))^\alpha (H_4(|\sin \frac{\eta}{2}|))^{1-\alpha} d\eta &\leq C \int_{\mathbb{T}} |K'_0(\frac{b}{2}|\sin \frac{\eta}{2})| |\sin \frac{\eta}{2}|^{\alpha+(1-\alpha)^2} d\eta \\
&\leq C \int_0^\pi |K'_0(\frac{b}{2\pi}\eta)| \eta^{1-\alpha+\alpha^2} d\eta \\
&\leq C \int_0^\pi |K_0(t)| t^{-\alpha+\alpha^2} dt + C.
\end{aligned} \tag{4.31}$$

Hence, according to (4.29), (4.31) and Lemma 6.4, we find

$$\left\| \mathcal{I}_{02}[r](h) \right\|_{C^{1-\alpha}} \leq C \|h\|_{C^{2-\alpha}}. \tag{4.32}$$

For $\mathcal{I}_{03}[r](h)$, we set

$$\mathbf{k}_3(\theta, \eta) \triangleq \nabla_{\mathbf{x}} K_0(|R(\theta)e^{i\theta} - R(\eta)e^{i\eta}|) \sin(\theta - \eta).$$

In a similar way as for deriving (4.28) and (4.30), we have

$$|\mathbf{k}_3(\theta, \theta + \eta)| \leq C H_3(|\sin \frac{\eta}{2}|), \quad |\partial_\theta(\mathbf{k}_3(\theta, \theta + \eta))| \leq C H_4(|\sin \frac{\eta}{2}|).$$

Lemma 6.4 and (4.29), (4.31) guarantee that

$$\left\| \int_{\mathbb{T}} \mathbf{k}_3(\theta, \eta) \frac{r'(\eta)}{R(\eta)} d\eta \right\|_{C^{1-\alpha}} \leq C \left\| \frac{r'(\eta)}{R(\eta)} \right\|_{C^{1-\alpha}} \leq C. \tag{4.33}$$

Hence, it follows from (4.25) and the product laws in $C^{1-\alpha}(\mathbb{T})$ that

$$\|\mathcal{I}_{03}[r](h)\|_{C^{1-\alpha}} \leq C \|h\|_{C^{1-\alpha}}. \tag{4.34}$$

Putting together (4.32) and (4.34) yields

$$\left\| \partial_\theta \mathcal{I}_0[r]h \right\|_{C^{1-\alpha}} \leq C \|h\|_{C^{2-\alpha}}. \tag{4.35}$$

Let us now move to the term $\mathcal{I}_1$ defined in (4.22). Then one gets

$$\begin{aligned} \partial_\theta \mathcal{I}_1[r]h(\theta) &= \int_{\mathbb{T}} \int_0^{R(\eta)} \nabla_{\mathbf{x}} K_1(R(\theta)e^{i\theta}, \rho e^{i\eta}) \rho d\rho d\eta \cdot \partial_\theta \left(\frac{h(\theta)}{R(\theta)} e^{i\theta} \right) \\ &\quad + \partial_\theta (R(\theta)e^{i\theta}) \cdot \int_{\mathbb{T}} \int_0^{R(\eta)} \nabla_{\mathbf{x}}^2 K_1(R(\theta)e^{i\theta}, \rho e^{i\eta}) \rho d\rho d\eta \cdot \left(\frac{h(\theta)}{R(\theta)} e^{i\theta} \right) \\ &\quad + \partial_\theta (R(\theta)e^{i\theta}) \cdot \int_{\mathbb{T}} \nabla_{\mathbf{x}} K_1(R(\theta)e^{i\theta}, R(\eta)e^{i\eta}) h(\eta) d\eta \\ &\triangleq \mathcal{I}_{11}[r](h)(\theta) + \mathcal{I}_{12}[r](h)(\theta) + \mathcal{I}_{13}[r](h)(\theta). \end{aligned} \quad (4.36)$$

Since $K_1 \in C_{\text{loc}}^4(\mathbf{D}^2)$, arguing for the estimate of $F_{01}[r]$ in (4.20), one can easily show that

$$\begin{aligned} \|\mathcal{I}_{11}[r](h)\|_{C^{1-\alpha}} &\leq C \left\| \int_{\mathbb{T}} \int_0^{R(\eta)} \nabla_{\mathbf{x}} K_1(R(\theta)e^{i\theta}, \rho e^{i\eta}) \rho d\rho d\eta \right\|_{C^1} \left\| \partial_\theta \left(\frac{h(\theta)}{R(\theta)} e^{i\theta} \right) \right\|_{C^{1-\alpha}} \\ &\leq C \|h\|_{C^{2-\alpha}}. \end{aligned}$$

In a similar way, we find

$$\begin{aligned} \|\partial_\theta \mathcal{I}_1[r]h\|_{C^{1-\alpha}} &\leq \|\mathcal{I}_{11}[r](h)\|_{C^{1-\alpha}} + \|\mathcal{I}_{12}[r](h)\|_{C^{1-\alpha}} + \|\mathcal{I}_{13}[r](h)\|_{C^{1-\alpha}} \\ &\leq C \|h\|_{C^{2-\alpha}}. \end{aligned} \quad (4.37)$$

Therefore, by collecting (4.21)-(4.35) and (4.37) we infer that

$$\|\partial_r F(\Omega, r)h\|_{C^{1-\alpha}} \leq C \|h\|_{C^{2-\alpha}}.$$

The next goal is to prove that for a given $\Omega \in \mathbb{R}$, the mapping $r \mapsto \partial_r F(\Omega, r) \in \mathcal{L}(\mathbf{X}_m, \mathbf{Y}_m)$ is continuous. Thanks to (4.21), (4.23), (4.36), it suffices to show that, for every $r_1, r_2 \in \mathbf{B}_{\epsilon_0}$ as $\|r_1 - r_2\|_{C^{2-\alpha}} \rightarrow 0$,

$$\sup_{\|h\|_{C^{2-\alpha}} \leq 1} \left(\sum_{j=0}^1 \|\mathcal{I}_{0j}[r_1](h) - \mathcal{I}_{0j}[r_2](h)\|_{C^{1-\alpha}} + \sum_{j=1}^3 \|\mathcal{I}_{1j}[r_1](h) - \mathcal{I}_{1j}[r_2](h)\|_{C^{1-\alpha}} \right) \rightarrow 0. \quad (4.38)$$

Denote by $R_j(\theta) \triangleq \sqrt{b^2 + 2r_j(\theta)}$, $j = 1, 2$. For $\mathcal{I}_{00}$ given by (4.23), we get

$$\|\mathcal{I}_{00}[r_1](h) - \mathcal{I}_{00}[r_2](h)\|_{C^{1-\alpha}} \leq C \left\| \int_{\mathbb{T}} \mathbf{k}_4(\theta, \eta) \partial_\eta (R_1(\eta)e^{i\eta}) d\eta \right\|_{C^{1-\alpha}} \left\| \partial_\theta \left(\frac{h(\theta)}{R_1(\theta)} e^{i\theta} \right) \right\|_{C^{1-\alpha}}$$

$$\begin{aligned}
& + C \left\| \int_{\mathbb{T}} K_0(|X_2(\theta, \eta)|) \partial_\eta ((R_1(\eta) - R_2(\eta)) e^{i\eta}) d\eta \right\|_{C^{1-\alpha}} \left\| \partial_\theta \left(\frac{h(\theta)}{R_1(\theta)} e^{i\theta} \right) \right\|_{C^{1-\alpha}} \\
& + C \left\| \int_{\mathbb{T}} K_0(|X_2(\theta, \eta)|) \partial_\eta (R_2(\eta) e^{i\eta}) d\eta \right\|_{C^{1-\alpha}} \left\| \partial_\theta \left(\frac{h(\theta)}{R_1(\theta)} e^{i\theta} \right) - \partial_\theta \left(\frac{h(\theta)}{R_2(\theta)} e^{i\theta} \right) \right\|_{C^{1-\alpha}},
\end{aligned}$$

where

$$X_j(\theta, \eta) \triangleq R_j(\theta) e^{i\theta} - R_j(\eta) e^{i\eta}, \quad j = 1, 2, \quad (4.39)$$

and

$$\mathbf{k}_4(\theta, \eta) \triangleq K_0(|X_1(\theta, \eta)|) - K_0(|X_2(\theta, \eta)|).$$

Hence, taking advantage of the estimates

$$\|R_1(\eta) - R_2(\eta)\|_{C^{2-\alpha}} \leq C \|r_1 - r_2\|_{C^{2-\alpha}},$$

and

$$\left\| \partial_\theta \left(\frac{h(\theta)}{R_1(\theta)} e^{i\theta} \right) - \partial_\theta \left(\frac{h(\theta)}{R_2(\theta)} e^{i\theta} \right) \right\|_{C^{1-\alpha}} \leq C \|h\|_{C^{2-\alpha}} \|r_1 - r_2\|_{C^{2-\alpha}},$$

together with (4.19) we deduce that

$$\begin{aligned}
\|\mathcal{I}_{00}[r_1](h) - \mathcal{I}_{00}[r_2](h)\|_{C^{1-\alpha}} & \leq C \|h\|_{C^{2-\alpha}} \|r_1 - r_2\|_{C^{2-\alpha}} \\
& + C \|h\|_{C^{2-\alpha}} \left\| \int_{\mathbb{T}} \mathbf{k}_4(\theta, \eta) \partial_\eta (R_1(\eta) e^{i\eta}) d\eta \right\|_{C^{1-\alpha}}. \quad (4.40)
\end{aligned}$$

The subsequent objective is to estimate $\mathbf{k}_4$. Without losing generality, we assume that $|X_1(\theta, \theta + \eta)| \leq |X_2(\theta, \theta + \eta)|$. Then, according to Lemma 6.1-(2) we deduce the inequality

$$|\mathbf{k}_4(\theta, \theta + \eta)| \leq (|X_1(\theta, \theta + \eta)| - |X_2(\theta, \theta + \eta)|) K'_0(|X_1(\theta, \theta + \eta)|).$$

Using the triangle inequality together with (4.11) yields

$$|\mathbf{k}_4(\theta, \theta + \eta)| \leq (|X_1(\theta, \theta + \eta) - X_2(\theta, \theta + \eta)|) |K'_0(b|\sin \frac{\eta}{2}|)|.$$

Applying Taylor's formula and using the 2π -periodicity, imply that we can assume $|\eta| \leq \pi$,

$$\begin{aligned}
|X_1(\theta, \theta + \eta) - X_2(\theta, \theta + \eta)| & = \left| \int_0^1 \partial_\theta \left(R_1(\theta + \tau\eta) e^{i(\theta + \tau\eta)} - R_2(\theta + \tau\eta) e^{i(\theta + \tau\eta)} \right) \cdot \eta d\tau \right| \\
& \leq C_0 \|R_1 - R_2\|_{C^1} |\eta| \leq C_0 \|r_1 - r_2\|_{C^1} \left| \sin \frac{\eta}{2} \right|. \quad (4.41)
\end{aligned}$$

Therefore, we deduce that

$$|\mathbf{k}_4(\theta, \theta + \eta)| \leq C \|r_1 - r_2\|_{C^1} |K'_0(\tfrac{1}{2}b) \sin \tfrac{\eta}{2}| |\sin \tfrac{\eta}{2}| \triangleq C \|r_1 - r_2\|_{C^1} H_3(|\sin \tfrac{\eta}{2}|). \quad (4.42)$$

Now, we shall estimate the derivative $\partial_\theta(\mathbf{k}_4(\theta, \theta + \eta))$ which takes the form

$$\begin{aligned} \partial_\theta(\mathbf{k}_4(\theta, \theta + \eta)) &= \left(K'_0(|X_1(\theta, \theta + \eta)|) - K'_0(|X_2(\theta, \theta + \eta)|) \right) \frac{X_1(\theta, \theta + \eta)}{|X_1(\theta, \theta + \eta)|} \cdot \partial_\theta(X_1(\theta, \theta + \eta)) \\ &\quad + K'_0(|X_2(\theta, \theta + \eta)|) \left(\frac{X_1(\theta, \theta + \eta)}{|X_1(\theta, \theta + \eta)|} - \frac{X_2(\theta, \theta + \eta)}{|X_2(\theta, \theta + \eta)|} \right) \cdot \partial_\theta(X_1(\theta, \theta + \eta)) \\ &\quad + K'_0(|X_2(\theta, \theta + \eta)|) \frac{X_2(\theta, \theta + \eta)}{|X_2(\theta, \theta + \eta)|} \cdot \left(\partial_\theta(X_1(\theta, \theta + \eta)) - \partial_\theta(X_2(\theta, \theta + \eta)) \right). \end{aligned}$$

As $R_1 \in C^{2-\alpha}(\mathbb{T})$, then we deduce that for $|\eta| \leq \pi$,

$$\begin{aligned} |\partial_\theta(X_1(\theta, \theta + \eta))| &\leq |R'_1(\theta)e^{i\theta} - R'_1(\theta + \eta)e^{i(\theta + \eta)}| + |R_1(\theta)ie^{i\theta} - R_1(\theta + \eta)ie^{i(\theta + \eta)}| \\ &\leq C(\|R'_1\|_{C^{1-\alpha}} + \|R_1\|_{C^{1-\alpha}})|\eta|^{1-\alpha} \\ &\leq C\|r_1\|_{C^{2-\alpha}}|\sin \tfrac{\eta}{2}|^{1-\alpha}. \end{aligned}$$

Similarly, we get

$$\begin{aligned} &|\partial_\theta(X_1(\theta, \theta + \eta)) - \partial_\theta(X_2(\theta, \theta + \eta))| \\ &\leq |(R'_1(\theta) - R'_2(\theta)) - (R'_1(\theta + \eta) - R'_2(\theta + \eta))e^{i\eta}| \\ &\quad + |(R_1(\theta) - R_2(\theta)) - (R_1(\theta + \eta) - R_2(\theta + \eta))e^{i\eta}| \\ &\leq C(\|R'_1 - R'_2\|_{C^{1-\alpha}} + \|R_1 - R_2\|_{C^{1-\alpha}})|\eta|^{1-\alpha} \\ &\leq C\|r_1 - r_2\|_{C^{2-\alpha}}|\sin \tfrac{\eta}{2}|^{1-\alpha}. \end{aligned}$$

We may assume $|X_1(\theta, \theta + \eta)| \leq |X_2(\theta, \theta + \eta)|$. Then, applying Lemma 6.1-(2) with the triangle inequality allows us to get

$$\begin{aligned} &|K'_0(|X_1(\theta, \theta + \eta)|) - K'_0(|X_2(\theta, \theta + \eta)|)| \\ &\leq (|X_1(\theta, \theta + \eta) - X_2(\theta, \theta + \eta)|) |K''_0(|X_1(\theta, \theta + \eta)|)|. \end{aligned}$$

Thus, we obtain by virtue of (4.41), the monotonicity of $|K''_0|$ and (4.11),

$$|K'_0(|X_1(\theta, \theta + \eta)|) - K'_0(|X_2(\theta, \theta + \eta)|)| \leq C_0 \|r_1 - r_2\|_{C^1} |\sin \tfrac{\eta}{2}| |K''_0(b|\sin \tfrac{\eta}{2}|)|.$$

Using Lemma 6.1-(1) gives

$$|K'_0(|X_1(\theta, \theta + \eta)|) - K'_0(|X_2(\theta, \theta + \eta)|)| \leq C_0 \|r_1 - r_2\|_{C^1} |K'_0(\tfrac{1}{2}b|\sin \tfrac{\eta}{2}|)|.$$

Putting together the foregoing estimates, we get by straightforward computations

$$\begin{aligned} |\partial_\theta(\mathbf{k}_4(\theta, \theta + \eta))| &\leq C \|r_1 - r_2\|_{C^{2-\alpha}} |K'_0(\tfrac{1}{2}b|\sin \tfrac{\eta}{2})| |\sin \tfrac{\eta}{2}|^{1-\alpha} \\ &\triangleq C \|r_1 - r_2\|_{C^{2-\alpha}} H_4(|\sin \tfrac{\eta}{2}|). \end{aligned} \quad (4.43)$$

Hence, (4.42), (4.43), (4.29), (4.31) and Lemma 6.4 ensure that

$$\begin{aligned} \left\| \int_{\mathbb{T}} \mathbf{k}_4(\theta, \eta) \partial_\eta(R_1(\eta)e^{i\eta}) d\eta \right\|_{C^{1-\alpha}} &\leq C \|r_1 - r_2\|_{C^{2-\alpha}} \|\partial_\eta(R_1(\eta)e^{i\eta})\|_{C^{1-\alpha}} \\ &\leq C \|r_1 - r_2\|_{C^{2-\alpha}}, \end{aligned}$$

and consequently,

$$\|\mathcal{I}_{00}[r_1](h) - \mathcal{I}_{00}[r_2](h)\|_{C^{1-\alpha}} \leq C \|r_1 - r_2\|_{C^{2-\alpha}} \|h\|_{C^{2-\alpha}}. \quad (4.44)$$

For $\mathcal{I}_{01}[r](h)$ given by (4.23), using (4.32) and the estimates that $\|R_i(\theta)e^{i\theta}\|_{C^1(\mathbb{T})} \leq C$ and

$$\left\| \partial_\theta(R_1(\theta)e^{i\theta}) - \partial_\theta(R_2(\theta)e^{i\theta}) \right\|_{C^{1-\alpha}} \leq C \|r_1 - r_2\|_{C^{2-\alpha}},$$

we find in a similar way as for deriving (4.40)

$$\begin{aligned} \|\mathcal{I}_{01}[r_1](h) - \mathcal{I}_{01}[r_2](h)\|_{C^{1-\alpha}} &\leq C \|h\|_{C^{2-\alpha}} \|r_1 - r_2\|_{C^{2-\alpha}} \\ &\quad + C \sum_{k=2}^3 \|\mathcal{I}_{0k}[r_1](h) - \mathcal{I}_{0k}[r_2](h)\|_{C^{1-\alpha}}. \end{aligned} \quad (4.45)$$

From (4.25), we denote

$$\|\mathcal{I}_{02}[r_1](h) - \mathcal{I}_{02}[r_2](h)\|_{C^{1-\alpha}} = \left\| \int_{\mathbb{T}} \mathbf{k}_5(\theta, \eta) d\eta \right\|_{C^{1-\alpha}}, \quad (4.46)$$

where

$$\mathbf{w}_{2j}(\theta, \eta) \triangleq h(\eta) - \frac{R_j(\eta) \cos(\theta - \eta)}{R_j(\theta)} h(\theta), \quad j = 1, 2,$$

and

$$\mathbf{k}_5(\theta, \eta) \triangleq \nabla_{\mathbf{x}} K_0(X_1(\theta, \eta)) \mathbf{w}_{21}(\theta, \eta) - \nabla_{\mathbf{x}} K_0(X_2(\theta, \eta)) \mathbf{w}_{22}(\theta, \eta).$$

We rewrite $\mathbf{k}_5(\theta, \theta + \eta)$ as follows

$$\begin{aligned} \mathbf{k}_5(\theta, \theta + \eta) &= \left(K'_0(|X_1(\theta, \theta + \eta)|) - K'_0(|X_2(\theta, \theta + \eta)|) \right) \frac{X_1(\theta, \theta + \eta)}{|X_1(\theta, \theta + \eta)|} \mathbf{w}_{21}(\theta, \theta + \eta) \\ &\quad + K'_0(|X_2(\theta, \theta + \eta)|) \left(\frac{X_1(\theta, \theta + \eta)}{|X_1(\theta, \theta + \eta)|} - \frac{X_2(\theta, \theta + \eta)}{|X_2(\theta, \theta + \eta)|} \right) \mathbf{w}_{21}(\theta, \theta + \eta) \\ &\quad + \nabla_{\mathbf{x}} K_0(|X_2(\theta, \theta + \eta)|) \left(\mathbf{w}_{21}(\theta, \theta + \eta) - \mathbf{w}_{22}(\theta, \theta + \eta) \right). \end{aligned}$$

By the identity

$$\begin{aligned} \mathbf{w}_{21}(\theta, \theta + \eta) - \mathbf{w}_{22}(\theta, \theta + \eta) &= h(\theta) \cos(\eta) \left(\frac{R_2(\theta + \eta)}{R_2(\theta)} - \frac{R_1(\theta + \eta)}{R_1(\theta)} \right) \\ &= h(\theta) \cos(\eta) \frac{R_1(\theta)((R_2 - R_1)(\theta + \eta) - (R_2 - R_1)(\theta)) + (R_2(\theta) - R_1(\theta))(R_1(\theta) - R_1(\theta + \eta))}{R_1(\theta)R_2(\theta)}, \end{aligned}$$

we deduce that

$$|\mathbf{w}_{21}(\theta, \theta + \eta) - \mathbf{w}_{22}(\theta, \theta + \eta)| \leq C \|h\|_{L^\infty} \|r_1 - r_2\|_{C^1} \left| \sin \frac{\eta}{2} \right|, \quad (4.47)$$

and

$$\left| \partial_\theta (\mathbf{w}_{21}(\theta, \theta + \eta)) - \partial_\theta (\mathbf{w}_{22}(\theta, \theta + \eta)) \right| \leq C \|h\|_{C^1} \|r_1 - r_2\|_{C^{2-\alpha}} \left| \sin \frac{\eta}{2} \right|^{1-\alpha}. \quad (4.48)$$

Arguing as in (4.28) and (4.43), and using (4.26), (4.47) and Lemma 6.1-(1), we infer that

$$\begin{aligned} |\mathbf{k}_5(\theta, \theta + \eta)| &\leq C \|r_1 - r_2\|_{C^1} \|h\|_{C^1} \left(|K'_0(b|\sin \frac{\eta}{2}|)| \left| \sin \frac{\eta}{2} \right| + |K''_0(b|\sin \frac{\eta}{2}|)| \left| \sin \frac{\eta}{2} \right|^2 \right) \\ &\leq C \|r_1 - r_2\|_{C^1} \|h\|_{C^1} |K'_0(\frac{1}{2}b|\sin \frac{\eta}{2}|)| \left| \sin \frac{\eta}{2} \right| \\ &= C \|r_1 - r_2\|_{C^1} \|h\|_{C^1} H_3(|\sin \frac{\eta}{2}|). \end{aligned}$$

In a similar way to (4.30) and (4.43), and after some tedious computations, we arrive at

$$\begin{aligned} |\partial_\theta (\mathbf{k}_5(\theta, \theta + \eta))| &\leq C \|r_1 - r_2\|_{C^{2-\alpha}} \|h\|_{C^{2-\alpha}} \left(\sum_{j=1}^3 |K_0^{(j)}(b|\sin \frac{\eta}{2}|)| \left| \sin \frac{\eta}{2} \right|^{j-\alpha} \right) \\ &\leq C \|r_1 - r_2\|_{C^{2-\alpha}} \|h\|_{C^{2-\alpha}} |K'_0(\frac{1}{2}b|\sin \frac{\eta}{2}|)| \left| \sin \frac{\eta}{2} \right|^{1-\alpha} \\ &= C \|r_1 - r_2\|_{C^{2-\alpha}} \|h\|_{C^{2-\alpha}} H_4(|\sin \frac{\eta}{2}|). \end{aligned}$$

Hence, (4.29), (4.31) and Lemma 6.4 imply that

$$\left\| \int_{\mathbb{T}} \mathbf{k}_5(\theta, \eta) d\eta \right\|_{C^{1-\alpha}} \leq C \|r_1 - r_2\|_{C^{2-\alpha}} \|h\|_{C^{2-\alpha}}.$$

Putting it together with (4.46) yields

$$\|\mathcal{I}_{02}[r_1](h) - \mathcal{I}_{02}[r_2](h)\|_{C^{1-\alpha}} \leq C \|r_1 - r_2\|_{C^{2-\alpha}} \|h\|_{C^{2-\alpha}}. \quad (4.49)$$

The estimate of $\mathcal{I}_{03}[r_1](h) - \mathcal{I}_{03}[r_2](h)$ is quite similar to the preceding one, using in particular (4.33), and we shall just state the final result omitting the details,

$$\|\mathcal{I}_{03}[r_1](h) - \mathcal{I}_{03}[r_2](h)\|_{C^{1-\alpha}} \leq C\|r_1 - r_2\|_{C^{2-\alpha}}\|h\|_{C^{2-\alpha}}. \quad (4.50)$$

Next, the estimation of $\mathcal{I}_{1j}[r_1](h) - \mathcal{I}_{1j}[r_2](h)$ ($j = 1, 2, 3$) is more straightforward and one obtains the following

$$\sum_{j=1,2,3} \|\mathcal{I}_{1j}[r_1](h) - \mathcal{I}_{1j}[r_2](h)\|_{C^{1-\alpha}} \leq C\|r_1 - r_2\|_{C^{2-\alpha}}\|h\|_{C^{2-\alpha}}. \quad (4.51)$$

In the sequel, we only give the proof of the estimate for $\mathcal{I}_{11}$, since the remaining terms are similar. Indeed, noting from (4.36) that

$$\begin{aligned} \mathcal{I}_{11}[r_1](h)(\theta) - \mathcal{I}_{11}[r_2](h)(\theta) &= \int_{\mathbb{T}} \int_{R_2(\eta)}^{R_1(\eta)} \nabla_{\mathbf{x}} K_1(R_1(\theta)e^{i\theta}, \rho e^{i\eta}) \rho d\rho d\eta \cdot \partial_{\theta} \left(\frac{h(\theta)}{R_1(\theta)} e^{i\theta} \right) \\ &\quad + (R_1(\theta)e^{i\theta} - R_2(\theta)e^{i\theta}) \cdot \Pi(\theta) \cdot \partial_{\theta} \left(\frac{h(\theta)}{R_1(\theta)} e^{i\theta} \right) \\ &\quad + \int_{\mathbb{T}} \int_0^{R_2(\eta)} \nabla_{\mathbf{x}} K_1(R_2(\theta)e^{i\theta}, \rho e^{i\eta}) \rho d\rho d\eta \cdot \left(\partial_{\theta} \left(\frac{h(\theta)}{R_1(\theta)} e^{i\theta} \right) - \partial_{\theta} \left(\frac{h(\theta)}{R_2(\theta)} e^{i\theta} \right) \right), \end{aligned}$$

with

$$\theta \in \mathbb{R} \mapsto \Pi(\theta) \triangleq \int_0^1 \int_{\mathbb{T}} \int_0^{R_2(\eta)} \nabla_{\mathbf{x}}^2 K_1(sR_1(\theta)e^{i\theta} + (1-s)R_2(\theta)e^{i\theta}, \rho e^{i\eta}) \rho d\rho d\eta ds \in C^1(\mathbb{T}),$$

using the C_{loc}^4 -smoothness of K_1 yields

$$\|\mathcal{I}_{11}[r_1](h) - \mathcal{I}_{11}[r_2](h)\|_{C^{1-\alpha}(\mathbb{T})} \leq C\|r_1 - r_2\|_{C^{2-\alpha}}\|h\|_{C^{2-\alpha}}.$$

Therefore, gathering the above estimates, we conclude (4.38), allowing us to get the desired result on the continuity of $\partial_r F(\Omega, r)$.

(3) Since $\partial_{\Omega} \partial_r F(\Omega, r) h(\theta) = h'(\theta)$, the regularity result follows immediately. $\square$

4.2. Spectral study

In this subsection, we focus on the spectral study of the linearized operator at zero, given by $\partial_r F(\Omega, 0)$. Consider

$$\theta \in \mathbb{R} \mapsto h(\theta) = \sum_{n=1}^{\infty} a_n \cos(nm\theta) \in \mathbf{X}_m, \quad a_n \in \mathbb{R}.$$

Then, according to (2.5), (2.9)-(2.11) and (2.13)-(2.14), we have

$$\begin{aligned}\partial_r F(\Omega, 0)h(\theta) &= (\Omega + V[0])h'(\theta) + \mathcal{L}[0](h')(\theta) \\ &= -\sum_{n=1}^{\infty} a_n (\Omega - \Omega_{nm,b}) nm \sin(nm\theta),\end{aligned}\quad (4.52)$$

where $\Omega_{n,b}$ ($n \in \mathbb{N}^*$) satisfies

$$\Omega_{n,b} = \Omega_{n,b}^0 + \Omega_{n,b}^1, \quad (4.53)$$

with

$$\begin{aligned}\Omega_{n,b}^0 &\triangleq \int_{\mathbb{T}} K_0(2b|\sin \frac{\eta}{2}|) \cos \eta \, d\eta - \int_{\mathbb{T}} K_0(2b|\sin \frac{\eta}{2}|) \cos(n\eta) d\eta \\ &= \lambda_{1,b} - \lambda_{n,b},\end{aligned}\quad (4.54)$$

and

$$\Omega_{n,b}^1 \triangleq -b^{-1} \int_0^{2\pi} \int_0^b \partial_{\rho_1} G_1(b, 0, \rho, \eta) \rho d\rho d\eta - \int_{\mathbb{T}} K_1(b, be^{i\eta}) \cos(n\eta) d\eta. \quad (4.55)$$

In particular, if $K(\mathbf{x}, \mathbf{y}) = K_0(|\mathbf{x} - \mathbf{y}|)$, then $\Omega_{n,b} = \Omega_{n,b}^0 = \lambda_{1,b} - \lambda_{n,b}$ with $\lambda_{n,b}$ given by (2.13).

Lemma 2.1 and the results in Section 3 imply the following crucial properties of $\Omega_{n,b}^0$.

Lemma 4.1. *Let K_0 be a smooth function satisfying assumptions (A1)-(A2), see (1.7) and (1.8). Then the following statements hold true.*

(1) *For any $n \in \mathbb{N}^*$, we have*

$$\lambda_{1,b} - \frac{8n^2}{4n^2 - 1} \int_0^\infty \frac{b}{n^2 + (bx)^2} d\mu(x) \leq \Omega_{n,b}^0 \leq \lambda_{1,b} - \frac{8n^2}{4n^2 + 1} \int_0^\infty \frac{b}{n^2 + (bx)^2} d\mu(x),$$

and

$$\lim_{n \rightarrow \infty} \Omega_{n,b}^0 = \int_{\mathbb{T}} K_0(2b|\sin \frac{\eta}{2}|) \cos \eta \, d\eta.$$

(2) *The map $n \in \mathbb{N}^* \mapsto \Omega_{n,b}^0$ is strictly increasing and*

$$\frac{1}{2}D_n \leq \Omega_{n+1,b}^0 - \Omega_{n,b}^0 \leq 4D_n,$$

with

$$D_n \triangleq \int_0^\infty \frac{b}{n^2 + (bx)^2} \frac{2n+1}{(n+1)^2 + (bx)^2} d\mu(x).$$

Proof of Lemma 4.1. In light of (4.54) and Lemma 2.1, we have a useful formula for $\Omega_{n,b}^0$ in terms of ϕ_n given by (2.16),

$$\Omega_{n,b}^0 = \lambda_{1,b} - \lambda_{n,b} = 2 \int_0^\infty (\phi_1(bx) - \phi_n(bx)) \frac{d\mu(x)}{x}. \quad (4.56)$$

Hence, the statement (1) follows directly from Lemma 3.2.

For the estimate of the point (2), it can be deduced from (3.22) and (4.56). In addition, since μ is a nonnegative measure and is not zero measure, there exist some $0 < d < \infty$ and $c_* > 0$ such that $\mu([0, d]) \geq c_* > 0$. Then, we obtain the strict monotonicity of $(\Omega_{n,b}^0)_{n \in \mathbb{N}^*}$, that is,

$$\begin{aligned} \Omega_{n+1,b}^0 - \Omega_{n,b}^0 &\geq \frac{1}{2} \int_0^d \frac{b}{n^2 + (bx)^2} \frac{2n+1}{(n+1)^2 + (bx)^2} d\mu(x) \\ &\geq \frac{c_*}{2} \frac{b}{n^2 + (bd)^2} \frac{2n+1}{(n+1)^2 + (bd)^2} \geq \frac{c'_*}{n^3}, \end{aligned} \quad (4.57)$$

with $c'_* > 0$ depending only on c_* , d and b . $\square$

Next, we intend to show the monotonicity of the sequence $(\Omega_{n,b})$ for large modes.

Lemma 4.2. Consider the general case (1.10) with K_0 and K_1 satisfying assumptions (A1)-(A4). Then there exist $m_0 \in \mathbb{N}^*$ and $C > 0$ such that for any $m \geq m_0$ and $n \geq 1$,

$$\Omega_{(n+1)m,b} - \Omega_{nm,b} \geq \frac{C}{(nm)^3}.$$

In addition,

$$\lim_{n \rightarrow \infty} \Omega_{nm,b} = \int_{\mathbb{T}} K_0(2b|\sin \frac{\eta}{2}|) \cos \eta d\eta - b^{-1} \int_0^{2\pi} \int_0^b \partial_{\rho_1} G_1(b, 0, \rho, \eta) \rho d\rho d\eta.$$

Proof of Lemma 4.2. Since the kernel K_1 belongs to $C_{\text{loc}}^3(\mathbf{D}^2)$ and $\eta \mapsto K_1(b, be^{i\eta})$ to $C^3(\mathbb{T})$. Then, using integration by parts we infer that

$$\begin{aligned} \left| \int_{\mathbb{T}} K_1(b, be^{i\eta}) \cos(nm\eta) d\eta \right| &= \left| \frac{1}{(nm)^3} \int_0^{2\pi} \partial_\eta^3(K_1(b, be^{i\eta})) \sin(nm\eta) d\eta \right| \\ &\leq \frac{C_2}{(nm)^3}, \end{aligned} \quad (4.58)$$

with some $C_2 > 0$. Hence, in view of (4.53), (4.55), (4.57) and (4.58), we find

$$\begin{aligned} \Omega_{(n+1)m,b} - \Omega_{nm,b} &= \Omega_{(n+1)m,b}^0 - \Omega_{nm,b}^0 + \Omega_{(n+1)m,b}^1 - \Omega_{nm,b}^1 \\ &\geq \sum_{k=nm}^{(n+1)m-1} \frac{c'_*}{k^3} - \frac{2C_2}{(nm)^3} \\ &\geq \frac{c'_*}{(n+1)^3 m^2} - \frac{C_2}{(nm)^3}. \end{aligned}$$

Choosing some $m_0 > \frac{20C_2}{c'_*}$, we show the first result on the lower bound of $\Omega_{(n+1)m,b} - \Omega_{nm,b}$.

Next, using Riemann-Lebesgue's lemma combined with Lemma 4.1 allows us to get the convergence result. This ends the proof of the desired result. $\square$

Now, we are in a position to show the main result on the spectral study of $\partial_r F(\Omega, 0)$, by showing the validity of all the requirements in Crandall-Rabinowitz's theorem. The function spaces that will be used below are described in (4.7) and (4.8).

Proposition 4.2. *Assume that the assumptions of Theorem 1.1 or those of Theorem 1.2 are satisfied. Then the following statements hold true.*

- (1) *The kernel of $\partial_r F(\Omega, 0) : \mathbf{X}_m \rightarrow \mathbf{Y}_m$ is non-trivial if and only if $\Omega = \Omega_{\ell m, b}$ for some $\ell \in \mathbb{N}^*$. In this case, it is a one-dimensional vector space generated by $\theta \mapsto \cos(\ell m \theta)$.*
- (2) *The range of $\partial_r F(\Omega_{\ell m, b}, 0)$ is closed and is of co-dimension one. It is given by*

$$\text{Range}(\partial_r F(\Omega_{\ell m, b}, 0)) = \left\{ r \in C^{1-\alpha}(\mathbb{T}) : r(\theta) = \sum_{n \geq 1, n \neq \ell} a_n \sin(nm\theta), a_n \in \mathbb{R} \right\}.$$

- (3) *Transversality condition:*

$$\partial_\Omega \partial_r F(\Omega_{\ell m, b}, 0)(\cos(\ell m \theta)) \notin R(\partial_r F(\Omega_{\ell m, b}, 0)).$$

Proof of Proposition 4.2. (1) The proof of statement (1) is a direct consequence of (4.52) and the strict monotonicity of $n \in \mathbb{N}^* \mapsto \Omega_{nm,b}$, seen in Lemma 4.1.

- (2) From (4.52), it is obvious to see that

$$R(\partial_r F(\Omega_{\ell m, b}, 0)) \subset \left\{ r \in C^{1-\alpha}(\mathbb{T}) : r(\theta) = \sum_{n \geq 1, n \neq \ell} a_n \sin(nm\theta), a_n \in \mathbb{R} \right\}.$$

Next, we prove the converse inclusion relationship. For any $r \in C^{1-\alpha}(\mathbb{T})$ satisfying

$$r(\theta) = \sum_{\substack{n \geq 1 \\ n \neq \ell}} b_n \sin(nm\theta),$$

we have to find some $h \in \mathbf{X}_m$ such that $\partial_r F(\Omega_{\ell m, b}, 0)h = r$. In view of (4.52), we formally get

$$h(\theta) = \sum_{\substack{n \geq 1 \\ n \neq \ell}} \frac{b_n}{(\Omega_{nm, b} - \Omega_{\ell m, b})nm} \cos(nm\theta),$$

and we need to prove that $h \in C^{2-\alpha}(\mathbb{T})$. First, we write

$$h(\theta) = \sum_{\substack{k \geq 1 \\ k \neq \ell m}} \frac{1}{\Omega_{k, b} - \Omega_{\ell m, b}} \tilde{b}_k \cos(k\theta), \quad (4.59)$$

where

$$\tilde{b}_k \triangleq \begin{cases} \frac{b_{k/m}}{k}, & \text{for } k \in m\mathbb{N}^*, \\ 0, & \text{for } k \notin m\mathbb{N}^*. \end{cases}$$

Notice that one easily gets that

$$\theta \mapsto \sum_{\substack{k \geq 1 \\ k \neq m\ell}} \tilde{b}_k \cos(k\theta) \in C^{2-\alpha}(\mathbb{T}).$$

Since the map $k \in \mathbb{N}^* \mapsto (\Omega_{k, b})_{k \in \mathbb{N}^*}$ is strictly increasing, we have

$$\sup_{k \neq m\ell} \frac{1}{|\Omega_{k, b} - \Omega_{\ell m, b}|} < \infty,$$

and

$$\sup_{\substack{k \neq m\ell \\ k \neq m\ell-1}} \left| \frac{1}{(\Omega_{k+1, b} - \Omega_{\ell m, b})(\Omega_{k, b} - \Omega_{\ell m, b})} \right| < \infty.$$

In order to show $h \in C^{2-\alpha}(\mathbb{T})$, by applying Lemma 6.5, we only need to prove that

$$\sup_{k \geq 1} k |\Omega_{k+1, b} - \Omega_{k, b}| < \infty. \quad (4.60)$$

Indeed, if the case (1.6) is considered, by virtue of (4.54), (2.13), the decreasing property of $|K'_0|$, (4.12)-(4.13) and Lemma 6.2, we infer from integration by parts that

$$\begin{aligned}
|\Omega_{k+1,b} - \Omega_{k,b}| &= 2 \left| \int_0^\pi K_0(2b \sin \eta) e^{2ik\eta} (e^{2i\eta} - 1) d\eta \right| \\
&\leq 2 \left| \int_0^{\frac{\pi}{2}} K_0(2b \sin \eta) e^{2ik\eta} (e^{2i\eta} - 1) d\eta \right| \\
&\quad + 2 \left| \int_0^{\frac{\pi}{2}} K_0(2b \sin \eta) e^{-2ik\eta} (e^{-2i\eta} - 1) d\eta \right| \\
&\leq \frac{4}{k} \int_0^{\frac{\pi}{2}} |K_0(2b \sin \eta)| d\eta + \frac{4b}{k} \int_0^{\frac{\pi}{2}} |K'_0(2b \sin \eta)| |e^{2i\eta} - 1| d\eta \\
&\leq \frac{4}{k} \int_0^{\frac{\pi}{2}} \left(|K_0(\frac{4}{\pi} b\eta)| + |K_0(2b\eta)| \right) d\eta + \frac{8b}{k} \int_0^{\frac{\pi}{2}} |K'_0(\frac{4}{\pi} b\eta)| \eta d\eta \\
&\leq \frac{C}{k},
\end{aligned}$$

where in the last line, we have applied (1.8) and Lemma 6.1. For the general case (1.10), we combine (4.53), (4.58) and the above inequality, leading to (4.60). Hence, we conclude that $h \in C^{2-\alpha}(\mathbb{T})$ and the proof of the range characterization follows immediately.

(3) Due to the fact $\partial_\Omega \partial_r F(\Omega_{\ell m, b}, 0)h = h'$, we find

$$\begin{aligned}
\partial_\Omega \partial_r F(\Omega_{\ell m, b}, 0) \cos(\ell m \theta) &= -\ell m \sin(\ell m \theta) \\
&\notin \text{Range}(\partial_r F(\Omega_{\ell m, b}, 0)),
\end{aligned}$$

as claimed. This ends the proof of Proposition 4.2. $\square$

5. Applications to geophysical flows

In this section, we examine special cases of (1.1)-(1.2) that cover crucial models encountered in geophysical flows. Through this exploration, we observe that our comprehensive framework often leads to known results on the construction of V-states in the simply connected cases. Furthermore, we derive new identities on special functions as a byproduct of our asymptotic description of the spectrum seen in Corollary 3.1.

5.1. 2D Euler equation in the whole space

Consider the 2D incompressible Euler equation in the whole plane. It corresponds to equation (1.1) with $\mathbf{D} = \mathbb{R}^2$ and $\psi = (-\Delta)^{-1}\omega$. Equivalently, the stream function ψ satisfies (1.2) with

$$K(\mathbf{x}, \mathbf{y}) = K_0(|\mathbf{x} - \mathbf{y}|) = -\frac{1}{2\pi} \log |\mathbf{x} - \mathbf{y}|.$$

Although $K_0(t) = -\frac{1}{2\pi} \log t$, $t > 0$ does not have a definite sign, the function $-K'_0(t) = \frac{1}{2\pi} \frac{1}{t}$ is completely monotone and has the following representation

$$-K'_0(t) = \frac{1}{2\pi} \frac{1}{t} = \frac{1}{2\pi} \int_0^\infty e^{-tx} dx = \int_0^\infty e^{-tx} d\mu(x),$$

that is, the associated non-negative measure μ is given by $d\mu(x) = \frac{1}{2\pi} dx$. Moreover, K_0 satisfies assumption (1.8) with any $\alpha \in (0, 1)$. Thus, assumptions (A1)-(A2) are verified and Theorem 1.1 can be applied in this case. This gives Burbea's result proved in [8]. On the other hand, in view of (1.9) (or (4.56)) and through straightforward calculus, using for instance the identity [43, 4.384] we have

$$\begin{aligned} \lambda_{n,1} &= \int_{\mathbb{T}} K_0\left(2 \sin \frac{\eta}{2}\right) \cos(n\eta) d\eta = -\frac{1}{2\pi} \int_0^{2\pi} \log\left(2 \sin \frac{\eta}{2}\right) \cos(n\eta) d\eta \\ &= -\frac{\log 2}{2\pi} \int_0^{2\pi} \cos(n\eta) d\eta - \frac{1}{2\pi} \int_0^{2\pi} \log\left(\sin \frac{\eta}{2}\right) \cos(n\eta) d\eta = \frac{1}{2n}, \end{aligned} \quad (5.1)$$

and

$$\Omega_{n,1}^0 = \lambda_{1,1} - \lambda_{n,1} = \frac{1}{2} \left(1 - \frac{1}{n}\right). \quad (5.2)$$

This is identical to the result in [8,56]. Using Corollary 3.1, we deduce that

$$\lambda_{n,1} = \sum_{k=0}^N \frac{A_k}{n^{2k+1}} + \varepsilon_{n,N},$$

where A_k is independent of n given by (the function Ψ_k is defined by (3.10)-(3.11))

$$A_k = \frac{1}{\pi} \int_0^\infty \frac{\Psi_k(x)}{x} dx, \quad k \in \mathbb{N}, \quad (5.3)$$

and

$$|\varepsilon_{n,N}| \leq C_{N,\delta} \frac{1}{n^{2N+\frac{5}{3}}} \int_0^\infty \frac{x^{\delta-1}}{1+\frac{x}{n}} dx \leq C'_{N,\delta} \frac{1}{n^{2N+\frac{5}{3}-\delta}}, \quad \delta \in (0, \frac{1}{3}).$$

From (5.1), we infer that

$$A_0 = \frac{1}{2}, \quad A_k = 0, \quad \forall k \in \mathbb{N}^*. \quad (5.4)$$

Note that the relations $A_0 = \frac{1}{2}$ and $A_1 = A_2 = 0$ can be easily justified from the formula (5.3), but it turns out that it is not trivial to show the general case $A_k = 0$ from (5.3) by a direct calculation.

5.2. gSQG equation in the whole space

The generalized surface quasi-geostrophic equation in the plane, denoted by gSQG equation, corresponds to

$$K(\mathbf{x}, \mathbf{y}) = K_0(|\mathbf{x} - \mathbf{y}|) = c_\beta |\mathbf{x} - \mathbf{y}|^{-\beta}, \quad c_\beta = \frac{\Gamma(\frac{\beta}{2})}{\pi^{2-\beta} \Gamma(1-\frac{\beta}{2})}, \quad \beta \in (0, 1).$$

Obviously, the function $t \in (0, \infty) \mapsto K_0(t) = c_\beta t^{-\beta}$ satisfies the fact that $-K'_0$ is completely monotone with

$$-K'_0(t) = \beta c_\beta t^{-\beta-1} = \frac{c_\beta}{\Gamma(\beta)} \int_0^\infty e^{-tx} x^\beta dx = \int_0^\infty e^{-tx} d\mu(x),$$

with the nonnegative measure μ given by $d\mu(x) = \frac{c_\beta}{\Gamma(\beta)} x^\beta dx$. In addition, condition (1.8) is satisfied for any $\alpha \in (0, 1 - \beta]$. Consequently, Theorem 1.1 can be applied in this case leading to the result of [45], with $b = 1$.

Now, let us discuss some identities that will mainly follow from Corollary 3.1. The explicit computation of the spectrum, which will be described in more detail in the following, was conducted in [45]. For the sake of completeness, we shall outline the main steps. By using (2.13) and the following identity, see for example [74, p. 4],

$$\forall \beta > -1, \quad \forall \gamma \in \mathbb{R}, \quad \int_0^\pi (\sin \eta)^\beta e^{i\gamma\eta} d\eta = \frac{\pi e^{i\frac{\gamma\pi}{2}} \Gamma(\beta+1)}{2^\beta \Gamma(1 + \frac{\beta+\gamma}{2}) \Gamma(1 + \frac{\beta-\gamma}{2})}, \quad (5.5)$$

we deduce that

$$\begin{aligned} \lambda_{n,1} &= c_\beta \int_0^{2\pi} \frac{1}{|2 \sin \frac{\eta}{2}|^\beta} \cos(n\eta) d\eta = 2c_\beta \int_0^\pi \frac{1}{|2 \sin \eta|^\beta} \cos(2n\eta) d\eta \\ &= \frac{\Gamma(\frac{\beta}{2})}{2^{1-\beta} \Gamma(1 - \frac{\beta}{2})} \frac{(-1)^n \Gamma(1 - \beta)}{\Gamma(n+1 - \frac{\beta}{2}) \Gamma(1 - n - \frac{\beta}{2})} \\ &= \frac{\Gamma(1 - \beta)}{2^{1-\beta} \Gamma^2(1 - \frac{\beta}{2})} \frac{\Gamma(n + \frac{\beta}{2})}{\Gamma(n+1 - \frac{\beta}{2})}, \end{aligned}$$

where in the last line we have used the identity that (using the relation $\Gamma(1+\mathbf{z}) = \mathbf{z}\Gamma(\mathbf{z})$)

$$\frac{(-1)^n}{\Gamma(1-n-\frac{\beta}{2})} = \frac{\Gamma(\frac{\beta}{2}+n)}{\Gamma(1-\frac{\beta}{2})\Gamma(\frac{\beta}{2})}.$$

Thus from (1.9) we have

$$\Omega_{n,1}^0 = \lambda_{1,1} - \lambda_{n,1} = \frac{\Gamma(1-\beta)}{2^{1-\beta}\Gamma^2(1-\frac{\beta}{2})} \left(\frac{\Gamma(1+\frac{\beta}{2})}{\Gamma(2-\frac{\beta}{2})} - \frac{\Gamma(n+\frac{\beta}{2})}{\Gamma(n+1-\frac{\beta}{2})} \right), \quad (5.6)$$

which recovers the rotating angular velocity of gSQG equation proposed in [12,45].

Thanks to (2.15), Lemma 3.2 and the fact that (e.g. see the identity [43, 3.241.2])

$$\int_0^\infty \frac{s^\beta}{1+s^2} ds = \frac{\pi}{2} \frac{1}{\sin(\frac{1+\beta}{2}\pi)} = \frac{\pi}{2} \frac{1}{\cos(\frac{\beta\pi}{2})},$$

we can deduce that

$$\frac{4n^2}{4n^2+1} \frac{A_{\beta,0}}{n^{1-\beta}} \leq \lambda_{n,1} \leq \frac{4n^2}{4n^2-1} \frac{A_{\beta,0}}{n^{1-\beta}}, \quad (5.7)$$

with

$$A_{\beta,0} = \frac{2c_\beta}{\Gamma(\beta)} \int_0^\infty \frac{s^\beta}{1+s^2} ds = \frac{\Gamma(\frac{\beta}{2})}{2^{2-\beta}\Gamma(\beta)\Gamma(1-\frac{\beta}{2})} \frac{1}{\cos(\frac{\beta\pi}{2})} = \frac{\Gamma(1-\beta)}{2^{1-\beta}\Gamma^2(1-\frac{\beta}{2})}, \quad (5.8)$$

where in the last inequality we used the reflection formula of the Gamma function

$$\Gamma(x)\Gamma(1-x) = \frac{\pi}{\sin(\pi x)}, \quad x \notin \mathbb{Z}.$$

Using Proposition 3.2 and Corollary 3.1, we can easily deduce the formula of $\lambda_{n,1}$:

$$\lambda_{n,1} = \frac{2c_\beta}{\Gamma(\beta)} \sum_{k=0}^N \frac{1}{n^{2k+1}} \int_0^\infty \Psi_k(\frac{x}{n}) x^{\beta-1} dx + \varepsilon_{n,N} = \sum_{k=0}^N \frac{A_{\beta,k}}{n^{2k+1-\beta}} + \varepsilon_{n,N},$$

where Ψ_k is given by (3.10)-(3.11) and

$$A_{\beta,k} = \frac{2c_\beta}{\Gamma(\beta)} \int_0^\infty \Psi_k(x) x^{\beta-1} dx,$$

and

$$|\varepsilon_{n,N}| \leq \frac{c_\beta C_N}{\Gamma(\beta)} \frac{1}{n^{2N+\frac{5}{3}}} \int_0^\infty \frac{x^{\beta-1}}{1+\frac{x}{n}} dx \leq \frac{C_{N,\beta}}{n^{2N+\frac{5}{3}-\beta}}.$$

Note that for $k = 0$, $A_{\beta,0}$ has the explicit formula shown by (5.8). Therefore, we infer from the formula (5.6), the following asymptotic expansion of the Wallis quotient

$$\frac{\Gamma(n + \frac{\beta}{2})}{\Gamma(n + 1 - \frac{\beta}{2})} = \frac{1}{n^{1-\beta}} + \sum_{k=1}^N \frac{A_{\beta,k}}{A_{\beta,0}} \frac{1}{n^{2k+1-\beta}} + O\left(\frac{1}{n^{2N+\frac{5}{3}-\beta}}\right). \quad (5.9)$$

On the other hand, recall that the Wallis quotient has the following expansion formula; see, for instance, [73, p. 34] or [9, Eq. (6.4)],

$$\frac{\Gamma(\mathbf{z} + a)}{\Gamma(\mathbf{z} + 1 - a)} = \mathbf{z}^{2a-1} \sum_{k=0}^N \frac{B_{2k}^{(2a)}(a)(1-2a)_{2k}}{(2k)!} \mathbf{z}^{-2k} + O(\mathbf{z}^{-2(N+1)+2a-1}), \quad |\arg \mathbf{z}| < \pi,$$

where $B_n^{(s)}(t)$ stands for the generalized Bernoulli polynomials given by the following generating function

$$\frac{x^s e^{tx}}{(e^x - 1)^s} = \sum_{n=0}^{\infty} B_n^{(s)}(t) \frac{x^n}{n!},$$

and $(t)_n$ is the Pochhammer symbol defined as

$$(t)_n \triangleq \begin{cases} t(t+1) \cdots (t+n-1), & \text{if } n \in \mathbb{N}^*, \\ 1, & \text{if } n = 0. \end{cases}$$

Thus, we also have

$$\frac{\Gamma(n + \frac{\beta}{2})}{\Gamma(n + 1 - \frac{\beta}{2})} = \sum_{k=0}^N \frac{B_{2k}^{(\beta)}(\frac{\beta}{2})(1-\beta)_{2k}}{(2k)!} \frac{1}{n^{2k+1-\beta}} + O(n^{-2(N+1)+\beta-1}). \quad (5.10)$$

By comparing (5.9) and (5.10), we deduce the following interesting identity:

$$\frac{B_{2k}^{(\beta)}(\frac{\beta}{2})(1-\beta)_{2k}}{(2k)!} A_{\beta,0} = A_{\beta,k}, \quad \forall k \in \mathbb{N}.$$

In addition, owing to (3.22) and (2.15), we find

$$\lambda_{n,1} - \lambda_{n+1,1} \approx \frac{2c_\beta}{\Gamma(\beta)} \int_0^\infty \frac{(2n+1)x^\beta}{(n^2+x^2)((n+1)^2+x^2)} dx \approx_\beta \frac{1}{n^{2-\beta}}. \quad (5.11)$$

Finally, applying Lemma 3.3 with $f(x) = \frac{c_\beta}{\Gamma(\beta)} x^{\beta-1}$ gives the convexity of $\{\lambda_{n,1}\}_{n \geq 2}$, that is,

$$\lambda_{n+1,1} + \lambda_{n-1,1} - 2\lambda_{n,1} \geq 0, \quad \forall n \geq 2.$$

5.3. QGSW equation in the whole space

Consider the QGSW equation in the whole plane, then it reduces to the equation (1.1) with $\mathbf{D} = \mathbb{R}^2$ and the stream function $\psi = (-\Delta + \varepsilon^2)^{-1}\omega$, with $\varepsilon > 0$ the deformation radius. According to [29], the kernel involved in (1.2) takes the form

$$K(\mathbf{x}, \mathbf{y}) = K_0(|\mathbf{x} - \mathbf{y}|) = \frac{1}{2\pi} \mathbf{K}_0(\varepsilon|\mathbf{x} - \mathbf{y}|),$$

where $\mathbf{K}_0$ is the modified Bessel function defined in Subsection 6.3. In view of (6.17),

$$K_0(t) = \frac{1}{2\pi} \mathbf{K}_0(\varepsilon t) = \frac{1}{2\pi} \int_1^\infty \frac{e^{-\varepsilon xt}}{\sqrt{x^2 - 1}} dx,$$

we obviously note that $-K'_0$ is completely monotone and by change of variables

$$-K'_0(t) = \frac{1}{2\pi} \int_1^\infty \frac{\varepsilon x e^{-\varepsilon xt}}{\sqrt{x^2 - 1}} dx = \frac{1}{2\pi} \int_\varepsilon^\infty e^{-tx} \frac{x}{\sqrt{x^2 - \varepsilon^2}} dx = \int_0^\infty e^{-tx} d\mu(x),$$

with the nonnegative measure μ given by

$$d\mu(x) = \frac{1}{2\pi} \frac{x}{\sqrt{x^2 - \varepsilon^2}} \mathbf{1}_{\{x > \varepsilon\}} dx.$$

In addition, for $0 \leq \alpha < 1$,

$$\begin{aligned} \int_0^{a_0} |K_0(t)| t^{-\alpha+\alpha^2} dt &= \frac{1}{2\pi} \int_1^\infty \frac{1}{\sqrt{x^2 - 1}} \int_0^{a_0} e^{-\varepsilon xt} t^{-\alpha+\alpha^2} dt dx \\ &\leq C_\alpha \int_1^\infty \frac{1}{\sqrt{x^2 - 1}} (\varepsilon x)^{-(1-\alpha+\alpha^2)} dx < \infty, \end{aligned}$$

which ensures that the condition (1.8) is verified. Hence, Theorem 1.1 can be applied in this case with any $\alpha \in (0, 1)$ that yields the result of [29]. Now, let us explore some other consequences.

Using (1.9) and the identity (6.19), we can easily recover the result in [29], that is,

$$\begin{aligned} \Omega_{n,1}^0 &= \lambda_{1,1} - \lambda_{n,1} = \frac{1}{2\pi} \int_{-\pi}^\pi \mathbf{K}_0(|2\varepsilon \sin \frac{\eta}{2}|) \cos \eta d\eta - \frac{1}{2\pi} \int_{-\pi}^\pi \mathbf{K}_0(|2\varepsilon \sin \frac{\eta}{2}|) \cos(n\eta) d\eta \\ &= \frac{2}{\pi} \int_0^{\frac{\pi}{2}} \mathbf{K}_0(2\varepsilon \sin \eta) \cos(2\eta) d\eta - \frac{2}{\pi} \int_0^{\frac{\pi}{2}} \mathbf{K}_0(2\varepsilon \sin \eta) \cos(2n\eta) d\eta \end{aligned}$$

$$\begin{aligned}
&= -\frac{2}{\pi} \int_0^{\frac{\pi}{2}} \mathbf{K}_0(2\varepsilon \cos \eta) \cos(2\eta) d\eta - \frac{2(-1)^n}{\pi} \int_0^{\frac{\pi}{2}} \mathbf{K}_0(2\varepsilon \cos \eta) \cos(2n\eta) d\eta \\
&= \mathbf{I}_1(\varepsilon) \mathbf{K}_1(\varepsilon) - \mathbf{I}_n(\varepsilon) \mathbf{K}_n(\varepsilon),
\end{aligned} \tag{5.12}$$

where $\mathbf{I}_n$ and $\mathbf{K}_n$ are the modified Bessel functions of the first and second kind (see Subsection 6.3).

Lemma 3.2 and (2.15) yield

$$\frac{4n^2}{4n^2+1} \frac{1}{\pi} \int_{\varepsilon}^{\infty} \frac{1}{\sqrt{x^2-\varepsilon^2}} \frac{x}{n^2+x^2} dx \leq \lambda_{n,1} \leq \frac{4n^2}{4n^2-1} \frac{1}{\pi} \int_{\varepsilon}^{\infty} \frac{1}{\sqrt{x^2-\varepsilon^2}} \frac{x}{n^2+x^2} dx.$$

From the explicit value

$$\int_{\varepsilon}^{\infty} \frac{1}{\sqrt{x^2-\varepsilon^2}} \frac{x}{n^2+x^2} dx = \frac{\pi}{2} \frac{1}{\sqrt{n^2+\varepsilon^2}},$$

we find that for $n \in \mathbb{N}^*$,

$$\frac{2n^2}{4n^2+1} \frac{1}{\sqrt{n^2+\varepsilon^2}} \leq \lambda_{n,1} \leq \frac{2n^2}{4n^2-1} \frac{1}{\sqrt{n^2+\varepsilon^2}}. \tag{5.13}$$

Inequality (3.22) implies that for $n \in \mathbb{N}^*$,

$$\begin{aligned}
\lambda_{n,1} - \lambda_{n+1,1} &\approx \int_{\varepsilon}^{\infty} \frac{x}{\sqrt{x^2-\varepsilon^2}} \frac{2n+1}{(n^2+x^2)((n+1)^2+x^2)} dx \\
&\approx \int_{\varepsilon}^{\infty} \frac{x}{\sqrt{x^2-\varepsilon^2}} \frac{1}{n^2+x^2} dx - \int_{\varepsilon}^{\infty} \frac{x}{\sqrt{x^2-\varepsilon^2}} \frac{1}{(n+1)^2+x^2} dx \\
&\approx \frac{1}{\sqrt{n^2+\varepsilon^2}} - \frac{1}{\sqrt{(n+1)^2+\varepsilon^2}}.
\end{aligned} \tag{5.14}$$

According to Corollary 3.1, we infer that

$$\begin{aligned}
\lambda_{n,1} &= \mathbf{I}_n(\varepsilon) \mathbf{K}_n(\varepsilon) = \frac{1}{\pi} \sum_{k=0}^N \frac{1}{n^{2k+1}} \int_{\varepsilon}^{\infty} \Psi_k\left(\frac{x}{n}\right) \frac{1}{\sqrt{x^2-\varepsilon^2}} dx + \varepsilon_{n,N} \\
&= \frac{1}{\pi} \sum_{k=0}^N \frac{1}{n^{2k+1}} \int_{\frac{\varepsilon}{n}}^{\infty} \Psi_k(x) \frac{1}{\sqrt{x^2-\varepsilon^2/n^2}} dx + \varepsilon_{n,N},
\end{aligned} \tag{5.15}$$

where Ψ_k is given by (3.10)-(3.11) and

$$|\varepsilon_{n,N}| \leq \frac{C_N}{n^{2N+\frac{5}{3}}} \int_{\frac{\varepsilon}{n}}^{\infty} \frac{1}{1+\frac{x}{n}} \frac{1}{\sqrt{x^2-\varepsilon^2}} dx \leq \frac{C_{N,\varepsilon}(\log n+1)}{n^{2N+\frac{5}{3}}},$$

with some constant $C_{N,\varepsilon} > 0$ independent of n . Note that the first term on the right-hand side of (5.15) is $\frac{1}{2\sqrt{n^2+\varepsilon^2}}$, and a direct calculation gives $\Psi_1(x) = \frac{x(x^4-6x^2+1)}{4(1+x^2)^4}$,

$$\begin{aligned} \int_{\frac{\varepsilon}{n}}^{\infty} \Psi_1(x) \frac{1}{\sqrt{x^2-\varepsilon^2}/n^2} dx &= \frac{n^3}{(\sqrt{n^2+\varepsilon^2})^3} \left(\frac{\pi}{16} - \frac{3\pi}{8} \frac{n^2}{n^2+\varepsilon^2} + \frac{5\pi}{16} \frac{n^4}{(n^2+\varepsilon^2)^2} \right) \\ &= \frac{n^3}{(\sqrt{n^2+\varepsilon^2})^3} \left(-\frac{\pi}{8} \frac{\varepsilon^2}{n^2+\varepsilon^2} + \frac{5\pi}{16} \frac{\varepsilon^4}{(n^2+\varepsilon^2)^2} \right), \end{aligned}$$

and

$$\int_{\frac{\varepsilon}{n}}^{\infty} \Psi_2(x) \frac{1}{\sqrt{x^2-\varepsilon^2}/n^2} dx = O\left(\frac{\varepsilon^2}{n^2+\varepsilon^2}\right).$$

Thus, we have

$$\lambda_{n,1} = \mathbf{I}_n(\varepsilon) \mathbf{K}_n(\varepsilon) = \frac{1}{2\sqrt{n^2+\varepsilon^2}} + O\left(\frac{1}{n^5}\right). \quad (5.16)$$

Finally, let us make a remark on the convexity of the spectrum $(\lambda_{n,1})_{n \geq 2}$. First, Lemma 3.3 does not apply to this case due to the jump in the measure density at ε , but through some numerical experiments, one can conjecture that for $n \geq 2$,

$$\lambda_{n+1,1} + \lambda_{n-1,1} - 2\lambda_{n,1} \geq 0.$$

So far we do not know how to rigorously prove this result, but as a simple application of (5.16), we can show the convexity result for every $n \geq n_\varepsilon$ with some $n_\varepsilon \in \mathbb{N}$ sufficiently large.

5.4. Euler- α equation in the whole space

The Euler- α equation is a regularization of the 2D Euler equation, and it has been introduced in the context of averaged fluid models; see [59,60]. Taking into account its vorticity form in the whole plane, it corresponds to equation (1.1) with $\mathbf{D} = \mathbb{R}^2$ and the stream function $\psi = (-\Delta)^{-1}\omega - (-\Delta + \frac{1}{\alpha^2})^{-1}\omega$, for $\alpha > 0$.

The kernel involved in (1.2) takes the form of

$$K(\mathbf{x}, \mathbf{y}) = K_0(|\mathbf{x} - \mathbf{y}|) = -\frac{1}{2\pi} \log |\mathbf{x} - \mathbf{y}| - \frac{1}{2\pi} \mathbf{K}_0\left(\frac{1}{\alpha} |\mathbf{x} - \mathbf{y}|\right).$$

Thus, $K_0(t) = -\frac{1}{2\pi} \log |t| - \frac{1}{2\pi} \mathbf{K}_0\left(\frac{1}{\alpha} |t|\right)$ satisfies

$$-K'_0(t) = \frac{1}{2\pi} \int_0^\infty e^{-tx} \left(1 - \frac{x\alpha \mathbf{1}_{\{x > \frac{1}{\alpha}\}}}{\sqrt{x^2\alpha^2 - 1}} \right) dx,$$

which implies that $-K'_0$ is not completely monotone and that Theorem 1.1 cannot be applied to any symmetry $m \in \mathbb{N}^*$. However, this theorem occurs for higher symmetry m . Indeed, using (1.9), (5.2), (5.12) and (5.16), we can deduce that $(\Omega_{n,1})$ is strictly increasing for every $n \geq n_\alpha$ with some $n_\alpha \in \mathbb{N}$ large enough. Hence, we may check that all the assumptions of Crandall-Rabinowitz's theorem work well. Note that in a recent work [85] the strict monotonicity of $(\Omega_{n,1})$ is satisfied for all the range $n \in \mathbb{N}^*$, and the author obtained the existence of m -fold symmetric V-states for the Euler- α equation. Actually, the monotonicity follows directly from the explicit formula of the spectrum which takes the following form

$$\Omega_{n,1} = \frac{2n-1}{2n} - \left(\mathbf{I}_1\left(\frac{1}{\alpha}\right) \mathbf{K}_1\left(\frac{1}{\alpha}\right) - \mathbf{I}_n\left(\frac{1}{\alpha}\right) \mathbf{K}_n\left(\frac{1}{\alpha}\right) \right).$$

5.5. 2D Euler equation in the unit disc

Consider the 2D incompressible Euler equation in the vorticity form in the unit disc $\mathbb{D}$ with rigid boundary condition (the non-penetration boundary condition), that is, the equation (1.1) in $\mathbf{D} = \mathbb{D}$ with the stream function ψ solving the Dirichlet problem in the unit disc

$$-\Delta\psi = \omega, \quad \text{in } \mathbb{D}, \quad \psi|_{\partial\mathbb{D}} = 0. \quad (5.17)$$

It is classical that the expression formula (1.2) holds with the Green function K given by

$$K(\mathbf{x}, \mathbf{y}) = -\frac{1}{2\pi} \log \frac{|\mathbf{x}-\mathbf{y}|}{|1-\mathbf{x}\bar{\mathbf{y}}|} = -\frac{1}{2\pi} \log |\mathbf{x}-\mathbf{y}| + \frac{1}{2\pi} \log |1-\mathbf{x}\bar{\mathbf{y}}|, \quad \mathbf{x}, \mathbf{y} \in \mathbb{D}.$$

Clearly, $t \in (0, \infty) \mapsto K_0(t) = -\frac{1}{2\pi} \log t$ satisfies assumptions (A1)-(A2) with $\alpha \in (0, 1)$, and the perturbative kernel $(\mathbf{x}, \mathbf{y}) \in \mathbb{D}^2 \mapsto K_1(\mathbf{x}, \mathbf{y}) = \frac{1}{2\pi} \log |1-\mathbf{x}\bar{\mathbf{y}}|$ is smooth and satisfies assumptions (A3)-(A4). Hence, Theorem 1.2 can be applied to the study of V-states around the Rankine vortices $\mathbf{1}_{b\mathbb{D}}$ ($0 < b < 1$) leading to the bifurcation for large symmetry m . Actually, as we shall see below, we are able to retrieve all the symmetry $m \in \mathbb{N}^*$. This allows us to replicate the full result obtained in [25]. To start with, we note that the quantity

$$G_1(\rho_1, \theta, \rho_2, \eta) = K_1(\rho_1 e^{i\theta}, \rho_2 e^{i\eta}) = \frac{1}{2\pi} \log (|\rho_1 \rho_2 e^{i\theta} - e^{i\eta}|),$$

satisfies the property that $\partial_{\rho_1} G_1(\rho_1, \theta, \rho_2, \eta) = \frac{\rho_2}{\rho_1} \partial_{\rho_2} G_1(\rho_1, \theta, \rho_2, \eta)$, and owing to (1.11), (5.1) and the following fact (see e.g. 4.397 of [43])

$$\int_0^{2\pi} \log(1 - 2a \cos \eta + a^2) \cos(n\eta) d\eta = -\frac{2\pi}{n} a^n, \quad |a| < 1,$$

$$\int_0^{2\pi} \log(1 - 2a \cos \eta + a^2) d\eta = 0, \quad |a| \leq 1,$$

we obtain through integration by parts

$$\begin{aligned} \Omega_{n,b} &= \frac{n-1}{2n} - b^{-1} \int_0^{2\pi} \int_0^b \partial_{\rho_1} G_1(b, 0, \rho, \eta) \rho d\rho d\eta - \int_0^{2\pi} G_1(b, 0, b, \eta) \cos(n\eta) d\eta \\ &= \frac{n-1}{2n} - b^{-2} \int_0^{2\pi} \int_0^b \partial_{\rho} G_1(b, 0, \rho, \eta) \rho^2 d\rho d\eta \\ &\quad - \frac{1}{4\pi} \int_0^{2\pi} \log(b^4 + 1 - 2b^2 \cos \eta) \cos(n\eta) d\eta \\ &= \frac{n-1 + b^{2n}}{2n} - \int_0^{2\pi} G_1(b, 0, b, \eta) d\eta + 2b^{-2} \int_0^{2\pi} \int_0^b G_1(b, 0, \rho, \eta) \rho d\rho d\eta \end{aligned}$$

implying that

$$\begin{aligned} \Omega_{n,b} &= \frac{n-1 + b^{2n}}{2n} - \frac{1}{4\pi} \int_0^{2\pi} \log(b^4 + 1 - 2b^2 \cos \eta) d\eta \\ &\quad + \frac{b^{-2}}{2\pi} \int_0^b \int_0^{2\pi} \log(b^2 \rho^2 + 1 - 2b\rho \cos \eta) d\eta \rho d\rho \\ &= \frac{n-1 + b^{2n}}{2n}. \end{aligned}$$

This formula coincides with the rotating angular velocity established in [25]. Direct calculation shows that $(\Omega_{n,1})_{n \in \mathbb{N}^*}$ is strictly increasing, thus we can remove the restriction on m in Theorem 1.2 and recover the existence of m -fold symmetric V-states with $m \in \mathbb{N}^*$ for the 2D Euler equation in the unit disc as in [25, Theorem 1].

5.6. gSQG equation in the unit disc

If we consider the gSQG equation in the unit disc $\mathbb{D}$ with rigid boundary, it corresponds to the equation (1.1) with $\mathbf{D} = \mathbb{D}$ and the stream function ψ solving

$$\psi = (-\Delta)^{-1+\frac{\beta}{2}}\omega, \quad \text{in } \mathbb{D}, \quad \psi|_{\partial\mathbb{D}} = 0, \quad \beta \in (0, 1).$$

Equivalently, ψ satisfies

$$\psi(\mathbf{x}) = \frac{1}{\Gamma(1-\frac{\beta}{2})} \int_0^\infty t^{-\frac{\beta}{2}} e^{t\Delta} \omega(\mathbf{x}) dt = \int_{\mathbb{D}} K(\mathbf{x}, \mathbf{y}) \omega(\mathbf{y}) d\mathbf{y}, \quad (5.18)$$

where the Laplacian Δ is defined in $\mathbb{D}$ with the Dirichlet boundary condition. According to [58, Lemma 2.3], the spectral Green function K satisfies

$$K(\mathbf{x}, \mathbf{y}) = K_0(|\mathbf{x} - \mathbf{y}|) + K_1(\mathbf{x}, \mathbf{y}) = \frac{\Gamma(\frac{\beta}{2})}{2^{2-\beta}\pi\Gamma(1-\frac{\beta}{2})} |\mathbf{x} - \mathbf{y}|^{-\beta} + K_1(\mathbf{x}, \mathbf{y}),$$

and $K_1 \in C^\infty(\mathbb{D} \times \mathbb{D})$. In view of Lemma 2.4 of [58], $K_1(\mathbf{x}, \mathbf{y})$ satisfies assumptions (A3)-(A4). Hence, Theorem 1.2 can be applied in this case to show the existence of m -fold symmetric rotating patch solutions around the trivial solution $\mathbf{1}_{b\mathbb{D}}$ ($0 < b < 1$) with sufficiently large m , which is one of the main results in [58]. On the other hand, by virtue of Lemma 6.6 and (5.18), we have

$$K(\mathbf{x}, \mathbf{y}) = \sum_{n \in \mathbb{N}, k \in \mathbb{N}^*} x_{n,k}^{\beta-2} \left(\phi_{n,k}^{(1)}(\mathbf{x}) \phi_{n,k}^{(1)}(\mathbf{y}) + \phi_{n,k}^{(2)}(\mathbf{x}) \phi_{n,k}^{(2)}(\mathbf{y}) \right).$$

For $\mathbf{x} = \rho_1 e^{i\theta} \in \mathbb{D}$, $\mathbf{y} = \rho_2 e^{i\eta} \in \mathbb{D}$, and using the notation (4.1), we also have

$$K(\mathbf{x}, \mathbf{y}) = G(\rho_1, \theta, \rho_2, \eta) = \sum_{n \in \mathbb{N}, k \in \mathbb{N}^*} x_{n,k}^{\beta-2} A_{n,k}^2 J_n(x_{n,k} \rho_1) J_n(x_{n,k} \rho_2) \cos(n(\theta - \eta)). \quad (5.19)$$

Recall that in Subsection 2.2 the spectrum $\Omega_{n,b} = -V[0] - \Lambda_{n,b}$ with

$$V[0] = b^{-1} \int_0^{2\pi} \int_0^b \left(\nabla_{\mathbf{x}} K(b e^{i\theta}, \rho e^{i\eta}) \cdot e^{i\theta} \right) \rho d\rho d\eta, \quad \Lambda_{n,b} = \int_0^{2\pi} K(b, b e^{i\eta}) e^{in\eta} d\eta, \quad (5.20)$$

we can argue as [58] to show that

$$V[0] = -2 \sum_{k \geq 1} x_{0,k}^{\beta-2} \frac{J_1^2(x_{0,k} b)}{J_1^2(x_{0,k})}, \quad \Lambda_{n,b} = 2 \sum_{k \geq 1} x_{n,k}^{\beta-2} \frac{J_n^2(x_{n,k} b)}{J_{n+1}^2(x_{n,k})}.$$

Using Sneddon's formula, Lemma 5.1 of [58] proves the strict monotonicity of $n \mapsto \Omega_{n,b}$ in the small b case or the small α case, and further implies the existence of m -fold symmetric V-states around $\mathbf{1}_{b\mathbb{D}}$ ($0 < b < 1$) in both cases.

5.7. QGSW equation in the unit disc

Consider the QGSW model in the unit disc $\mathbb{D}$ with rigid boundary, which corresponds to equation (1.1) with $\mathbf{D} = \mathbb{D}$ and the relationship between ψ and ω can be expressed by

$$\psi = (-\Delta + \varepsilon^2)^{-1}\omega,$$

which denotes the unique solution to the following Dirichlet problem,

$$(-\Delta + \varepsilon^2)\psi = \omega \text{ in } \mathbb{D}, \quad \psi|_{\partial\mathbb{D}} = 0.$$

In order to describe the associated Green function, we need to solve the equation for every $\mathbf{x} \in \mathbb{D}$,

$$-\Delta_{\mathbf{y}}K(\mathbf{x}, \mathbf{y}) + \varepsilon^2K(\mathbf{x}, \mathbf{y}) = \delta_{\mathbf{x}}(\mathbf{y}) \text{ in } \mathbb{D}, \quad K(\mathbf{x}, \cdot)|_{\partial\mathbb{D}} = 0,$$

where $\delta_{\mathbf{x}}(\mathbf{y})$ is the Dirac measure centered at the point $\mathbf{x}$. According to the spectral theory of elliptic problems, for example [30, Sec. 6.2 and Sec. 6.3], we infer that $(-\Delta + \varepsilon^2)^{-1}$ is well-defined and bounded from $L^2(\mathbb{D})$ to $H^2(\mathbb{D})$. In addition, we can split the kernel as follows

$$K(\mathbf{x}, \mathbf{y}) = K_0(|\mathbf{x} - \mathbf{y}|) + K_1(\mathbf{x}, \mathbf{y}) \quad \text{with} \quad K_0(|\mathbf{x} - \mathbf{y}|) = \frac{1}{2\pi}\mathbf{K}_0(\varepsilon|\mathbf{x} - \mathbf{y}|),$$

and K_1 solves the elliptic problem

$$-\Delta_{\mathbf{y}}K_1(\mathbf{x}, \mathbf{y}) + \varepsilon^2K_1(\mathbf{x}, \mathbf{y}) = 0, \text{ in } \mathbb{D}, \quad K_1(\mathbf{x}, \mathbf{y})|_{\mathbf{y} \in \partial\mathbb{D}} = -\frac{1}{2\pi}\mathbf{K}_0(\varepsilon|\mathbf{x} - \mathbf{y}|).$$

Since $\mathbf{K}_0$ is smooth except for $\mathbf{x} = 0$, by the classical regularity theory of elliptic PDE, we see that $\mathbf{y} \in \mathbb{D} \mapsto K_1(\mathbf{x}, \mathbf{y})$ is smooth for any $\mathbf{x} \in \mathbb{D}$. Following exactly the same argument as in [30, p.39], we find $K(\mathbf{x}, \mathbf{y}) = K(\mathbf{y}, \mathbf{x})$. Thus, K_1 belongs to $C^\infty(\mathbb{D} \times \mathbb{D})$ and, moreover, the geometric properties (1.3) and (1.4) in $\mathbb{D}$ can be easily checked by arguing as [58, Lemma 2.4]. Hence, for every $\varepsilon > 0$ and $b \in (0, 1)$, we can apply Theorem 1.2 to show the existence of m -fold symmetric V-states for the QGSW equation in the unit disc $\mathbb{D}$ with m large enough.

On the other hand, according to the work developed on the unit disc $\mathbb{D}$, we actually obtain an explicit formula for K expressed by series in terms of the eigenvalue-eigenfunction pairs of the spectral problem (6.20). According to the spectral theory of second-order elliptic PDE, see, e.g., [30, Sec. 6.5], the eigenfunctions $\{\phi_{n,k}^{(1)}, \phi_{n,k}^{(2)}\}_{n \in \mathbb{N}, k \in \mathbb{N}^*}$ in Lemma 6.6 form an orthonormal basis in $L^2(\mathbb{D})$ and belong to $H_0^1(\mathbb{D})$. Then, via a simple calculation, we infer that

$$\psi(\mathbf{x}) = \int_{\mathbb{D}} \sum_{n \in \mathbb{N}, k \in \mathbb{N}^*} \frac{1}{x_{n,k}^2 + \varepsilon^2} \left(\phi_{n,k}^{(1)}(\mathbf{x})\phi_{n,k}^{(1)}(\mathbf{y}) + \phi_{n,k}^{(2)}(\mathbf{x})\phi_{n,k}^{(2)}(\mathbf{y}) \right) \omega(\mathbf{y}) d\mathbf{y}.$$

Upon comparison with (1.2), it results in

$$K(\mathbf{x}, \mathbf{y}) = \sum_{n \in \mathbb{N}, k \in \mathbb{N}^*} \frac{1}{x_{n,k}^2 + \varepsilon^2} \left(\phi_{n,k}^{(1)}(\mathbf{x}) \phi_{n,k}^{(1)}(\mathbf{y}) + \phi_{n,k}^{(2)}(\mathbf{x}) \phi_{n,k}^{(2)}(\mathbf{y}) \right). \quad (5.21)$$

Similarly to (5.19), we find

$$K(\mathbf{x}, \mathbf{y}) = G(\rho_1, \theta, \rho_2, \eta) = \sum_{n \in \mathbb{N}, k \in \mathbb{N}^*} \frac{1}{x_{n,k}^2 + \varepsilon^2} A_{n,k}^2 J_n(x_{n,k} \rho_1) J_n(x_{n,k} \rho_2) \cos(n(\theta - \eta)), \quad (5.22)$$

with $A_{n,k}$ given by (6.22). Concerning the spectrum $\Omega_{n,b} = -V[0] - \Lambda_{n,b}$, in view of (5.20) and (5.22), and arguing as [58, Eq. (113)], we can show that

$$V[0] = -2 \sum_{k \in \mathbb{N}^*} \frac{1}{x_{0,k}^2 + \varepsilon^2} \frac{J_1^2(bx_{0,k})}{J_1^2(x_{0,k})}, \quad (5.23)$$

and

$$\begin{aligned} \Lambda_{n,b} &= \int_0^{2\pi} \left(\sum_{\ell \in \mathbb{N}, k \in \mathbb{N}^*} \frac{1}{x_{\ell,k}^2 + \varepsilon^2} A_{\ell,k}^2 J_\ell^2(bx_{\ell,k}) \cos(\ell\eta) \right) e^{in\eta} d\eta \\ &= 2 \sum_{k \in \mathbb{N}^*} \frac{1}{x_{n,k}^2 + \varepsilon^2} \frac{J_n^2(bx_{n,k})}{J_{n+1}^2(x_{n,k})}. \end{aligned} \quad (5.24)$$

Interestingly, similar to Sneddon's formula used in [58], there is also a suitable summation formula for (5.24). Choosing $X = Y = b$, $\nu = n$ and $\mathbf{z} = \varepsilon$ in the Kneser-Sommerfeld expansion (6.23), we obtain

$$\Lambda_{n,b} = \mathbf{I}_n(b\varepsilon) \mathbf{K}_n(b\varepsilon) - \frac{\mathbf{K}_n(\varepsilon)}{\mathbf{I}_n(\varepsilon)} \mathbf{I}_n^2(b\varepsilon). \quad (5.25)$$

As $\varepsilon \rightarrow 0$, noting that $\sum_{k=1}^{\infty} \frac{1}{x_{0,k}^2} \frac{J_1^2(bx_{0,k})}{J_1^2(x_{0,k})} = \frac{1}{4}$ (using Sneddon's formula, e.g. see [58, Eq. (29)]), and applying the asymptotics of $\mathbf{I}_n(x)$ and $\mathbf{K}_n(x)$ in (6.18), we deduce that the spectrum $\Omega_{n,b}$ in QGSW equation in the unit disc satisfies

$$\lim_{\varepsilon \rightarrow 0} \Omega_{n,b} = 2 \sum_{k=1}^{\infty} \frac{1}{x_{2,k}^2} \frac{J_1^2(bx_{0,k})}{J_1^2(x_{0,k})} - \lim_{\varepsilon \rightarrow 0} \Lambda_{n,b} = \frac{1}{2} - \frac{1 - b^{2n}}{2n},$$

which coincides with the spectrum of the 2D Euler equation in the unit disc.

The monotonicity of $(\Lambda_{n,b})_{n \in \mathbb{N}^*}$ given by (5.25) for every $b \in (0, 1)$ and $\varepsilon > 0$ is a crucial property and seems not easy to achieve. In the following, we show that for every

$\varepsilon > 0$ and $b \in (0, b_*)$ with some small $b_* \in (0, \frac{1}{2})$ depending only on ε , such a sequence $(\Lambda_{n,b})_{n \in \mathbb{N}^*}$ is strictly decreasing with respect to n . Notice that

$$\begin{aligned} \Lambda_{n,b} - \Lambda_{n+1,b} &= \left(\mathbf{I}_n(b\varepsilon) \mathbf{K}_n(b\varepsilon) - \mathbf{I}_{n+1}(b\varepsilon) \mathbf{K}_{n+1}(b\varepsilon) \right) \left(1 - \frac{\mathbf{I}_n(\varepsilon) \mathbf{K}_n(\varepsilon)}{\mathbf{I}_n(b\varepsilon) \mathbf{K}_n(b\varepsilon)} \frac{\mathbf{I}_n^2(b\varepsilon)}{\mathbf{I}_n^2(\varepsilon)} \right) \\ &\quad - \mathbf{I}_{n+1}(b\varepsilon) \mathbf{K}_{n+1}(b\varepsilon) \left(\frac{\mathbf{I}_n(\varepsilon) \mathbf{K}_n(\varepsilon)}{\mathbf{I}_n(b\varepsilon) \mathbf{K}_n(b\varepsilon)} \frac{\mathbf{I}_n^2(b\varepsilon)}{\mathbf{I}_n^2(\varepsilon)} - \frac{\mathbf{I}_{n+1}(\varepsilon) \mathbf{K}_{n+1}(\varepsilon)}{\mathbf{I}_{n+1}(b\varepsilon) \mathbf{K}_{n+1}(b\varepsilon)} \frac{\mathbf{I}_{n+1}^2(b\varepsilon)}{\mathbf{I}_{n+1}^2(\varepsilon)} \right). \end{aligned}$$

Using (5.13), (5.14) and the following facts

$$\forall n \in \mathbb{N}^*, \quad x > 0, \quad b \in (0, 1), \quad \mathbf{K}_n(x) > 0, \quad \text{and} \quad 0 < \mathbf{I}_n(bx) \leq b^n \mathbf{I}_n(x),$$

we deduce that

$$\frac{\mathbf{I}_n(\varepsilon) \mathbf{K}_n(\varepsilon)}{\mathbf{I}_n(b\varepsilon) \mathbf{K}_n(b\varepsilon)} \leq \frac{4n^2 + 1}{4n^2 - 1} \frac{\sqrt{n^2 + (b\varepsilon)^2}}{\sqrt{n^2 + \varepsilon^2}} \leq \frac{5}{3},$$

and

$$\begin{aligned} \Lambda_{n,b} - \Lambda_{n+1,b} &\geq \frac{1}{4} \left(\frac{1}{\sqrt{n^2 + (b\varepsilon)^2}} - \frac{1}{\sqrt{(n+1)^2 + (b\varepsilon)^2}} \right) \left(1 - \frac{5}{3} b^{2n} \right) - \frac{8}{17} \frac{1}{\sqrt{(n+1)^2 + (b\varepsilon)^2}} \frac{5}{3} b^{2n} \\ &\geq \frac{2n+1}{\sqrt{n^2 + (b\varepsilon)^2} ((n+1)^2 + (b\varepsilon)^2)} \left(\frac{1}{8} - \frac{5}{6} b^{2n} - \frac{40}{51} \frac{(n+1)^2 + (b\varepsilon)^2}{2n+1} b^{2n} \right). \end{aligned}$$

Thus, by taking $b \leq \frac{1}{2}$ and setting $\sup_{n \in \mathbb{N}^*} \frac{((n+1)^2 + (\varepsilon/2)^2)}{2n+1} \frac{1}{2^n} = C(\varepsilon)$, we deduce that

$$\Lambda_{n,b} - \Lambda_{n+1,b} \geq \frac{2n+1}{\sqrt{n^2 + (b\varepsilon)^2} ((n+1)^2 + (b\varepsilon)^2)} \left(\frac{1}{8} - \frac{5}{6} b^{2n} - \frac{40}{51} C(\varepsilon) b^n \right).$$

Hence, there exists a small constant $b_* \in (0, \frac{1}{2})$ depending only on ε so that for all $\varepsilon > 0$ and $b \in (0, b_*)$ the sequence $(\Lambda_{n,b})_{n \in \mathbb{N}^*}$ is strictly decreasing with respect to n . With this property at hand, and for all $\varepsilon > 0$ and $b \in (0, b_*)$, we can show the existence of m -fold ($m \in \mathbb{N}^*$) symmetric rotating solutions around $\mathbf{1}_{b\mathbb{D}}(\mathbf{x})$ for the QGSW equation in the unit disc. This result is completely new, in contrast to the models discussed above.

6. Tools

In this section, we collect some useful results used throughout the paper.

6.1. Completely monotone functions

This subsection is devoted to outlining some properties of completely monotone functions. We start with the following definition.

Definition 6.1. A function $f : (0, \infty) \rightarrow \mathbb{R}$ is said to be completely monotone if it is of class C^∞ and satisfies

$$(-1)^n f^{(n)}(t) \geq 0 \quad \forall t > 0, \quad \forall n \in \mathbb{N}.$$

The typical example is $f(t) = t^{-\alpha}$, with $\alpha \geq 0$. One can refer, for instance, to [86] for various examples of completely monotone functions.

The following result is fundamental in the theory of completely monotone functions. It gives a useful characterization through the Laplace transform of the Borel measure. For more details, see, for example, Theorem 1.4 in [86].

Theorem 6.1 (Bernstein's theorem). Let $g : (0, \infty) \mapsto \mathbb{R}$ be a completely monotone function. Then it is the Laplace transform of a unique nonnegative measure μ on $[0, \infty)$, that is,

$$\forall t > 0, \quad g(t) = \int_0^\infty e^{-tx} d\mu(x) \triangleq \mathcal{L}(\mu)(t).$$

Conversely, whenever $\mathcal{L}(\mu)(t) < \infty$ for every $t > 0$, the function $t \mapsto \mathcal{L}(\mu)$ is a completely monotone function.

The next goal is to discuss useful pointwise estimates on completely monotone functions.

Lemma 6.1. The following assertions are true.

(1) Let $f : (0, \infty) \mapsto \mathbb{R}$ be a completely monotone function. Then, for any $n \in \mathbb{N}$ and $\alpha \in (0, 1)$ we have

$$\forall t > 0, \quad t^n |f^{(n)}(t)| \leq \left(\frac{n}{1-\alpha}\right)^n f(\alpha t).$$

(2) Consider $f : (0, \infty) \mapsto \mathbb{R}$ such that $-f'$ is completely monotone. Then, we have

$$\forall 0 < t_1 \leq t_2, \quad 0 \leq f(t_1) - f(t_2) \leq (t_1 - t_2)f'(t_1).$$

Proof of Lemma 6.1. (1) By differentiation, we get

$$t^n f^{(n)}(t) = (-1)^n \int_0^\infty (tx)^n e^{-tx} d\mu(x).$$

Now, we use the inequality

$$\forall s \geq 0, \quad s^n \leq \left(\frac{n}{1-\alpha}\right)^n e^{(1-\alpha)s},$$

to get

$$\begin{aligned} t^n |f^{(n)}(t)| &\leq \left(\frac{n}{1-\alpha}\right)^n \int_0^\infty e^{-\alpha tx} d\mu(x) \\ &\leq \left(\frac{n}{1-\alpha}\right)^n f(\alpha t). \end{aligned}$$

(2) Using the identity (2.18) yields for any $a > 0$ and $t > 0$,

$$f(t) = f(a) + \int_0^\infty \frac{e^{-tx} - e^{-ax}}{x} d\mu(x).$$

Let $0 < t_1 \leq t_2$, then

$$f(t_1) - f(t_2) = \int_0^\infty e^{-t_1 x} \frac{1 - e^{-(t_2 - t_1)x}}{x} d\mu(x).$$

At this stage we use the inequality

$$\forall s \geq 0, \quad 0 \leq 1 - e^{-s} \leq s,$$

which implies that

$$0 \leq f(t_1) - f(t_2) \leq (t_2 - t_1) \int_0^\infty e^{-t_1 x} d\mu(x) = (t_1 - t_2) f'(t_1).$$

This achieves the proof of the desired result. $\square$

Next, we intend to discuss the propagation of higher regularity/integrability of completely monotone functions, which will be used later.

Lemma 6.2. *The following statements hold true.*

(1) Assume that f is a completely monotone function satisfying

$$\int_0^{t_0} |f(t)|t^\beta dt < \infty, \quad \text{for some } \beta \in (-1, \infty) \text{ and } t_0 > 0, \quad (6.1)$$

then we have

$$\int_0^{t_0} |f^{(k)}(t)|t^{k+\beta} dt \leq C_{k,\beta} \int_0^{t_0} |f(t)|t^\beta dt, \quad \forall k \in \mathbb{N}.$$

(2) Assume that f is a smooth function satisfying (6.1) and f' has a constant sign, then we have

$$\int_0^{t_0} |f'(t)|t^{1+\beta} dt \leq (1+\beta) \int_0^{t_0} |f(t)|t^\beta dt + |f(t_0)|t_0^{1+\beta}. \quad (6.2)$$

Proof of Lemma 6.2. (1) By virtue of Lemma 6.1-(1) and the decreasing property of f , we infer

$$\int_0^{t_0} |f^{(k)}(t)|t^{k+\beta} dt \leq C_k \int_0^{t_0} f\left(\frac{t}{2}\right)t^\beta dt \leq C_{k,\beta} \int_0^{t_0} f(t)t^\beta dt.$$

(2) Without loss of generality, we may suppose that f' is non-positive, then f is non-increasing. If $\lim_{t \rightarrow 0^+} f(t) < \infty$, then $\lim_{t \rightarrow 0^+} f(t)t^{1+\beta} = 0$. However if $\lim_{t \rightarrow 0^+} f(t) = \infty$, then $f(t) > 0$ for sufficiently small $t > 0$, and thus

$$0 \leq \lim_{t \rightarrow 0^+} f(t)t^{1+\beta} \leq \lim_{t \rightarrow 0^+} (1+\beta) \int_0^t f(s)s^\beta ds = 0.$$

Using integration by parts we see that

$$\begin{aligned} - \int_0^{t_0} f'(t)t^{1+\beta} dt &= -t_0^{1+\beta} f(t_0) + \lim_{t \rightarrow 0^+} t^{1+\beta} f(t) + (1+\beta) \int_0^{t_0} f(t)t^\beta dt \\ &\leq |f(t_0)|t_0^{1+\beta} + (1+\beta) \int_0^{t_0} |f(t)|t^\beta dt, \end{aligned}$$

which yields the desired estimate (6.2). $\square$

Next, we shall discuss a result which will be used frequently in the paper.

Lemma 6.3. *If $-f'$ is a completely monotone function on $(0, \infty)$ and f satisfies (6.1), then for any $\alpha \in [0, 1]$, $m, n \in \mathbb{N}^*$, $t_0, c_1, c_2 > 0$, we have*

$$\int_0^{t_0} |f^{(m)}(c_1 t)|^\alpha |f^{(n)}(c_2 t)|^{1-\alpha} t^{m\alpha+n(1-\alpha)+\beta} dt \leq C \left(\int_0^{t_0} |f(t)| t^\beta dt + |f(t_0)| t_0^{1+\beta} \right), \quad (6.3)$$

and

$$\int_0^{t_0} |f(c_1 t)|^\alpha |f^{(n)}(c_2 t)|^{1-\alpha} t^{n(1-\alpha)+\beta} dt \leq C \left(\int_0^{(c_1 \vee 1)t_0} |f(t)| t^\beta dt + |f(t_0)| t_0^{1+\beta} \right), \quad (6.4)$$

with $c_1 \vee 1 \triangleq \max\{c_1, 1\}$ and the constant $C > 0$ depends on $m, n, \alpha, \beta, c_1, c_2$.

Proof of Lemma 6.3. First, by virtue of Lemma 6.2, we have that for $k \in \mathbb{N}^*$,

$$\int_0^{t_0} |f^{(k)}(t)| t^{k+\beta} dt \leq C \int_0^{t_0} (-f'(t)) t^{1+\beta} dt \leq C \left(\int_0^{t_0} |f(t)| t^\beta dt + t_0^{1+\beta} |f(t_0)| \right). \quad (6.5)$$

Since $(-1)^k f^{(k)}$ is non-negative and non-increasing, then we get for $k \in \mathbb{N}^*$ and $c_1 > 0$,

$$\begin{aligned} \int_0^{t_0} |f^{(k)}(c_1 t)| t^{k+\beta} dt &\leq \begin{cases} \int_0^{t_0} (-1)^k f^{(k)}(t) t^{k+\beta} dt, & \text{if } c_1 \geq 1, \\ c_1^{-k-1-\beta} \int_0^{c_1 t_0} (-1)^k f^{(k)}(t) t^{k+\beta} dt, & \text{if } c_1 \leq 1, \end{cases} \\ &\leq \max\{1, c_1^{-k-1-\beta}\} \int_0^{t_0} |f^{(k)}(t)| t^{k+\beta} dt, \end{aligned} \quad (6.6)$$

and

$$\int_0^{t_0} |f(c_1 t)| t^\beta dt = c_1^{-\beta-1} \int_0^{c_1 t_0} |f(t)| t^\beta dt.$$

Choosing $c = \frac{1}{2} \min\{c_1, c_2\}$, by Lemma 6.1-(1) and the fact that $|f'(t)|$ is non-increasing, we have

$$|f^{(m)}(c_1 t)|^\alpha |f^{(n)}(c_2 t)|^{1-\alpha} t^{m\alpha+n(1-\alpha)+\beta} \leq C_{m,n} |f'(ct)| t^\beta.$$

Then (6.3) is a consequence of (6.5) and (6.6).

Now, we move to the proof of (6.4). Using Hölder's inequality and Lemma 6.1, we can deduce that for every $n \in \mathbb{N}^*$,

$$\begin{aligned}
& \int_0^{t_0} |f(c_1 t)|^\alpha |f^{(n)}(c_2 t)|^{1-\alpha} t^{n(1-\alpha)+\beta} dt \\
& \leq \left(\int_0^{t_0} |f(c_1 t) t^\beta| dt \right)^\alpha \left(\int_0^{t_0} |f^{(n)}(c_2 t) t^{n+\beta}| dt \right)^{1-\alpha} \\
& \leq C \left(\int_0^{c_1 t_0} |f(t) t^\beta| dt \right)^\alpha \left(\int_0^{t_0} |f^{(n)}(t) t^{n+\beta}| dt \right)^{1-\alpha} \\
& \leq C \left(|t_0^{1+\beta} f(t_0)| + \int_0^{(c_1 \vee 1)t_0} |f(t) t^\beta| dt \right),
\end{aligned}$$

which corresponds to (6.4). $\square$

6.2. Boundedness property of some operators on the torus

In this subsection, we give useful estimates for the following integral operator

$$\mathcal{T}f(\theta) \triangleq \int_{\mathbb{T}} \mathbb{K}(\theta, \eta) f(\eta) d\eta, \quad (6.7)$$

where $\mathbb{T} = \mathbb{R}/2\pi\mathbb{Z}$ is the torus, $\mathbb{K} : \mathbb{T} \times \mathbb{T} \rightarrow \mathbb{C}$ is the kernel function, and f is a 2π -periodic function.

Lemma 6.4. *Let $\alpha \in (0, 1)$, $n \in \mathbb{N}^*$. Assume the existence of $C > 0$ and functions $H_1(\cdot), \dots, H_{n+1}(\cdot)$ satisfying*

$$\begin{aligned}
& \int_{\mathbb{T}} H_k(|\sin \frac{\eta}{2}|) d\eta \leq C, \quad \forall k = 1, 2, \dots, n, \\
& \int_{\mathbb{T}} |H_n(|\sin \frac{\eta}{2}|)|^\alpha |H_{n+1}(|\sin \frac{\eta}{2}|)|^{1-\alpha} d\eta \leq C,
\end{aligned} \quad (6.8)$$

such that $\mathbb{K} : \mathbb{T} \times \mathbb{T} \rightarrow \mathbb{C}$ satisfies the following properties.

(1) $\mathbb{K}$ is measurable on $\mathbb{T} \times \mathbb{T} \setminus \{(\theta, \theta), \theta \in \mathbb{T}\}$ and

$$|\mathbb{K}(\theta, \theta + \eta)| \leq H_1(|\sin \frac{\eta}{2}|).$$

(2) For each $\eta \in \mathbb{T}$, the mapping $\theta \mapsto \mathbb{K}(\theta, \theta + \eta)$ is n -times differentiable in $\mathbb{T}$ and

$$|\partial_\theta^k(\mathbb{K}(\theta, \theta + \eta))| \leq H_{k+1}(|\sin \frac{\eta}{2}|), \quad \forall k = 1, \dots, n.$$

Then the linear integral operator $\mathcal{T}$ given by (6.7) is continuous from $C^{n-\alpha}(\mathbb{T})$ to $C^{n-\alpha}(\mathbb{T})$ and

$$\|\mathcal{T}f\|_{C^{n-\alpha}(\mathbb{T})} \leq C_n C \|f\|_{C^{n-\alpha}(\mathbb{T})}. \quad (6.9)$$

Proof of Lemma 6.4. The proof is by the induction method. Making change of variables gives

$$\mathcal{T}f(\theta) = \int_{\mathbb{T}} \mathbb{K}(\theta, \theta + \eta) f(\theta + \eta) d\eta. \quad (6.10)$$

First, we start with the case $n = 1$. From (6.8) we have

$$\begin{aligned} |\mathcal{T}f(\theta)| &\leq \|f\|_{L^\infty} \int_{\mathbb{T}} |\mathbb{K}(\theta, \theta + \eta)| d\eta \\ &\leq \|f\|_{L^\infty} \int_{\mathbb{T}} H_1(|\sin \frac{\eta}{2}|) d\eta \leq C \|f\|_{L^\infty}. \end{aligned}$$

By using interpolation inequalities together with the mean value theorem, we infer

$$\begin{aligned} &|\mathbb{K}(\theta_1, \theta_1 + \eta) - \mathbb{K}(\theta_2, \theta_2 + \eta)| \\ &\leq \left(|\mathbb{K}(\theta_1, \theta_1 + \eta)|^\alpha + |\mathbb{K}(\theta_2, \theta_2 + \eta)|^\alpha \right) |\mathbb{K}(\theta_1, \theta_1 + \eta) - \mathbb{K}(\theta_2, \theta_2 + \eta)|^{1-\alpha} \\ &\leq \left(|\mathbb{K}(\theta_1, \theta_1 + \eta)|^\alpha + |\mathbb{K}(\theta_2, \theta_2 + \eta)|^\alpha \right) \left(\int_0^1 |\partial_{\theta_\tau}(\mathbb{K}(\theta_\tau, \theta_\tau + \eta))| d\tau \right)^{1-\alpha} |\theta_1 - \theta_2|^{1-\alpha} \\ &\leq 2 |H_1(|\sin \frac{\eta}{2}|)|^\alpha |H_2(|\sin \frac{\eta}{2}|)|^{1-\alpha} |\theta_1 - \theta_2|^{1-\alpha}, \end{aligned}$$

with $\theta_\tau = \tau\theta_1 + (1 - \tau)\theta_2$. It follows that

$$\begin{aligned} |\mathcal{T}f(\theta_1) - \mathcal{T}f(\theta_2)| &\leq \int_{\mathbb{T}} |\mathbb{K}(\theta_1, \theta_1 + \eta)| |f(\theta_1 + \eta) - f(\theta_2 + \eta)| d\eta \\ &\quad + \int_{\mathbb{T}} |\mathbb{K}(\theta_1, \theta_1 + \eta) - \mathbb{K}(\theta_2, \theta_2 + \eta)| f(\theta_2 + \eta) d\eta. \end{aligned}$$

Therefore,

$$\begin{aligned} |\mathcal{T}f(\theta_1) - \mathcal{T}f(\theta_2)| &\leq |\theta_1 - \theta_2|^{1-\alpha} \|f\|_{C^{1-\alpha}} \int_{\mathbb{T}} H_1(|\sin \frac{\eta}{2}|) d\eta \\ &\quad + 2 |\theta_1 - \theta_2|^{1-\alpha} \|f\|_\infty \int_{\mathbb{T}} |H_1(|\sin \frac{\eta}{2}|)|^\alpha |H_2(|\sin \frac{\eta}{2}|)|^{1-\alpha} d\eta \end{aligned}$$

$$\leq 2C|\theta_1 - \theta_2|^{1-\alpha}\|f\|_{C^{1-\alpha}}.$$

Hence, combining the above estimates yields the desired inequality (6.9) with $n = 1$.

Now, assuming that Lemma 6.4 is true for $n = j$ and for the operator $\mathcal{T}$ given by (6.10), we prove that it also holds for $n = j + 1$. Observe that

$$\partial_\theta(\mathcal{T}f)(\theta) = \int_{\mathbb{T}} \partial_\theta(\mathbb{K}(\theta, \theta + \eta))f(\theta + \eta)d\eta + \int_{\mathbb{T}} \mathbb{K}(\theta, \theta + \eta)\partial_\theta f(\theta + \eta)d\eta. \quad (6.11)$$

In view of the fact that

$$\begin{aligned} \int_{\mathbb{T}} |H_j(|\sin \frac{\eta}{2}|)|^\alpha |H_{j+1}(|\sin \frac{\eta}{2}|)|^{1-\alpha} d\eta &\leq \int_{\mathbb{T}} |H_j(|\sin \frac{\eta}{2}|)| d\eta + \int_{\mathbb{T}} |H_{j+1}(|\sin \frac{\eta}{2}|)| d\eta \\ &\leq C, \end{aligned}$$

and by the inductive hypothesis, we have $\|\mathcal{T}(f)\|_{C^{j-\alpha}(\mathbb{T})} \leq C_j C \|f\|_{C^{j-\alpha}(\mathbb{T})}$ and

$$\left\| \int_{\mathbb{T}} \mathbb{K}(\theta, \theta + \eta) \partial_\theta f(\theta + \eta) d\eta \right\|_{C^{j-\alpha}(\mathbb{T})} = \|\mathcal{T}(\partial_\eta f)\|_{C^{j-\alpha}(\mathbb{T})} \leq C_j C \|f\|_{C^{j+1-\alpha}(\mathbb{T})}.$$

Noting that $\tilde{\mathbb{K}}(\theta, \theta + \eta) = \partial_\theta(\mathbb{K}(\theta, \theta + \eta))$ satisfies

$$|\partial_\theta^k \tilde{\mathbb{K}}(\theta, \theta + \eta)| \leq H_{k+2}(|\sin \frac{\eta}{2}|) \triangleq \tilde{H}_{k+1}(|\sin \frac{\eta}{2}|), \quad \forall k = 0, 1, \dots, j,$$

and $\tilde{H}_k$ ($k = 1, 2, \dots, j + 1$) satisfies (6.8) with $n = j$ and $\tilde{H}_k$ in place of H_k , we use the induction hypothesis to deduce that

$$\begin{aligned} \left\| \int_{\mathbb{T}} \partial_\theta(\mathbb{K}(\theta, \theta + \eta))f(\theta + \eta)d\eta \right\|_{C^{j-\alpha}(\mathbb{T})} &= \left\| \int_{\mathbb{T}} \tilde{\mathbb{K}}(\theta, \theta + \eta)f(\theta + \eta)d\eta \right\|_{C^{j-\alpha}(\mathbb{T})} \\ &\leq C_j C \|f\|_{C^{j-\alpha}(\mathbb{T})}. \end{aligned}$$

Hence, we prove that

$$\|\mathcal{T}f\|_{C^{j+1-\alpha}(\mathbb{T})} = \|\mathcal{T}f\|_{C^{j-\alpha}(\mathbb{T})} + \|\partial_\theta \mathcal{T}f\|_{C^{j-\alpha}(\mathbb{T})} \leq 3C_j C \|f\|_{C^{j+1-\alpha}(\mathbb{T})}.$$

The induction method guarantees that Lemma 6.4 holds for every $n \in \mathbb{N}^*$ and $\alpha \in (0, 1)$. $\square$

In the study of the linearized operator done before, we used the following Mikhlin multiplier type theorem for an operator defined on a periodic function, see for instance [3, Theorem 4.5].

Lemma 6.5. Given $\{a_n\}_{n \in \mathbb{Z}}$ and $h \in L^1(\mathbb{T})$, and define the operator

$$Th(\theta) = \sum_{n \in \mathbb{Z}} a_n \widehat{h}(n) e^{in\theta},$$

where $\widehat{h}(n) = \int_{\mathbb{T}} h(\theta) e^{-in\theta} d\theta$ is the n -th Fourier coefficient of h . Assume that

$$\sup_{n \in \mathbb{Z}} |a_n| < \infty, \quad \text{and} \quad \sup_{n \in \mathbb{Z}} |n(a_{n+1} - a_n)| < \infty,$$

then the operator T is bounded in $C^{k+\alpha}(\mathbb{T})$, for any $k \in \mathbb{N}$ and $\alpha \in (0, 1)$.

6.3. Bessel functions and Hankel transform

In this subsection, we collect some useful properties about Bessel functions and Hankel transform. We recall for instance from [90, Chapter 3] that

$$J_\nu(\mathbf{z}) = \sum_{n=0}^{\infty} \frac{(-1)^n \left(\frac{\mathbf{z}}{2}\right)^{\nu+2n}}{n! \Gamma(\nu + n + 1)}, \quad \forall \mathbf{z}, \nu \in \mathbb{C}, \quad (6.12)$$

$$J_{\nu-1}(\mathbf{z}) - J_{\nu+1}(\mathbf{z}) = 2J'_\nu(\mathbf{z}), \quad \forall \mathbf{z}, \nu \in \mathbb{C}, \quad (6.13)$$

$$\frac{d}{d\mathbf{z}} (\mathbf{z}^\nu J_\nu(\mathbf{z})) = \mathbf{z}^\nu J_{\nu-1}(\mathbf{z}), \quad \forall \mathbf{z}, \nu \in \mathbb{C}. \quad (6.14)$$

In particular, when $\nu = n$ is an integer, then we have according to [90, Chapter 2]

$$J_n(\mathbf{z}) = \frac{1}{2\pi} \int_0^{2\pi} \cos(n\theta - \mathbf{z} \sin \theta) d\theta, \quad \forall \mathbf{z} \in \mathbb{C},$$

$$J_{-n}(\mathbf{z}) = (-1)^n J_n(\mathbf{z}), \quad \forall \mathbf{z} \in \mathbb{C}.$$

Next, we shall introduce Bessel functions of imaginary argument also called modified Bessel functions of first and second kind, see for instance [74, p.66],

$$\mathbf{I}_\nu(\mathbf{z}) = \sum_{n=0}^{\infty} \frac{\left(\frac{\mathbf{z}}{2}\right)^{\nu+2n}}{n! \Gamma(\nu + n + 1)}, \quad \nu \in \mathbb{C}, |\arg(\mathbf{z})| < \pi, \quad (6.15)$$

and

$$\mathbf{K}_\nu(\mathbf{z}) = \frac{\pi}{2} \frac{\mathbf{I}_{-\nu}(\mathbf{z}) - \mathbf{I}_\nu(\mathbf{z})}{\sin(\nu\pi)}, \quad \nu \in \mathbb{C} \setminus \mathbb{Z}, |\arg(\mathbf{z})| < \pi.$$

When $\nu = j \in \mathbb{Z}$, $\mathbf{K}_j$ is defined through the formula $\mathbf{K}_j(\mathbf{z}) = \lim_{\nu \rightarrow j} \mathbf{K}_\nu(\mathbf{z})$. From [43, 8.432.1], we recall the following integral representation

$$\mathbf{K}_\nu(\mathbf{z}) = \int_0^\infty e^{-\mathbf{z} \cosh s} \cosh(\nu s) ds, \quad \forall \nu \in \mathbb{C}, |\arg(\mathbf{z})| < \frac{\pi}{2}. \quad (6.16)$$

Another useful identity that can be found in [43, 8.432.3] deals with the representation in terms of Laplace transform,

$$\mathbf{K}_\nu(\mathbf{z}) = \frac{(\frac{\mathbf{z}}{2})^\nu \Gamma(\frac{1}{2})}{\Gamma(\nu + \frac{1}{2})} \int_1^\infty e^{-s\mathbf{z}} (s^2 - 1)^{\nu - \frac{1}{2}} ds > 0, \quad \operatorname{Re}(\nu + \frac{1}{2}) > 0, |\arg(\mathbf{z})| < \frac{\pi}{2}. \quad (6.17)$$

For $\mathbf{I}_n(x)$ and $\mathbf{K}_n(x)$, we have the asymptotic expansion of small argument (e.g. see [1, p. 375])

$$\forall n \in \mathbb{N}^*, \quad \mathbf{I}_n(x) \underset{x \rightarrow 0}{\sim} \frac{(\frac{1}{2}x)^n}{\Gamma(n+1)}, \quad \text{and} \quad \mathbf{K}_n(x) \underset{x \rightarrow 0}{\sim} \frac{\Gamma(n)}{2(\frac{1}{2}x)^n}. \quad (6.18)$$

The following Nicholson's integral representation of $\mathbf{I}_n(\mathbf{z})\mathbf{K}_n(\mathbf{z})$ is useful in the sequel, see for instance [90, page 441]. For $n \in \mathbb{N}$,

$$\mathbf{I}_n(\mathbf{z})\mathbf{K}_n(\mathbf{z}) = \frac{2(-1)^n}{\pi} \int_0^{\frac{\pi}{2}} \mathbf{K}_0(2\mathbf{z} \cos \theta) \cos(2n\theta) d\theta. \quad (6.19)$$

The following useful result states that the eigenvalues and eigenfunctions of the spectral Laplacian $-\Delta$ on the unit disc $\mathbb{D} \subset \mathbb{R}^2$ have precise expression formula through Bessel functions (e.g., see Section 5.5 of Chapter V in [22]).

Lemma 6.6. *The eigenvalues and the eigenfunctions solving the spectral problem*

$$\text{for } j \geq 1, \quad -\Delta \phi_j = \lambda_j \phi_j, \quad \phi_j|_{\partial \mathbb{D}} = 0, \quad \int_{\mathbb{D}} \phi_j^2(x) dx = 1, \quad (6.20)$$

are described by double index families $(\lambda_{n,k})_{n \in \mathbb{N}, k \in \mathbb{N}^*}$ and $((\phi_{n,k}^{(1)}, \phi_{n,k}^{(2)}))_{n \in \mathbb{N}, k \in \mathbb{N}^*}$ such that

$$\lambda_{n,k} = x_{n,k}^2, \quad \phi_{n,k}^{(1)}(x) = J_n(x_{n,k}|x|)A_{n,k} \cos(n\theta), \quad \phi_{n,k}^{(2)}(x) = J_n(x_{n,k}|x|)A_{n,k} \sin(n\theta), \quad (6.21)$$

where

$$\pi A_{0,k}^2 = \frac{1}{J_1^2(x_{0,k})} \quad \text{and} \quad \pi A_{n,k}^2 = \frac{2}{J_{n+1}^2(x_{n,k})}, \quad \forall n \in \mathbb{N}^*, \quad (6.22)$$

and J_n denotes the Bessel function of order n and $(x_{n,k})_{k \in \mathbb{N}^*}$ are its zeroes.

We also have the following Kneser-Sommerfeld expansion (e.g., see [78, Eq. (12)] or [74, p. 134]) involving the zeros of Bessel functions:

$$\sum_{k=1}^{\infty} \frac{1}{\mathbf{z}^2 + x_{\nu,k}^2} \frac{J_{\nu}(X x_{\nu,k}) J_{\nu}(Y x_{\nu,k})}{J_{\nu+1}^2(x_{\nu,k})} = \frac{1}{2} \frac{\mathbf{I}_{\nu}(X \mathbf{z})}{\mathbf{I}_{\nu}(\mathbf{z})} \left(\mathbf{I}_{\nu}(z) \mathbf{K}_{\nu}(Y \mathbf{z}) - \mathbf{K}_{\nu}(\mathbf{z}) \mathbf{I}_{\nu}(Y \mathbf{z}) \right), \quad (6.23)$$

where $(x_{\nu,k})_{k \in \mathbb{N}^*}$ are k -th zeros of $J_{\nu}(x)$ on the positive real axis and $\nu \in \mathbb{C} \setminus \{-\mathbb{N}^*\}$, $0 \leq X \leq Y \leq 1$, $\mathbf{z} \in \mathbb{C}$.

In what follows, we shall discuss some basic properties of the Hankel transform, and we refer the readers, for instance, to [81, Chap. 9]. First, recall that the ν -th order Hankel transform of $f : (0, \infty) \rightarrow \mathbb{R}$ is defined as

$$\forall r > 0, \quad \mathcal{H}_{\nu} f(r) \triangleq \int_0^{\infty} x f(x) J_{\nu}(rx) dx. \quad (6.24)$$

This transformation is well defined, for example, when f is piecewise continuous and subject to the integrability condition $\int_0^{\infty} |f(r)| \sqrt{r} dr < \infty$. Furthermore, under the following assumptions that f is of class C^2 and

$$\lim_{x \rightarrow \infty} x^{\frac{1}{2}} f(x) = 0, \quad \lim_{x \rightarrow \infty} x^{\frac{1}{2}} f'(x) = 0, \quad \lim_{x \rightarrow 0} x f(x) = 0, \quad (6.25)$$

we have

$$\mathcal{H}_{\nu} \left(\left(\frac{d^2}{dx^2} + \frac{1}{x} \frac{d}{dx} - \frac{\nu^2}{x^2} \right) f(x) \right) = -r^2 \mathcal{H}_{\nu} f(r). \quad (6.26)$$

For $\nu > -\frac{1}{2}$, we also have

$$\mathcal{H}_{\nu}^2 f(x) = f(x). \quad (6.27)$$

6.4. Crandall-Rabinowitz's theorem

The Crandall-Rabinowitz theorem from local bifurcation theory plays a fundamental role in our paper, and for the proof we refer to [23].

Theorem 6.2 (Crandall-Rabinowitz's theorem). *Let X and Y be two Banach spaces, V a neighborhood of 0 in X and let $F : \mathbb{R} \times V \rightarrow Y$ be with the following properties:*

- (1) $F(\lambda, 0) = 0$ for any $\lambda \in \mathbb{R}$.
- (2) The partial derivatives $\partial_{\lambda} F$, $\partial_x F$ and $\partial_{\lambda} \partial_x F$ exist and are continuous.
- (3) $N(\mathcal{L}_0)$ and $Y/R(\mathcal{L}_0)$ are one-dimensional.

(4) *Transversality assumption:* $\partial_\lambda \partial_x F(0, 0)x_0 \notin R(\mathcal{L}_0)$, where

$$N(\mathcal{L}_0) = \text{span}\{x_0\}, \quad \mathcal{L}_0 \triangleq \partial_x F(0, 0).$$

If Z is any complement of $N(\mathcal{L}_0)$ in X , then there is a neighborhood U of $(0, 0)$ in $\mathbb{R} \times X$, an interval $(-a, a)$, and continuous functions $\varphi : (-a, a) \rightarrow \mathbb{R}$, $\psi : (-a, a) \rightarrow Z$ such that $\varphi(0) = 0$, $\psi(0) = 0$ and

$$F^{-1}(0) \cap U = \left\{ (\varphi(\xi), \xi x_0 + \xi \psi(\xi)) : |\xi| < a \right\} \cup \left\{ (\lambda, 0) : (\lambda, 0) \in U \right\}.$$

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